

# Thermodynamics

## Question1

10 mole of oxygen is heated at constant volume from  $30^{\circ}\text{C}$  to  $40^{\circ}\text{C}$ . The change in the internal energy of the gas is \_\_\_\_\_ cal. (The molecular specific heat of oxygen at constant pressure,  $C_p = 7\text{cal/mol.}^{\circ}\text{C}$  and  $R = 2\text{cal. /mol.}^{\circ}\text{C}$ .)

**JEE Main 2026 (Online) 21st January Morning Shift**

**Answer: 500**

### Solution:

For an ideal gas, according to Mayer's relation

$$C_p - C_v = R$$

Where R is the universal gas constant.

$$C_v = C_p - R$$

Given:

- $C_p = 7 \frac{\text{cal}}{\text{mol}^{\circ}\text{C}}$
- $R = 2 \frac{\text{cal}}{\text{mol}^{\circ}\text{C}}$

So, the molecular specific heat of oxygen at constant volume is,

$$C_v = 7 - 2 = 5 \frac{\text{cal}}{\text{mol}^{\circ}\text{C}}$$

Internal energy change is a state function that depends only on the temperature change. For n moles of an ideal gas, it is given by the formula:

$$\Delta U = nC_v \Delta T$$

The given values are,

- Number of moles  $n = 10 \text{ mol}$

- Initial temperature,  $T_1 = 30^\circ\text{C}$
- Final temperature  $T_2 = 40^\circ\text{C}$

So, the change in temperature is,  $\Delta T = T_2 - T_1 = 40 - 30 = 10^\circ\text{C}$

So, the change in internal energy is,

$$\Delta U = 10 \times 5 \times 10 = 500\text{cal}$$

Therefore, when 10 moles of oxygen is heated at constant volume from  $30^\circ\text{C}$  to  $40^\circ\text{C}$  internal energy increase by 500 cal.

Hence, the correct answer is 500 .

## Question2

**A diatomic gas ( $\gamma = 1.4$ ) does 100 J of work when it is expanded isobarically. Then the heat given to the gas \_\_\_\_\_ J.**

**JEE Main 2026 (Online) 21st January Evening Shift**

**Answer: 350**

### Solution:

An isobaric process occurs at constant pressure (P). For isobaric process :

The work done is  $W = P\Delta V = nR\Delta T$

The change in internal energy is  $\Delta U = nC_v\Delta T$

Heat added to the system is,  $Q = nC_p\Delta T$

The ratio of specific heats is defined as  $\gamma = \frac{C_p}{C_v}$ .

Also, from Mayer's relation  $C_p - C_v = R$ .

The ratio of Heat to Work is:

$$\frac{Q}{W} = \frac{nC_p\Delta T}{nR\Delta T} = \frac{C_p}{R}$$

Substituting  $R = C_p - C_v$  :

$$\frac{Q}{W} = \frac{C_p}{C_p - C_v}$$

$$\Rightarrow \frac{Q}{W} = \frac{\frac{C_p}{C_v}}{\left(\frac{C_p}{C_v}\right) - 1} = \frac{\gamma}{\gamma - 1}$$

The work done is given as,  $W = 100\text{ J}$  and the ratio of specific heats  $\gamma = 1.4$  (for a diatomic gas) So,

$$\frac{Q}{100} = \frac{1.4}{1.4-1}$$

$$\Rightarrow \frac{Q}{100} = \frac{1.4}{0.4}$$

$$\Rightarrow \frac{Q}{100} = 3.5$$

$$\Rightarrow Q = 350 \text{ J}$$

Therefore, the heat given to the gas is 350 J .

Hence, the correct option is **350**.

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## Question3

**An insulated cylinder of volume  $60 \text{ cm}^3$  is filled with a gas at  $27^\circ\text{C}$  and 2 atmospheric pressure. Then the gas is compressed making the final volume as  $20 \text{ cm}^3$  while allowing the temperature to rise to  $77^\circ\text{C}$  . The final pressure is \_\_\_\_\_ atmospheric pressure.**

**JEE Main 2026 (Online) 22nd January Evening Shift**

**Answer: 7**

**Solution:**

Initial state of the gas,

Initial Volume,  $V_1 = 60 \text{ cm}^3$

Initial Temperature,  $T_1 = 27^\circ\text{C} = (27 + 273)\text{K} = 300 \text{ K}$

Initial Pressure,  $P_1 = 2 \text{ atm}$

Final state of the gas is,

Final Volume,  $V_2 = 20 \text{ cm}^3$

Final Temperature,  $T_2 = 77^\circ\text{C} = (77 + 273)\text{K} = 350 \text{ K}$

We need to calculate the pressure of the gas at final state.

Using Ideal gas equation,  $PV = nRT \Rightarrow \frac{PV}{T} = nR = \text{constant}$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\Rightarrow P_2 = \frac{P_1 V_1 T_2}{T_1 V_2}$$

$$\Rightarrow P_2 = \frac{2 \times 60 \times 350}{300 \times 20} \text{ atm}$$

$$\Rightarrow P_2 = 7 \text{ atm}$$

## Question4

A gas of certain mass filled in a closed cylinder at a pressure of 3.23 kPa has temperature  $50^\circ\text{C}$ . The gas is now heated to double its temperature. The modified pressure is \_\_\_\_ Pa .

**JEE Main 2026 (Online) 24th January Morning Shift**

**Answer: 3730**

### Solution:

The problem states the gas is filled in a closed cylinder. That means the volume of the gas remains constant during the heating process (  $V = \text{constant}$  ).

For a fixed mass of gas at constant volume, the pressure is directly proportional to the absolute temperature.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Initial pressure is  $P_1 = 3.23\text{kPa} = 3.23 \times 1000 \text{ Pa} = 3230 \text{ Pa}$

Initial temperature is  $T_1 = 50^\circ\text{C} = (50 + 273)\text{K} = 323 \text{ K}$

The final temperature is  $T_2 = 2 \times T_1 = 2 \times 323 \text{ K} = 646 \text{ K}$

Using the relation  $\frac{P_1}{T_1} = \frac{P_2}{T_2}$  :

$$P_2 = P_1 \times \frac{T_2}{T_1}$$

$$\Rightarrow P_2 = 3230 \times \frac{646}{323}$$

$$P_2 = 3230 \times 2 = 6460 \text{ Pa}$$

Therefore, the modified pressure is 6460 Pa . Hence, the correct answer is 6460 .

**Other Solution :**



$V = \text{constant}$

so  $P \propto T$

$$T_i = 50^\circ\text{C} = 323 \text{ K}$$

$$T_f = 100^\circ\text{C} = 373 \text{ K}$$

$$\Rightarrow \frac{P_f}{P_i} = \frac{T_f}{T_i}$$

$$\Rightarrow \frac{P_f}{3.23\text{kPa}} = \frac{373}{323}$$

$$\Rightarrow P_f = 3730 \text{ Pa}$$

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## Question5

**When 300 J of heat given to an ideal gas with  $C_p = \frac{7}{2}R$  its temperature raises from  $20^\circ\text{C}$  to  $50^\circ\text{C}$  keeping its volume constant. The mass of the gas is (approximately) \_\_\_\_\_ g. ( $R = 8.314 \text{ J/mol. K}$ )**

**JEE Main 2026 (Online) 24th January Evening Shift**

**Answer: 481**

### Solution:

For a constant volume process, the heat supplied (  $Q$  ) is given by the formula :

$$Q = nC_v\Delta T$$

Where :

- 1.  $n$  = number of moles of the gas.
- 2.  $C_v$  = molar specific heat capacity at constant volume.
- 3.  $\Delta T$  = change in temperature  $50^\circ\text{C} - 20^\circ\text{C} = 30^\circ\text{C} = 30 \text{ K}$

We are given the molar specific heat at constant pressure,  $C_p = \frac{7}{2}R$ . Using Mayer's Relation ( $C_p - C_v = R$ ) :

$$C_v = C_p - R$$

$$\Rightarrow C_v = \frac{7}{2}R - R = \frac{5}{2}R$$

Using the formula of heat,

$$300 = n \times \left(\frac{5}{2} \times 8.314\right) \times 30$$

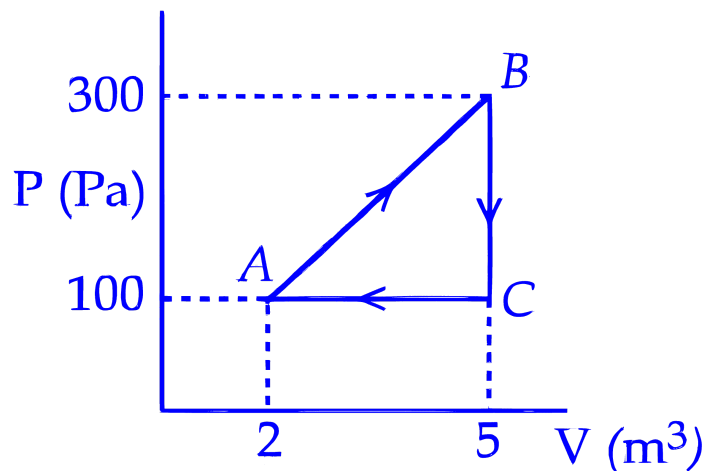
$$n = \frac{300 \times 2}{5 \times 8.314 \times 30} \approx 0.481 \text{ moles}$$

So, the amount of the gas is 0.481 moles. As the molar mass of the gas is not given in the question, we can't determine the mass of the gas directly.

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## Question6

A thermodynamic system is taken through the cyclic process  $ABC$  as shown in the figure. The total work done by the system during the cycle  $ABC$  is \_\_\_\_\_ J.

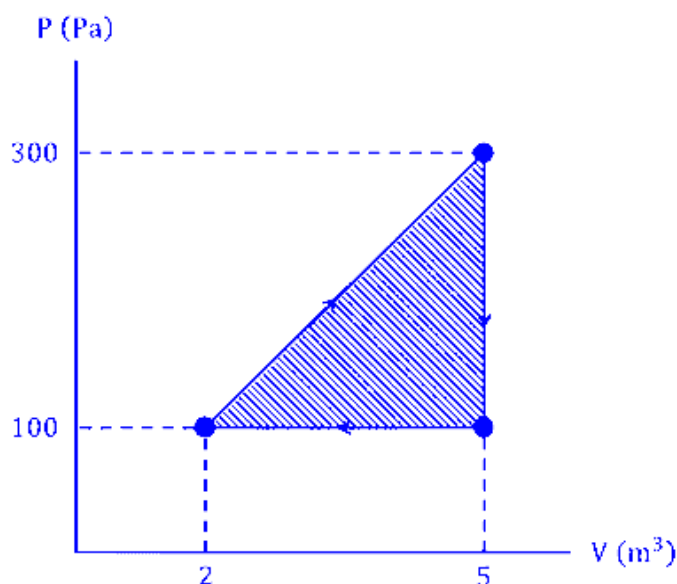


**JEE Main 2026 (Online) 28th January Evening Shift**

**Answer: 300**

**Solution:**

In a P-V diagram (Pressure vs. Volume), the total work done during a cyclic process is equivalent to the area enclosed by the cycle.



The cycle  $A \rightarrow B \rightarrow C \rightarrow A$  is moving in a clockwise direction.

In a clockwise cycle on a P-V diagram, the work done by the system is positive.

The area enclosed by the triangle ABC is calculated using the formula for the area of a right-angled triangle:

$$\text{Work Done} = \text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$\text{Base} = V_C - V_A = 5 \text{ m}^3 - 2 \text{ m}^3 = 3 \text{ m}^3$$

$$\text{Height} = P_B - P_C = 300 \text{ Pa} - 100 \text{ Pa} = 200 \text{ Pa}$$

Substituting the values into the area formula:

$$\text{Work Done} = \frac{1}{2} \times 3 \text{ m}^3 \times 200 \text{ Pa} = 300 \text{ J}$$

Therefore, the total work done by the system during the cycle ABC is 300 J . Hence, the correct answer is **300**.

## Question7

**A vessel contains  $0.15 \text{ m}^3$  of a gas at pressure 8 bar and temperature  $140^\circ\text{C}$  with  $c_p = 3R$  and  $c_v = 2R$ . It is expanded adiabatically till pressure falls to 1 bar. The work done during this process is \_\_\_\_\_ kJ. (R is gas constant)**

**JEE Main 2026 (Online) 2nd April Morning Shift**

**Answer: 120**

## Solution:

In an adiabatic process, there is no heat exchange with the surroundings ( $\Delta Q = 0$ ).

The adiabatic constant is the ratio of specific heats:

$$\gamma = \frac{c_p}{c_v} = \frac{3R}{2R} = 1.5$$

For an adiabatic process:

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

The initial pressure of the gas is  $P_1 = 8$  bar,

The initial volume is  $V_1 = 0.15 \text{ m}^3$ , and

The final pressure is  $P_2 = 1$  bar:

$$8 \times (0.15)^{1.5} = 1 \times V_2^{1.5}$$

$$V_2^{1.5} = 8 \times (0.15)^{1.5}$$

$$V_2 = 8^{2/3} \times 0.15 = (2^3)^{2/3} \times 0.15 = 4 \times 0.15 = 0.6 \text{ m}^3$$

So, the final volume of the gas is  $0.6 \text{ m}^3$

From the first law of thermodynamics,  $\Delta Q = \Delta U + W$ . Since  $\Delta Q = 0$ :

$$W = -\Delta U = -nc_v(T_2 - T_1)$$

Using the ideal gas relation  $PV = nRT$ ,

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

As  $1 \text{ bar} = 10^5 \text{ Pa}$ :

So,  $P_1 = 8 \times 10^5 \text{ Pa}$  and  $P_2 = 1 \times 10^5 \text{ Pa}$

Putting in the work done formula,

$$W = \frac{(8 \times 10^5 \times 0.15) - (1 \times 10^5 \times 0.6)}{1.5 - 1}$$

$$W = \frac{1.2 \times 10^5 - 0.6 \times 10^5}{0.5}$$

$$W = \frac{0.6 \times 10^5}{0.5} = 1.2 \times 10^5 \text{ J} = 120 \text{ kJ}$$

Therefore, the work done during this process is 120 kJ. Thus, the correct answer is 120.

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## Question8

**5 moles of unknown gas is heated at constant volume from 10°C to 20°C. The molar specific heat of this gas at constant pressure  $c_p = 8$  cal/mol.°C and  $R = 8.36$  J/mol.°C. The change in the internal energy of the gas is \_\_\_\_\_ calorie.**

## **JEE Main 2026 (Online) 2nd April Evening Shift**

**Answer: 300**

### **Solution:**

Using Mayer's relation for an ideal gas, the relationship between molar specific heat at constant pressure ( $C_p$ ) and constant volume ( $C_v$ ) is given by:

$$C_p - C_v = R$$

The gas constant  $R$  in is in terms of joules, we need to convert it to  $\frac{\text{cal}}{\text{mol} \cdot ^\circ\text{C}}$ .

We know that  $1 \text{ cal} \approx 4.18\text{J}$ .

$$R = \frac{8.36\text{J/mol} \cdot ^\circ\text{C}}{4.18\text{J/cal}} = \frac{2\text{cal}}{\text{mol} \cdot ^\circ\text{C}}$$

Now, substituting the values into Mayer's relation:

$$8 - C_v = 2$$

$$C_v = \frac{6\text{cal}}{\text{mol} \cdot ^\circ\text{C}}$$

The change in internal energy for  $n$  moles of gas when its temperature changes by  $\Delta T$  is:

$$\Delta U = n C_v \Delta T$$

The amount of gas is  $n = 5$  moles

The change in temperature is

$$\Delta T = T_{\text{final}} - T_{\text{initial}} = 20^\circ\text{C} - 10^\circ\text{C} = 10^\circ\text{C}$$

Putting the values,

$$\Delta U = 5 \times 6 \times 10$$

$$\Delta U = 300 \text{ calories}$$

Therefore, the change in internal energy of the gas is 300 Calories. Thus, the correct answer is 300.

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## **Question9**

The heat extracted out of  $x$  gram of water initially at  $50^{\circ}\text{C}$  to cool it down to  $0^{\circ}\text{C}$  is sufficient to evaporate  $(1000 - x)$  gram of water also initially at  $50^{\circ}\text{C}$ . The value of  $x$  (closest integer) is \_\_\_\_ .

(Take latent heat of water  $2256 \text{ kJ/kg} \cdot \text{K}$ , specific heat capacity of water  $4200 \text{ J/kg} \cdot \text{K}$  )

## JEE Main 2026 (Online) 5th April Evening Shift

**Answer: 922**

### Solution:

When  $x$  grams of water cools from  $50^{\circ}\text{C}$  to  $0^{\circ}\text{C}$ , the heat released is:

$$Q_1 = m_1 \cdot c \cdot \Delta T$$

Where,  $m_1 = x\text{g} = \frac{x}{1000} \text{ kg}$  is the mass of water,  $c = 4200 \text{ J/kg} \cdot \text{K}$  is the specific heat capacity of water, and  $\Delta T = 50 - 0 = 50^{\circ}\text{C}$

Putting the values,

$$Q_1 = \left(\frac{x}{1000}\right) \times 4200 \times 50$$

$$\Rightarrow Q_1 = x \times 4.2 \times 50 = 210xJ$$

The water to be evaporated is initially at  $50^{\circ}\text{C}$ . To evaporate it, it must first be heated to  $100^{\circ}\text{C}$  and then converted to steam.

$$\text{Let } m_2 = (1000 - x)\text{g} = \frac{1000-x}{1000} \text{ kg}.$$

Heating from  $50^{\circ}\text{C}$  to  $100^{\circ}\text{C}$

$$Q_{2a} = m_2 \cdot c \cdot (100 - 50)$$

$$Q_{2a} = \left(\frac{1000-x}{1000}\right) \times 4200 \times 50 = (1000 - x) \times 210 J$$

Latent heat of vaporization at  $100^{\circ}\text{C}$

$$Q_{2b} = m_2 \cdot L$$

Where the latent heat of water is  $L = 2256 \text{ kJ/kg} = 2,256,000 \text{ J/kg}$ .

$$Q_{2b} = \left(\frac{1000-x}{1000}\right) \times 2,256,000 = (1000 - x) \times 2256J$$

So, the total heat for evaporation is

$$Q_2 = Q_{2a} + Q_{2b} = (1000 - x) \times [210 + 2256] = (1000 - x) \times 2466J$$

Using conservation of heat energy, the total amount of heat gained is equal to total amount of heat rejected.

$$Q_1 = Q_2$$

$$210x = (1000 - x)2466$$

$$210x = 2,466,000 - 2466x$$

$$210x + 2466x = 2,466,000$$

$$2676x = 2,466,000$$

$$x = \frac{2,466,000}{2676} \approx 921.52$$

Therefore, the value of x (closest integer) is 922 .

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## Question10

**A gas based geyser heats water flowing at the rate of 5.0 litres per minute from  $27^{\circ}\text{C}$  to  $87^{\circ}\text{C}$ . The rate of consumption of the gas is \_\_\_\_\_ g/s.**

**(Take heat of combustion of gas =  $5.0 \times 10^4 \text{ J/g}$  ) specific heat capacity of water =  $4200 \text{ J/kg} \cdot ^{\circ}\text{C}$**

### JEE Main 2026 (Online) 21st January Morning Shift

**Options:**

A.

4.2

B.

2.1

C.

0.21

D.

0.42

**Answer: D**

**Solution:**

The heat energy (  $Q$  ) required to raise the temperature of a mass (  $m$  ) is given by  $Q = mc\Delta T$

Where  $c$  is the specific heat capacity and  $\Delta T$  is the change in temperature.

Temperature change is  $\Delta T = 87^\circ\text{C} - 27^\circ\text{C} = 60^\circ\text{C}$

Specific heat is  $c = 4200 \text{ J/kg}^\circ\text{C}$ .

The heat required per unit time is  $P_{\text{req}} = \frac{dQ}{dt} = \frac{dm}{dt} \cdot c \cdot \Delta T$

Water flows at  $5.0 \text{ litres / min}$ .

Since the density of water is  $1 \text{ kg/L}$ , the mass flow rate is  $5.0 = \frac{5.0}{60} \text{ kg/sec}$

$$P_{\text{req}} = \left( \frac{5.0}{60} \right) \times 4200 \times 60 \frac{\text{J}}{\text{s}} = 21000 \text{ J/s}$$

Let the rate of consumption of gas be  $R \text{ g/s}$ .

The heat produced per second is,

$$P_{\text{prod}} = R \times H$$

Given heat of combustion of gas  $H = 5.0 \times 10^4 \text{ J/g}$

$$P_{\text{prod}} = R \times 50000 \text{ J/s}$$

Assuming 100% efficiency (because no heat loss is given),  $P_{\text{prod}} = P_{\text{req}}$

$$R \times 50000 = 21000$$

$$\Rightarrow R = \frac{21000}{50000} = 0.42 \text{ g/s}$$

Therefore, the gas must be consumed at a rate of  $0.42 \text{ g/s}$  to maintain the temperature rise.

Hence, the correct option is (D).

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## Question11

**The r.m.s. speed of oxygen molecules at  $47^\circ\text{C}$  is equal to that of the hydrogen molecules kept at \_\_\_\_\_  $^\circ\text{C}$ . (Mass of oxygen molecule/mass of hydrogen molecule =  $32/2$ )**

**JEE Main 2026 (Online) 21st January Evening Shift**

**Options:**

A.



B.

-253

C.

-20

D.

-235

**Answer: B**

### Solution:

According to the Kinetic Theory of Gases, the r.m.s speed of an ideal gas is given by :

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Where T is the absolute temperature and M is the molecular mass (in kg ).

It is given that the r.m.s. speed of oxygen ( $\text{O}_2$ ) at  $T_{\text{O}_2}$  is equal to the r.m.s. speed of hydrogen ( $\text{H}_2$ ) at  $T_{\text{H}_2}$ .

$$V_{\text{rms}(\text{O}_2)} = V_{\text{rms}(\text{H}_2)}$$

$$\Rightarrow \sqrt{\frac{3RT_{\text{O}_2}}{M_{\text{O}_2}}} = \sqrt{\frac{3RT_{\text{H}_2}}{M_{\text{H}_2}}}$$

$$\Rightarrow \frac{T_{\text{O}_2}}{M_{\text{O}_2}} = \frac{T_{\text{H}_2}}{M_{\text{H}_2}}$$

Temperature of oxygen is  $T_{\text{O}_2} = 47 + 273 = 320 \text{ K}$ .

Molar masses are  $M_{\text{O}_2} = 32 \times 10^{-3} \text{ kg}$  and  $M_{\text{H}_2} = 2 \times 10^{-3} \text{ kg}$ .

$$\frac{320}{32} = \frac{T_{\text{H}_2}}{2}$$

$$\Rightarrow 10 = \frac{T_{\text{H}_2}}{2} \Rightarrow T_{\text{H}_2} = 20 \text{ K}$$

$$\Rightarrow T_{\text{H}_2} = 20 - 273$$

$$\Rightarrow T_{\text{H}_2} = -253^\circ \text{C}$$

Therefore, the rms speed of oxygen molecules at  $47^\circ \text{C}$  is equal to that of the hydrogen molecules kept at  $-253^\circ \text{C}$ .

Hence, the correct option is (B).

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## Question12

**The volume of an ideal gas increases 8 times and temperature becomes  $(1/4)^{\text{th}}$  of initial temperature during a reversible change. If there is no exchange of heat in this process ( $\Delta Q = 0$ ) then identify the gas from the following options (Assuming the gases given in the options are ideal gases) :**

### **JEE Main 2026 (Online) 22nd January Morning Shift**

**Options:**

A.

$\text{NH}_3$

B.

$\text{O}_2$

C.

$\text{CO}_2$

D.

He

**Answer: D**

**Solution:**

It is given that there is no exchange of heat in this process, which means  $\Delta Q = 0$ , i.e., this is an adiabatic process.

For an ideal gas undergoing a reversible adiabatic process, the relationship between Temperature (T) and Volume (V) is given by the equation:

$$T \cdot V^{\gamma-1} = \text{constant}$$

Where  $\gamma$  is the adiabatic index or the ratio of specific heats  $\left(\frac{C_p}{C_v}\right)$ . This  $\gamma$  value is different depending on the atomicity of the gases (monoatomic, diatomic, polyatomic).

$$T_1 \cdot V_1^{\gamma-1} = T_2 \cdot V_2^{\gamma-1}$$

Final volume ( $V_2$ ) is 8 times the initial volume ( $V_1$ )  $\Rightarrow V_2 = 8V_1$

Final temperature ( $T_2$ ) is  $(\frac{1}{4})^{\text{th}}$  the initial temperature ( $T_1$ )  $\Rightarrow T_2 = \frac{T_1}{4}$

$$T_1 \cdot V_1^{\gamma-1} = \left(\frac{T_1}{4}\right) \cdot (8V_1)^{\gamma-1}$$

$$\Rightarrow V_1^{\gamma-1} = \frac{1}{4} \cdot 8^{\gamma-1} \cdot V_1^{\gamma-1}$$

$$\Rightarrow 1 = \frac{1}{4} \cdot 8^{\gamma-1}$$

$$\Rightarrow 4 = 8^{\gamma-1}$$

$$\Rightarrow 2^2 = 2^{3(\gamma-1)}$$

$$\Rightarrow 2 = 3(\gamma - 1) \Rightarrow 2 = 3\gamma - 3$$

$$\Rightarrow 5 = 3\gamma$$

$$\Rightarrow \gamma = \frac{5}{3}$$

The value of  $\gamma$  is directly related to the degrees of freedom ( $f$ ) of the gas molecules by the formula :

$$\gamma = 1 + \frac{2}{f}$$

$$\Rightarrow \frac{5}{3} = 1 + \frac{2}{f}$$

$$\Rightarrow \frac{5}{3} - 1 = \frac{2}{f}$$

$$\Rightarrow \frac{2}{3} = \frac{2}{f}$$

$$\Rightarrow f = 3$$

A gas with 3 degrees of freedom (all translational, no rotational or vibrational modes activated) is a monoatomic gas.

$\text{NH}_3$  (Ammonia) is polyatomic molecule (non-linear).

$\text{O}_2$  (Oxygen) is a diatomic molecule.

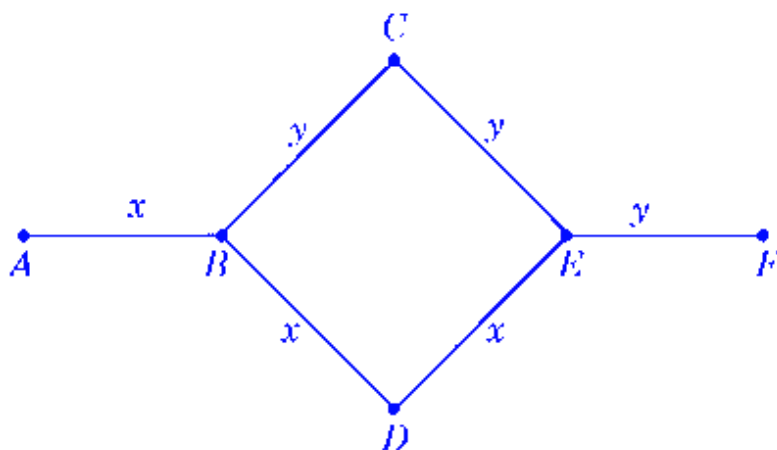
$\text{CO}_2$  (Carbon Dioxide) is polyatomic molecule (linear).

He (Helium) is a noble gas, exists as single atoms. It is a monoatomic gas. Therefore, the given gas is a monoatomic gas (He).

Hence, the correct option is (D).

## Question13

**Rods  $x$  and  $y$  of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points  $A$  and  $F$  are maintained at  $100^\circ\text{C}$  and  $40^\circ\text{C}$  respectively. Given the thermal conductivity of rod  $x$  is three times of that of rod  $y$ , the temperature at junction points  $B$  and  $E$  are (close to):**



## JEE Main 2026 (Online) 22nd January Morning Shift

**Options:**

A.

$60^{\circ}\text{C}$  and  $45^{\circ}\text{C}$  respectively

B.

$80^{\circ}\text{C}$  and  $70^{\circ}\text{C}$  respectively

C.

$89^{\circ}\text{C}$  and  $73^{\circ}\text{C}$  respectively

D.

$80^{\circ}\text{C}$  and  $60^{\circ}\text{C}$  respectively

**Answer: C**

**Solution:**

Using law of heat conduction the rate of heat flow  $\left(H = \frac{dQ}{dt}\right)$  through a material is proportional to the crosssectional area (A) and the temperature difference (  $\Delta T$  ), and inversely proportional to the length (L) :

$$H = \frac{kA\Delta T}{L}$$

Where k is the thermal conductivity of the material.

We can rearrange this equation to look like Ohm's Law for electricity  $\left(I = \frac{\Delta V}{R}\right)$  :

$$H = \frac{\Delta T}{L/kA}$$

Here, the term  $L/kA$  acts as the thermal resistance ( $R_{th}$ ) of the rod. Heat current ( $H$ ) is analogous to electrical current (I), and temperature difference (  $\Delta T$  ) is analogous to potential difference (  $\Delta V$  ).

$$R_{th} = \frac{L}{kA}$$

It is given that rods  $x$  and  $y$  have equal dimensions, meaning they have the same length ( $L$ ) and same crosssectional area ( $A$ ).

We are also given that the thermal conductivity of rod  $x$  ( $k_x$ ) is three times that of rod  $y$  ( $k_y$ ).

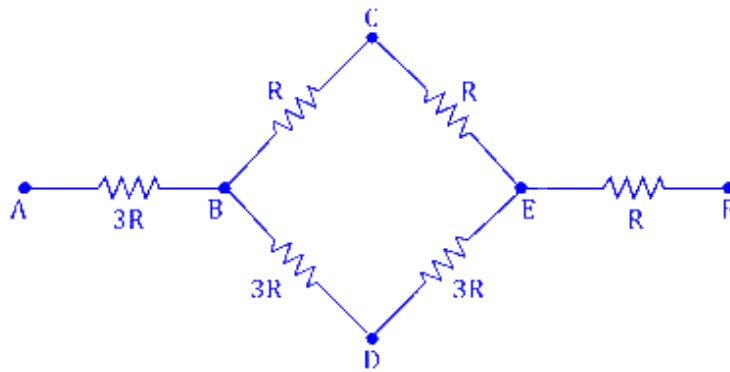
Let  $k_y = k$ , then  $k_x = 3k$ .

So, the thermal resistances of rod  $x$  and  $y$  will be,

$$R_y = \frac{L}{k_y A} = \frac{L}{kA} = R$$

$$\Rightarrow R_x = \frac{L}{k_x A} = \frac{L}{3kA} = \frac{1}{3} \left( \frac{L}{kA} \right) = \frac{R}{3}$$

We can now treat the given structure as an electrical circuit with specific resistances.



The heat current splits into two parallel paths  $BC$  and  $BD$  at  $B$ .

$$R_{BCE} = R_{BC} + R_{CE} = R + R = 2R$$

$$\Rightarrow R_{BDE} = R_{BD} + R_{DE} = \frac{R}{3} + \frac{R}{3} = \frac{2R}{3}$$

For parallel resistors  $R_1$  and  $R_2$ , the equivalent resistance is  $\frac{R_1 R_2}{R_1 + R_2}$  :

$$R_{BE} = \frac{(2R) \times \left(\frac{2R}{3}\right)}{(2R) + \left(\frac{2R}{3}\right)}$$

$$\Rightarrow R_{BE} = \frac{4R^2/3}{\frac{6R+2R}{3}} = \frac{4R^2}{8R} = \frac{R}{2}$$

From  $A$  to  $F$  with three equivalent resistors:  $R_{AB}$ ,  $R_{BE}$ , and  $R_{EF}$ .

The total equivalent resistance ( $R_{total}$ ) is :

$$R_{total} = R_{AB} + R_{BE} + R_{EF}$$

$$\Rightarrow R_{total} = \frac{R}{3} + \frac{R}{2} + R$$

$$\Rightarrow R_{total} = \frac{11R}{6}$$

We know the overall temperature difference between  $A$  and  $F$  :

$$\Delta T_{total} = T_A - T_F = 100 - 40 = 60^\circ\text{C}$$

The total steady-state heat current ( $H$ ) flowing through the entire system is :

$$H = \frac{\Delta T_{\text{total}}}{R_{\text{total}}} = \frac{60}{11R/6} = \frac{360}{11R}$$

Consider the segment from A to B :

$$H = \frac{T_A - T_B}{R_{AB}}$$

$$\Rightarrow \frac{360}{11R} = \frac{100 - T_B}{R/3}$$

$$\Rightarrow \frac{360}{11R} = \frac{3(100 - T_B)}{R}$$

$$\Rightarrow \frac{120}{11} = 100 - T_B$$

$$\Rightarrow T_B = 100 - \frac{120}{11} = \frac{1100-120}{11} = \frac{980}{11}$$

$$\Rightarrow T_B \approx 89.09^\circ\text{C}$$

Consider the segment from E to F :

$$H = \frac{T_E - T_F}{R_{EF}}$$

$$\Rightarrow \frac{360}{11R} = \frac{T_E - 40}{R}$$

$$\Rightarrow \frac{360}{11} = T_E - 40$$

$$\Rightarrow T_E = 40 + \frac{360}{11} = \frac{440+360}{11} = \frac{800}{11}$$

$$\Rightarrow T_E \approx 72.72^\circ\text{C}$$

Therefore, the temperatures at junction points B and E are close to  $89^\circ\text{C}$  and  $73^\circ\text{C}$  respectively.

Hence, the correct option is (C).

## Question14

Consider two boxes containing ideal gases  $A$  and  $B$  such that their temperatures, pressures and number densities are same. The molecular size of  $A$  is half of that of  $B$  and mass of molecule  $A$  is four times that of  $B$ . If the collision frequency in gas  $B$  is  $32 \times 10^{18}/\text{s}$  then collision frequency in gas  $A$  is \_\_\_\_\_ /s.

**JEE Main 2026 (Online) 22nd January Evening Shift**

**Options:**

A.

$$8 \times 10^{18}$$

B.

$$2 \times 10^{18}$$

C.

$$32 \times 10^{18}$$

D.

$$4 \times 10^{18}$$

**Answer: D**

## Solution:

For an ideal gas, the collision frequency of a molecule is given by the kinetic theory of gases :

$$Z = \sqrt{2} \pi d^2 v_{\text{avg}} n$$

Where :

- $d$  is the molecular diameter (or size).
- $V_{\text{avg}}$  is the average speed of the gas molecules.
- $n$  is the number density (number of molecules per unit volume).

The average speed ( $v_{\text{avg}}$ ) of gas molecules depends on the absolute temperature ( $T$ ) and the mass of a single molecule ( $m$ ) according to the formula :

$$v_{\text{avg}} = \sqrt{\frac{8kT}{\pi m}}$$

Where  $k$  is the Boltzmann constant.

Substituting this into the collision frequency formula gives :

$$Z = \sqrt{2} \pi d^2 \left( \sqrt{\frac{8kT}{\pi m}} \right) n$$

It is given that the temperatures ( $T$ ), pressures, and number densities ( $n$ ) are exactly the same for both gases A and B.

$$Z \propto \frac{d^2}{\sqrt{m}}$$

$$\Rightarrow \frac{Z_A}{Z_B} = \left( \frac{d_A}{d_B} \right)^2 \times \sqrt{\frac{m_B}{m_A}}$$

$$\text{Molecular size of A is half that of B} \Rightarrow d_A = \frac{1}{2} d_B \Rightarrow \frac{d_A}{d_B} = \frac{1}{2}$$

$$\text{Mass of molecule A is four times that of B} \Rightarrow m_A = 4 m_B \Rightarrow \frac{m_B}{m_A} = \frac{1}{4}$$

$$\text{Collision frequency in gas B is } Z_B = 32 \times 10^{18} / \text{s}.$$

$$\frac{Z_A}{Z_B} = \left(\frac{1}{2}\right)^2 \times \sqrt{\frac{1}{4}}$$

$$\Rightarrow \frac{Z_A}{Z_B} = \frac{1}{4} \times \frac{1}{2}$$

$$\Rightarrow \frac{Z_A}{Z_B} = \frac{1}{8} \Rightarrow Z_A = \frac{Z_B}{8} = \frac{32 \times 10^{18}}{8} = 4 \times 10^{18}/s$$

So, the collision frequency in gas A is  $4 \times 10^{18}/s$ .

---

## Question15

**An air bubble of volume  $2.9 \text{ cm}^3$  rises from the bottom of a swimming pool of 5 m deep. At the bottom of the pool water temperature is  $17^\circ\text{C}$ . The volume of the bubble when it reaches the surface, where the water temperature is  $27^\circ\text{C}$ , is \_\_\_\_\_  $\text{cm}^3$ .**

**(  $g = 10 \text{ m/s}^2$ , density of water =  $10^3 \text{ kg/m}^3$ , and 1 atm pressure is  $10^5 \text{ Pa}$  )**

**JEE Main 2026 (Online) 23rd January Evening Shift**

**Options:**

A.

2.0

B.

4.2

C.

3.0

D.

4.5

**Answer: D**

**Solution:**

Initial state (at the bottom)



$$\text{Volume} = V_1 = 2.9 \text{ cm}^3$$

$$\text{Temperature} = T_1 = 17^\circ\text{C} = 17 + 273 = 290 \text{ K}$$

$$\text{Pressure} = P_1$$

Total pressure at the bottom is atmospheric pressure ( $P_0$ ) plus the hydrostatic pressure ( $h\rho g$ ).

$$P_1 = 10^5 + 10^3 \times 10 \times 5 = 1.5 \times 10^5 \text{ Pa}$$

Final state (at the surface)

$$\text{Temperature} = T_2 = 27^\circ\text{C} = 27 + 273 = 300 \text{ K}$$

Pressure =  $P_2 = P_0$  (Only atmospheric pressure acts at the surface.)

$$P_2 = 10^5 \text{ Pa}$$

According to the Ideal Gas Law :

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\Rightarrow \frac{(1.5 \times 10^5) \times 2.9}{290} = \frac{(10^5) \times V_2}{300}$$

$$\Rightarrow \frac{1.5 \times 2.9}{290} = \frac{V_2}{300}$$

$$\Rightarrow V_2 = \frac{1.5 \times 2.9 \times 300}{290}$$

$$\Rightarrow V_2 = 4.5 \text{ cm}^3$$

So, as the bubble rises, the external pressure decreases significantly while the temperature increases slightly. Both factors contribute to the expansion of the air bubble, resulting in a final volume of  $4.5 \text{ cm}^3$ .

Therefore, the correct option is (D).

## Question 16

**The internal energy of a monoatomic gas is  $3nRT$ . One mole of helium is kept in a cylinder having internal cross section area of  $17 \text{ cm}^2$  and fitted with a light movable frictionless piston. The gas is heated slowly by supplying  $126 \text{ J}$  heat. If the temperature rises by  $4^\circ\text{C}$ , then the piston will move \_\_\_\_\_ cm.**

**(atmospheric pressure =  $10^5 \text{ Pa}$ )**

**JEE Main 2026 (Online) 23rd January Evening Shift**

### Options:

A.

1.55

B.

14.5

C.

15.5

D.

1.45

**Answer: C**

### Solution:

According to the first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

Where,

The heat supplied to a system  $= \Delta Q = 126 \text{ J}$

The change in internal energy  $= \Delta U = 3nRT$

The work done by the system  $= \Delta W$

Number of moles  $n = 1$  mole of Helium.

Change in temperature  $= \Delta T = 4^\circ\text{C} = 4 \text{ K}$

Since the piston is light, movable, and frictionless, the gas pressure remains equal to the atmospheric pressure  $= 10^5 \text{ Pa}$ .

$$\text{Area} = A = 17 \text{ cm}^2 = 17 \times 10^{-4} \text{ m}^2$$

Using the formula provided :

$$\Delta U = 3 \cdot n \cdot R \cdot \Delta T$$

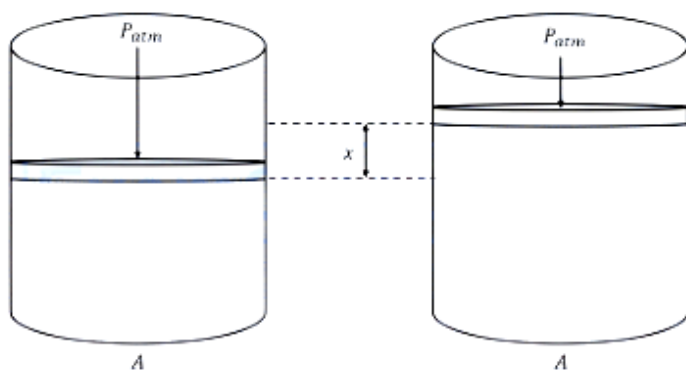
$$\Rightarrow \Delta U = 3 \cdot 1 \cdot 8.314 \cdot 4 \text{ J}$$

$$\Delta U = 99.768 \text{ J}$$

From the first law :

$$\Delta W = \Delta Q - \Delta U$$

$$\Rightarrow \Delta W = 126 - 99.768 = 26.232 \text{ J}$$



Work done by a gas at constant pressure is :

$$\Delta W = P \Delta V = P \times (Ax)$$

Where  $x$  is the distance the piston moves.

$$26.232 = 10^5 \times (17 \times 10^{-4}) \times x$$

$$\Rightarrow 26.232 = 170 \times x$$

$$\Rightarrow x = \frac{26.232}{170} \approx 0.1543 \text{ m}$$

$$\Rightarrow x = 0.1543 \times 100 \text{ cm} \approx 15.5 \text{ cm}$$

## Question17

**One mole of an ideal diatomic gas expands from volume  $V$  to  $2V$  isothermally at a temperature  $27^\circ\text{C}$  and does  $W$  joule of work. If the gas undergoes same magnitude of expansion adiabatically from  $27^\circ\text{C}$  doing the same amount of work  $W$ , then its final temperature will be (close to) \_\_\_\_\_  $^\circ\text{C}$ .**

$$(\log_e 2 = 0.693)$$

**JEE Main 2026 (Online) 23rd January Evening Shift**

**Options:**

A.

-56

B.

-117

C.

-30

D.

-189

**Answer: A**

## Solution:

**Case -1 :** Isothermal Expansion ( $\Delta T = 0$ )

For an ideal gas undergoing isothermal expansion from  $V_1$  to  $V_2$  :

$$W_{\text{iso}} = nRT \ln \left( \frac{V_2}{V_1} \right)$$

Given :

- $n = 1$  mole
- $T = 27^\circ\text{C} = 300$  K
- $\frac{V_2}{V_1} = \frac{2V}{V} = 2$
- $\ln 2 = 0.693$

$$W = 1 \times R \times 300 \times 0.693 = 207.9R$$

**Case - 2 :** Adiabatic Expansion (  $\Delta Q$  )

For an adiabatic process, the work done is given by:

$$W_{\text{adia}} = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

The gas is diatomic, so the adiabatic constant  $\gamma = \frac{7}{5} = 1.4$ .

$$T_1 = 300 \text{ K}$$

Work done is the same magnitude (  $W = 207.9R$  ).

$$207.9R = \frac{1 \times R(300 - T_2)}{1.4 - 1}$$

$$\Rightarrow 207.9 = \frac{300 - T_2}{0.4}$$

$$\Rightarrow 83.16 = 300 - T_2$$

$$\Rightarrow T_2 = 216.84 \text{ K}$$

$$\Rightarrow T(^{\circ}\text{C}) = 216.84 - 273.15$$

$$T(^{\circ}\text{C}) \approx -56.31^{\circ}\text{C}$$

The final temperature is approximately  $-56^{\circ}\text{C}$ .

Hence, the correct option is (A).

---

## Question18

Density of water at  $4^{\circ}\text{C}$  and  $20^{\circ}\text{C}$  are  $1000\text{ kg/m}^3$  and  $998\text{ kg/m}^3$  respectively. The increase in internal energy of  $4\text{ kg}$  of water when it is heated from  $4^{\circ}\text{C}$  to  $20^{\circ}\text{C}$  is \_\_\_\_\_ J.

(specific heat capacity of water =  $4.2\text{ J/kg.}$  and  $1\text{ atmospheric pressure} = 10^5\text{ Pa}$  )

**JEE Main 2026 (Online) 24th January Morning Shift**

Options:

A.

268799.2

B.

315826.2

C.

234699.2

D.

258700.8

**Answer: A**

**Solution:**

Heat supplied to water:

$$Q = mc\Delta T$$

Here,  $m = 4\text{ kg}$ ,  $c = 4.2 \times 10^3\text{ J kg}^{-1}\text{K}^{-1}$ , and  $\Delta T = 20 - 4 = 16\text{ K}$ .

So,

$$Q = 4 \times 4.2 \times 10^3 \times 16 = 268800\text{ J}$$

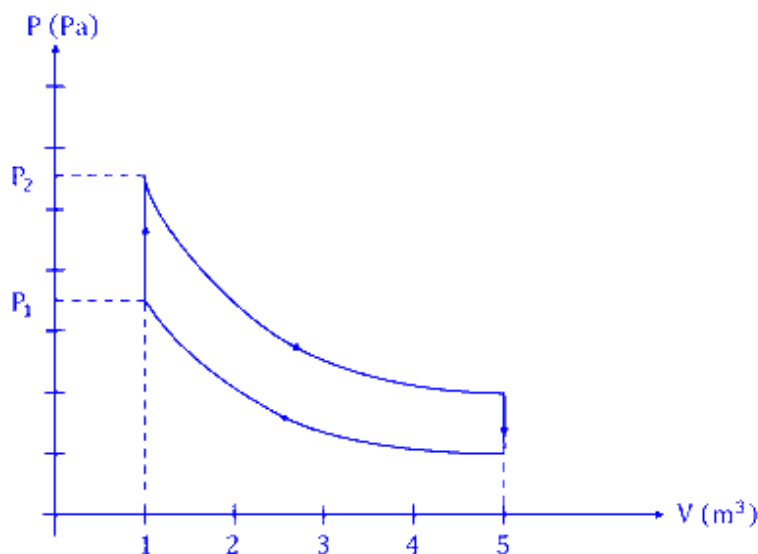
Since heating is done at atmospheric pressure, water expands and does work:

$$W = P\Delta V$$

$$V_1 = \frac{m}{\rho_1} = \frac{4}{1000} = 0.004 \text{ m}^3$$
$$V_2 = \frac{m}{\rho_2} = \frac{4}{998} = 0.004008016 \text{ m}^3$$
$$\Delta V = V_2 - V_1 = 0.004008016 - 0.004 = 8.016 \times 10^{-6} \text{ m}^3$$
$$W = 10^5 \times 8.016 \times 10^{-6} = 0.8016 \text{ J}$$
$$\Delta U = Q - W = 268800 - 0.8016 = 268799.1984 \text{ J} \approx 268799.2 \text{ J}$$

### Question19

**10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from  $P_1$  to  $P_2$  is  $\alpha$  Joule (  $P_1 = 21.7$  Pa and  $P_2 = 30$  Pa,  $C_v = 21$  J/K. mol,  $R = 8.3$  J/mol. K). The value of  $\alpha$  is \_\_\_\_\_ .**



**JEE Main 2026 (Online) 24th January Evening Shift**

**Options:**

A.

21

B.

28

C.

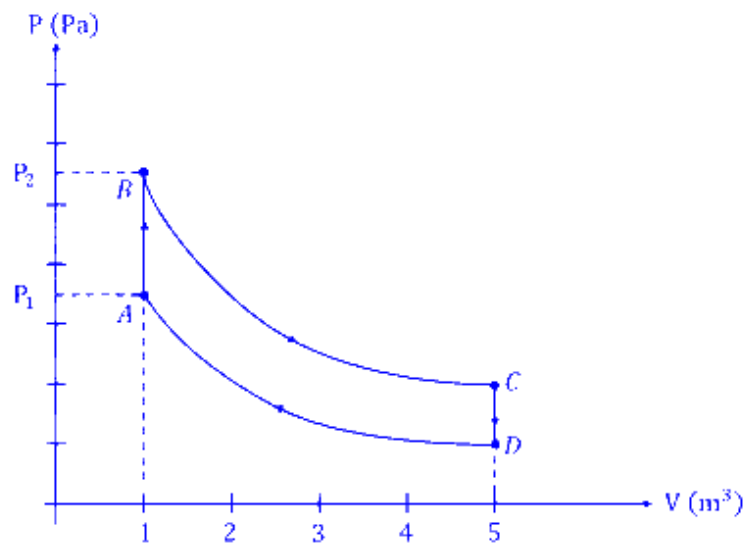
24

D.

15

**Answer: A**

**Solution:**



The path (AB) from state  $P_1$  to state  $P_2$  is a vertical line which represents isochoric process where volume remains constant.

Initial Pressure:  $P_1 = 21.7 \text{ Pa}$ , Initial volume:  $V_1 = 1 \text{ m}^3$ .

Final Pressure:  $P_2 = 30 \text{ Pa}$ , Final volume:  $V_2 = 1 \text{ m}^3$

For any isochoric process, since there is no change in volume (  $dV = 0$  ), the work done by the gas is zero :

$$W_{AB} = \int P dV = 0$$

Applying the 1<sup>st</sup> law of thermodynamics

The heat involved (  $Q = \alpha$  ) in a constant volume process is equal to the change in internal energy (  $\Delta U$  ) :

$$Q = \Delta U + W$$

$$\Rightarrow Q = \Delta U = nC_v \Delta T = \alpha \dots (i)$$

From the ideal gas law (  $PV = nRT$  ), we know that  $nT = \frac{PV}{R}$ . Therefore, the change

$$n\Delta T = \frac{P_2 V_2 - P_1 V_1}{R} \dots (ii)$$

From both the equations,

$$\alpha = C_v \left( \frac{P_2 V_2 - P_1 V_1}{R} \right)$$

Substituting the given values :

$$V_1 = V_2 = 1 \text{ m}^3; P_2 - P_1 = 30 - 21.7 = 8.3 \text{ Pa}; C_v = 21 \text{ J/Kmol}; R = 8.3 \text{ J/molK}$$

$$\alpha = 21 \times \left( \frac{(30 \times 1) - (21.7 \times 1)}{8.3} \right)$$

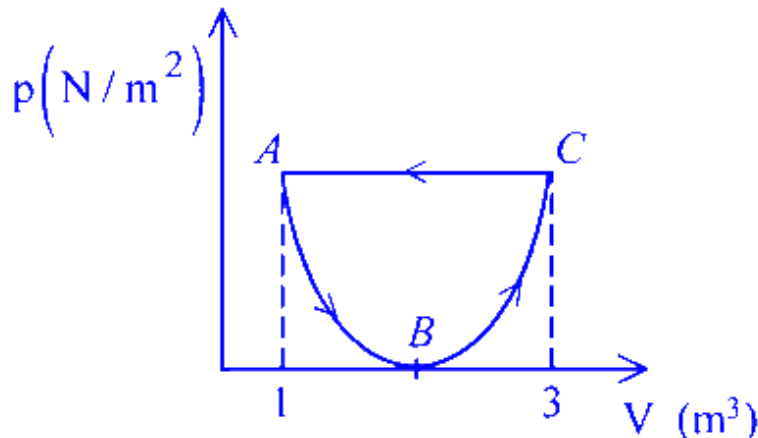
$$\Rightarrow \alpha = 21 \times \left( \frac{8.3}{8.3} \right)$$

Therefore, the heat involved in the process from  $P_1$  to  $P_2$  is 21 Joules.

Hence, the correct option is (A).

## Question20

In the following  $p - V$  diagram the equation of state along the curved path is given by  $(V - 2)^2 = 4ap$  where  $a$  is a constant. The total work done in the closed path is



**JEE Main 2026 (Online) 28th January Morning Shift**

**Options:**

A.

$$+ \frac{1}{3a}$$

B.



$$-\frac{1}{a}$$

C.

$$\frac{1}{2a}$$

D.

$$-\frac{1}{3a}$$

**Answer: D**

## Solution:

The work done in a cyclic process represented on a  $p - V$  diagram is equal to the area enclosed by the closed loop.

The path of the cycle is  $A \rightarrow B \rightarrow C \rightarrow A$  in an anti-clockwise direction.

In a  $p - V$  diagram, a clockwise cycle represents positive work done by the system ( $W < 0$ ).

The equation of curve is  $(V - 2)^2 = 4a p$  ( path  $A \rightarrow B \rightarrow C$  ).

- At  $V = 2, p = 0$  (Point B is on the  $V$  -axis).
- At  $V = 1$  and  $V = 3, p = \frac{(1)^2}{4a} = \frac{1}{4a}$ . This is the pressure at points A and C .

The total work done  $W$  is the area under the straight line minus the area under the curve:

$$W = \int_1^3 p dV - \text{Area}(CA)$$

Since the line CA is at  $p = \frac{1}{4a}$ , its area is :

$$\text{Area}(CA) = p \times \Delta V = \frac{1}{4}a \times (3 - 1) = \frac{2}{4a} = \frac{1}{2a}$$

Work done along the curve is,

$$\int_1^3 p dV = \int_1^3 \left[ \frac{(V-2)^2}{4a} \right] dV = \frac{1}{4a} \left[ \frac{(V-2)^3}{3} \right]_1^3$$

$$\Rightarrow \int_1^3 p dV = \frac{1}{4a} \left[ \frac{(3-2)^3}{3} - \frac{(1-2)^3}{3} \right] = \frac{1}{4a} \left[ \frac{1}{3} - \left( -\frac{1}{3} \right) \right] = \frac{1}{4a} \times \frac{2}{3} = \frac{1}{6a}$$

So, the net work done in the cycle is

$$W_{\text{net}} = \frac{1}{6a} - \frac{1}{2a} = \frac{1-3}{6a} = -\frac{1}{3a}$$

Therefore, the total work done in closed path is  $-\frac{1}{3a}$ . Hence, the correct option is (D).

## Question21

**10 kg of ice at  $-10^{\circ}\text{C}$  is added to 100 kg of water to lower its temperature from  $25^{\circ}\text{C}$ . Consider no heat exchange to surroundings. The decrement to the temperature of water is \_\_\_\_\_  $^{\circ}\text{C}$ .**

**(specific heat of ice =  $2100\text{ J/Kg. }^{\circ}\text{C}$ , specific heat of water =  $4200\text{ J/Kg. }^{\circ}\text{C}$ , latent heat of fusion of ice =  $3.36 \times 10^5\text{ J/Kg}$ )**

**JEE Main 2026 (Online) 28th January Morning Shift**

**Options:**

A.

15

B.

10

C.

6.67

D.

11.6

**Answer: B**

**Solution:**

Mass of the ice is  $m_i = 10\text{ kg}$ ,

Initial temperature of ice is  $T_i = -10^{\circ}\text{C}$ ,

The specific heat of ice is  $c_i = 2100 \frac{\text{J}}{\text{kg.}^{\circ}\text{C}}$ ,

The latent heat of fusion of ice is  $L_f = 3.36 \times 10^5\text{ J/kg}$

The mass of water is  $m_w = 100\text{ kg}$ ,

The initial temperature of water is  $T_w = 25^{\circ}\text{C}$ ,

The specific heat capacity of water is  $c_w = 4200 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}$

Let the final equilibrium temperature is  $T_f$ .

The heat gained by ice is for three process,

$$Q_{\text{gained}} = Q_1 \text{ (Heating ice to } 0^\circ\text{C)} + Q_2 \text{ (Melting ice at } 0^\circ\text{C)} + Q_3 \text{ (Heating the melted ice (water) to } T_f \text{)}$$

$$\Rightarrow Q_1 = m_i c_i \Delta T = 10 \times 2100 \times (0 - (-10)) = 210000 \text{ J}$$

$$\Rightarrow Q_2 = m_i L_f = 10 \times 3.36 \times 10^5 = 3360000 \text{ J}$$

$$\Rightarrow Q_3 = m_i c_w (T_f - 0) = 10 \times 4200 \times T_f = 42000 T_f$$

$$Q_{\text{gained}} = 210000 + 3360000 + 42000 T_f = 3570000 + 42000 T_f$$

The water cools from  $25^\circ\text{C}$  to  $T_f$ . So the heat lost by water is,

$$Q_{\text{loss}} = m_w c_w (25 - T_f) = 100 \times 4200 \times (25 - T_f) = 420000 (25 - T_f)$$

$$\Rightarrow Q_{\text{loss}} = 10500000 - 420000 T_f$$

Using conservation of energy,  $Q_{\text{gain}} = Q_{\text{loss}}$

$$3570000 + 42000 T_f = 10500000 - 420000 T_f$$

$$\Rightarrow 462000 T_f = 6930000$$

$$\Rightarrow T_f = \frac{6930000}{462000} = 15^\circ\text{C}$$

So the decrease in the temperature of the water:

$$\Delta T_{\text{water}} = T_{\text{initial}} - T_f = 25^\circ\text{C} - 15^\circ\text{C} = 10^\circ\text{C}$$

Therefore, the decrement to the temperature of water is  $10^\circ\text{C}$ .

Hence, the correct option is (B).

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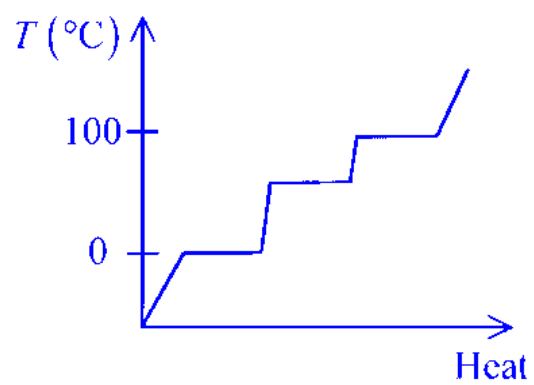
## Question22

**Which of the following best represents the temperature versus heat supplied graph for water, in the range of  $-20^\circ\text{C}$  to  $120^\circ\text{C}$ ?**

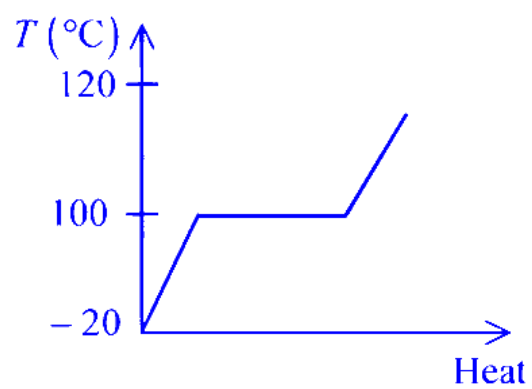
**JEE Main 2026 (Online) 28th January Morning Shift**

Options:

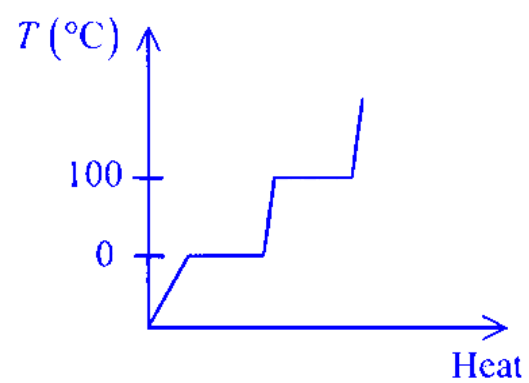
A.



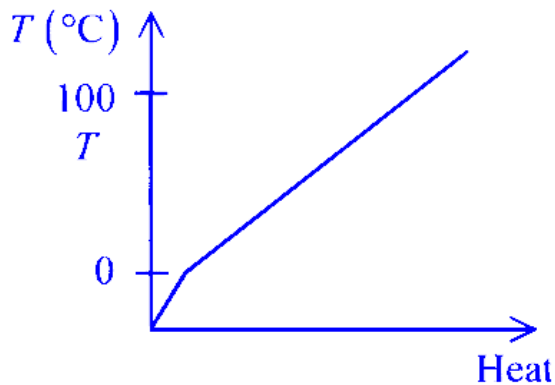
B.



C.



D.



**Answer: C**

## Solution:

To determine the correct graph, we must analyse the phase changes and temperature changes that occur when heat is supplied to water starting from  $-20^{\circ}\text{C}$ .

### Step 1 : Heating Ice ( $-20^{\circ}\text{C}$ to $0^{\circ}\text{C}$ )

Initially, the substance is ice. As heat is supplied, the temperature of the ice increases linearly from  $-20^{\circ}\text{C}$  to  $0^{\circ}\text{C}$ . This is represented by a sloping line on the graph.

$$Q_1 = m_{\text{ice}} c_{\text{ice}} (T + 20)$$

### Step 2 : Melting (Phase Change at $0^{\circ}\text{C}$ )

At  $0^{\circ}\text{C}$ , the ice begins to melt into water. During a phase change, the temperature remains constant even as heat is added (latent heat of fusion). This is represented by a horizontal line at  $0^{\circ}\text{C}$ .

$$Q_2 = m_{\text{ice}} L_f$$

### Step 3 : Heating Liquid Water ( $0^{\circ}\text{C}$ to $100^{\circ}\text{C}$ )

Once all the ice has melted, further heat increases the temperature of the liquid water linearly from  $0^{\circ}\text{C}$  up to its boiling point at  $100^{\circ}\text{C}$ . This is another sloping line.

$$Q_3 = m_{\text{water}} c_{\text{water}} T$$

### Step 4 : Boiling (Phase Change at $100^{\circ}\text{C}$ )

At  $100^{\circ}\text{C}$ , the water begins to turn into steam. The temperature stays constant during this process (latent heat of vaporization). This is represented by a second horizontal line at  $100^{\circ}\text{C}$ .

$$Q_4 = m_{\text{water}} L_v$$

### Step 5 : Heating Steam ( $100^{\circ}\text{C}$ to $120^{\circ}\text{C}$ )

After all the water has turned into steam, the temperature of the steam increases from  $100^{\circ}\text{C}$  toward  $120^{\circ}\text{C}$ . This is represented by a final sloping line.

$$Q_5 = m_{\text{vapour}} C_{\text{vapour}} (T - 100)$$

So, the correct graph must show :

1. 1. A start point below  $0^{\circ}\text{C}$  (at  $-20^{\circ}\text{C}$  ).
2. 2. A horizontal plateau at  $0^{\circ}\text{C}$  (melting).
3. 3. A horizontal plateau at  $100^{\circ}\text{C}$  (boiling).

Hence, the correct option is (C).

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## Question23

The mean free path of a molecule of diameter  $5 \times 10^{-10} \text{ m}$  at the temperature  $41^{\circ}\text{C}$  and pressure  $1.38 \times 10^5 \text{ Pa}$ , is given as \_\_\_\_\_ m. (Given  $k_B = 1.38 \times 10^{-23} \text{ J/K}$ ).

### JEE Main 2026 (Online) 28th January Evening Shift

**Options:**

A.

$$2\sqrt{2} \times 10^{-10}$$

B.

$$10\sqrt{2} \times 10^{-8}$$

C.

$$2\sqrt{2} \times 10^{-8}$$

D.

$$2 \times 10^{-8}$$

**Answer: C**

**Solution:**

The diameter of molecule is,  $d = 5 \times 10^{-10} \text{ m}$

Temperature,  $T 41^{\circ}\text{C} = (41 + 273)\text{K} = 314 \text{ K}$

Pressure,  $P = 1.38 \times 10^5 \text{ Pa}$

Boltzmann constant,  $k_B = 1.38 \times 10^{-23} \text{ J/K}$

The standard formula for the mean free path  $\lambda$  is:

$$\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P}$$

Substituting the values,

$$\lambda = \frac{(1.38 \times 10^{-23}) \times 314}{\sqrt{2} \times \pi \times (5 \times 10^{-10})^2 \times (1.38 \times 10^5)}$$

$$\Rightarrow \lambda = \frac{10^{-23} \times 314}{\sqrt{2} \times \pi \times 25 \times 10^{-20} \times 10^5}$$

$$\Rightarrow \lambda = \frac{314}{\sqrt{2} \times \pi \times 25 \times 10^{-8}}$$

Using  $\pi \approx 3.14$  :

$$\lambda = \frac{314}{\sqrt{2} \times 3.14 \times 25 \times 10^{-8}}$$

$$\Rightarrow \lambda = \frac{4}{\sqrt{2}} \times 10^{-8} = 2\sqrt{2} \times 10^{-8} \text{ m}$$

Hence, the correct option is D.

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## Question24

Heat is supplied to a diatomic gas at constant pressure. Then the ratio of  $\Delta Q : \Delta U : \Delta W$  is \_\_\_\_\_.

**JEE Main 2026 (Online) 2nd April Morning Shift**

**Options:**

A.

2 : 3 : 5

B.

5 : 3 : 2

C.

2 : 5 : 7

D.

7 : 5 : 2

**Answer: D**

## Solution:

For a diatomic gas (assuming no vibrational modes), the degrees of freedom  $f = 5$ .

The molar specific heat at constant volume for diatomic gas is

$$C_v = \frac{f}{2}R = \frac{5}{2}R$$

The molar specific heat at constant pressure is

$$C_p = C_v + R = \frac{5}{2}R + R = \frac{7}{2}R$$

The heat supplied at constant pressure is

$$\Delta Q = nC_p\Delta T = \frac{7}{2}nR\Delta T$$

The change in internal energy is

$$\Delta U = nC_v\Delta T = \frac{5}{2}nR\Delta T$$

The work done at constant pressure is

$$\Delta W = P\Delta V$$

From ideal gas equation,

$$PV = nRT \rightarrow P\Delta V = nR\Delta T$$

So, the work done at constant pressure is

$$\Delta W = P\Delta V = nR\Delta T = \frac{2}{2}nR\Delta T$$

So, the ratio  $\Delta Q : \Delta U : \Delta W$  is:

$$\frac{7}{2} : \frac{5}{2} : \frac{2}{2} \rightarrow 7 : 5 : 2$$

Therefore, the correct option is (D).

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## Question 25

**A mixture of carbon dioxide and oxygen has volume  $8310 \text{ cm}^3$ , temperature  $300 \text{ K}$ , pressure  $100 \text{ kPa}$  and mass  $13.2 \text{ g}$ . The number of moles of carbon dioxide and oxygen gases in the mixture respectively are \_\_\_\_\_.**

**(Assume both carbon dioxide and oxygen gases behave like ideal gases) [ $R = 8.31 \text{ J/mol K}$ ]**



## JEE Main 2026 (Online) 2nd April Evening Shift

### Options:

A.

0.15 and 0.18

B.

0.25 and 0.08

C.

0.21 and 0.12

D.

0.13 and 0.20

**Answer: C**

### Solution:

Using the ideal gas equation  $PV = nRT$

The pressure of gas is given as  $P = 100\text{kPa} = 10^5 \text{ Pa}$ .

The volume of gas is  $V = 8310 \text{ cm}^3 = 8310 \times 10^{-6} \text{ m}^3 = 8.31 \times 10^{-3} \text{ m}^3$

The temperature of gas is  $T = 300 \text{ K}$

The universal gas constant is  $R = 8.31 \text{ J/mol. K}$

So, the number of moles of gas is,

$$n_{\text{total}} = \frac{PV}{RT} = \frac{10^5 \times 8.31 \times 10^{-3}}{8.31 \times 300}$$

$$\Rightarrow n_{\text{total}} = \frac{831}{2493} = \frac{1}{3} \approx 0.333 \text{ moles}$$

Let  $n_1$  be the moles of  $\text{CO}_2$  and  $n_2$  be the moles of  $\text{O}_2$ ,  $n_1 + n_2 = 0.333 \dots\dots(i)$

Molar mass of  $\text{CO}_2$  is  $M_1 = 12 + (2 \times 16) = 44 \text{ g/mol}$

Molar mass of  $\text{O}_2$  is  $M_2 = 2 \times 16 = 32 \text{ g/mol}$

As the total mass of the mixture is  $13.2 \text{ g}$  :

$$n_1 M_1 + n_2 M_2 = 13.2$$

$$44n_1 + 32n_2 = 13.2 \dots\dots(ii)$$

From equation (i),  $n_2 = 0.333 - n_1$ .

Substituting this into (ii) :

$$44n_1 + 32(0.333 - n_1) = 13.2$$

$$\Rightarrow 44n_1 + 10.66 - 32n_1 = 13.2$$

$$\Rightarrow 12n_1 = 2.54$$

$$\Rightarrow n_1 \approx 0.21 \text{ moles}$$

$$\text{So, } n_2 = 0.333 - 0.21 = 0.12 \text{ moles}$$

Therefore, the number of moles of carbon dioxide and oxygen gases in the mixture respectively are 0.21, 0.12.

Hence, the correct option is (C).

## Question 26

An ideal gas undergoes a process maintaining relation between

pressure ( $P$ ) and volume ( $V$ ) as  $P = P_0 \left( 1 + \left( \frac{V_0}{V} \right)^2 \right)^{-1}$ , where  $P_0$

and  $V_0$  are constants. If two samples  $A$  and  $B$  (two moles each) with initial volumes  $V_0$  and  $3V_0$  respectively undergo above mentioned process and attain same pressure, then the difference at the temperatures of these samples,  $T_B - T_A$  is \_\_\_\_ .

(  $R$  = gas constant)

### JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

$$\frac{9P_0V_0}{8R}$$

B.

$$\frac{11P_0V_0}{10R}$$

C.

$$\frac{7P_0V_0}{6R}$$

D.

$$\frac{13P_0V_0}{11R}$$

**Answer: B**

**Solution:**

From the ideal gas law

$$PV = nRT \Rightarrow T = \frac{PV}{nR}$$

$$\text{The process for the given gas is } P = P_0 \left( 1 + \left( \frac{V_0}{V} \right)^2 \right)^{-1}$$

Substituting the given expression for P :

$$P = \frac{P_0}{1 + \frac{V_0^2}{V^2}} = \frac{P_0 V^2}{V^2 + V_0^2}$$

Now, substituting this P into the temperature equation :

$$T = \frac{1}{nR} \left( \frac{P_0 V^2}{V^2 + V_0^2} \right) V = \frac{P_0 V^3}{nR(V^2 + V_0^2)}$$

For sample A , the volume is  $V_A = V_0$  and the number of moles  $n = 2$ .

$$T_A = \frac{P_0(V_0)^3}{2R(V_0^2 + V_0^2)} = \frac{P_0 V_0^3}{2R(2V_0^2)} = \frac{P_0 V_0}{4R}$$

For sample B , the volume is  $V_B = 3 V_0$  and  $n = 2$ .

$$T_B = \frac{P_0(3V_0)^3}{2R((3V_0)^2 + V_0^2)} = \frac{P_0(27V_0^3)}{2R(9V_0^2 + V_0^2)} = \frac{27P_0 V_0^3}{2R(10V_0^2)} = \frac{27P_0 V_0}{20R}$$

So, the temperature difference is,

$$T_B - T_A = \frac{27P_0 V_0}{20R} - \frac{P_0 V_0}{4R}$$

$$\Rightarrow T_B - T_A = \frac{27P_0 V_0 - 5P_0 V_0}{20R} = \frac{22P_0 V_0}{20R} = \frac{11P_0 V_0}{10R}$$

Therefore, the correct option is (B).

## Question27

One gas of  $n_1$  mole of molecules at temperature  $T_1$ , volume  $V_1$ , and pressure  $P_1$ , and another gas of  $n_2$  mole of molecules at temperature  $T_2$ , volume  $V_2$ , and pressure  $P_2$ , are mixed resulting in pressure  $P$  and volume  $V$  of the mixture. The temperature of the mixture is \_\_\_\_ .

**JEE Main 2026 (Online) 4th April Morning Shift**

**Options:**

A.  $\frac{(T_1+T_2)}{2}$

B.  $\frac{T_1 T_2 P V}{T_2 P_1 V_1 + T_1 P_2 V_2}$

C.  $\frac{(T_2 P_1 V_1 + T_1 P_2 V_2)}{T_1 T_2 P V}$

D.  $\frac{[T_1 - T_2]}{2}$

**Answer: B**

**Solution:**

The ideal gas law is given by:  $PV = nRT \Rightarrow n = \frac{PVR}{T}$

Where R is the universal gas constant.

For the first gas :  $n_1 = \frac{P_1 V_1 R}{T_1}$

For the second gas :  $n_2 = \frac{P_2 V_2 R}{T_2}$

When the gases are mixed, the total number of moles  $n$  is the sum of the individual moles :

$$n = n_1 + n_2$$

Using the ideal gas law for the final mixture (pressure P , volume V , and temperature T ) :

$$\frac{PVR}{T} = \frac{P_1 V_1 R}{T_1} + \frac{P_2 V_2 R}{T_2}$$

$$\frac{PV}{T} = \frac{P_1 V_1}{T_1} + \frac{P_2 V_2}{T_2}$$

$$\frac{PV}{T} = \frac{P_1 V_1 T_2 + P_2 V_2 T_1}{T_1 T_2}$$

$$\frac{PV}{T} = \frac{P_1 V_1 T_2 + P_2 V_2 T_1}{T_1 T_2}$$

$$T = \frac{T_1 T_2 P V}{P_1 V_1 T_2 + P_2 V_2 T_1}$$

Therefore, the temperature of the mixture of two gases is  $\frac{T_1 T_2 P V}{P_1 V_1 T_2 + P_2 V_2 T_1}$ .

Thus, the correct option is (B).

## Question28

The temperature of a metal strip having coefficient of linear expansion  $\alpha$  is increased from  $T_1$  to  $T_2$  resulting in increase of its length by  $\Delta L_1$ . The temperature is further increased from  $T_2$  to  $T_3$  such that the increase in its length is  $\Delta L_2$ .

**Given  $T_3 + T_1 = 2T_2$  and  $T_2 - T_1 = \Delta T$ , the value of  $\Delta L_2$  is \_\_\_\_ .**

## **JEE Main 2026 (Online) 4th April Evening Shift**

**Options:**

A.

$$\Delta L_1 [1 + 2\alpha^2(\Delta T)^2]$$

B.

$$\Delta L_1 [1 + \alpha^2(\Delta T)^2]$$

C.

$$\Delta L_1 [1 + 2\alpha\Delta T]$$

D.

$$\Delta L_1 [1 + \alpha\Delta T]$$

**Answer: D**

## **Solution:**

When a metal of length  $L$  having coefficient of linear expansion  $\alpha$  is heated by temperature  $\Delta T$ , then the change in length is given as  $\Delta L = L\alpha\Delta T$

For the first interval of heating ( $T_1 \rightarrow T_2$ ) :

The increase in length  $\Delta L_1$  is given by  $\Delta L_1 = L_1\alpha(T_2 - T_1)$

It is given that  $T_2 - T_1 = \Delta T$  :

$$\Delta L_1 = L_1\alpha\Delta T \dots\dots (i)$$

The new length at temperature  $T_2$  is:

$$L_2 = L_1 + \Delta L_1 = L_1(1 + \alpha\Delta T) \dots\dots\dots (ii)$$

For the second interval of heating, ( $T_2 \rightarrow T_3$ )

The increase in length is  $\Delta L_2$  depends on the length at the start of this stage ( $L_2$ ) :

$$\Delta L_2 = L_2\alpha(T_3 - T_2)$$

From the given condition  $T_3 + T_1 = 2T_2$  :

$$T_3 - T_2 = T_2 - T_1$$

Since  $T_2 - T_1 = \Delta T$ , then  $T_3 - T_2 = \Delta T$ .

Substituting this into the  $\Delta L_2$  formula:

$$\Delta L_2 = L_2 \alpha \Delta T \dots\dots (iii)$$

Substitute the value of  $L_2$  from equation (ii) into equation (iii) :

$$\Delta L_2 = [L_1(1 + \alpha \Delta T)] \alpha \Delta T$$

$$\Rightarrow \Delta L_2 = (L_1 \alpha \Delta T)(1 + \alpha \Delta T)$$

From equation (i), we know  $L_1 \alpha \Delta T = \Delta L_1$ .

$$\Delta L_2 = \Delta L_1(1 + \alpha \Delta T)$$

Therefore, the value of  $\Delta L_2$  is  $\Delta L_1[1 + \alpha \Delta T]$ .

Thus, the correct option is (D).

-----

## Question29

**Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R**

**Assertion A : If the average kinetic energy of  $H_2$  and  $O_2$  molecules, kept in two different sized containers are same, then their temperatures will be same.**

**Reason R : The r.m.s speed of  $H_2$  and  $O_2$  molecules are same at same temperature.**

**Choose the correct answer from the options given below**

**JEE Main 2026 (Online) 4th April Evening Shift**

**Options:**

A.

Both **A** and **R** are true and **R** is the correct explanation of **A**

B.

Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

C.

A is true but **R** is false

D.

A is false but **R** is true

**Answer: C**

## Solution:

According to the kinetic theory of gases, the average translational kinetic energy (  $K_{\text{avg}}$  ) of a gas molecule is directly proportional to its absolute temperature (T).

$$K_{\text{avg}} = \frac{3}{2} k_B T$$

Where  $k_B$  is the Boltzmann constant.

So, the average kinetic energy depends only on the temperature and not on the mass of the molecule or the size/volume of the container.

Since  $K_{\text{avg}}$  is the same for both  $\text{H}_2$  and  $\text{O}_2$ , their temperatures (T) must be identical.

Therefore, Assertion A is true.

The root mean square (r. m. s.) speed (  $v_{\text{rms}}$  ) of a gas molecule is given by the formula:

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Where R is the universal gas constant, T is the absolute temperature, and M is the molar mass of the gas.

At the same temperature (T),  $v_{\text{rms}}$  is inversely proportional to the square root of the molar mass (M).

Since the molar mass of  $\text{H}_2$  (2 g/mol) is different from  $\text{O}_2$  (32 g/mol), their r.m.s. speeds will be different even at the same temperature.

Therefore, Reason R is false.

Hence, Assertion A is true, but Reason R is false.

So, the correct option is (C).

---

## Question30

**Consider the following statements:**

**A. Zeroth law of thermodynamics gives concept of temperature**

**B. First law of thermodynamics gives concept of internal energy**

**C. In isothermal expansion of ideal gas,  $\Delta Q \neq \Delta W$**

**D. Product of intensive and extensive variables is extensive**

**E. The ratio of any extensive variable to mass will be an extensive variable**

**Choose the correct combination of statements from the options given below:**

**JEE Main 2026 (Online) 5th April Morning Shift**

**Options:**

A.

C, D and E Only

B.

A, B and C Only

C.

A, B and D Only

D.

B, C and D Only

**Answer: C**

**Solution:**

The Zeroth Law states that if two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other, i.e., the Zeroth law gives idea about the property of temperature to be measured. Hence, statement A is true.

The First Law is the mathematical relation between heat (  $\Delta Q$  ), change in internal energy (  $\Delta U$  ) and work done by the system (  $\Delta W$  )

$$\Delta Q = \Delta U + \Delta W$$

This relation introduces internal energy (  $U$  ) as a state function which is based on energy conservation. Hence, statement B is true.

In an isothermal expansion of an ideal gas, temperature remains constant, so  $\Delta U = 0$ .

From the First Law:  $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = 0 + \Delta W \Rightarrow \Delta Q = \Delta W$ .

But as per statement C, in isothermal process  $\Delta Q \neq \Delta W$ , so statement C is False.



Intensive variables are the properties which do not depend on size, for example temperature and pressure.

Extensive variables are the properties depend on size, for example volume.

The product of intensive and extensive property  $P \times V$  represents work or energy, which depends with the size of the system. Therefore, the product is extensive.

When an extensive variable (like volume or heat capacity) is divided by mass, it becomes a specific quantity (like specific volume or specific heat). Specific quantities are intensive because they do not depend on the total mass of the system. Statement E says it remains extensive, so it is False.

Therefore, only A, B, and D are correct. Hence, the correct option is (C).

-----

## Question31

**Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R**

**Statement I : Change in internal energy of a system containing  $n$  mole of ideal gas can be written as  $\Delta U = nC_v (T_f - T_i) = \frac{nR}{\gamma-1} (T_f - T_i)$ ,**

**where  $\gamma = \frac{C_p}{C_v}$ ,  $T_i$  = initial temperature,  $T_f$  = final temperature.**

**Statement II : Relation between degree of freedom  $f$  and  $\gamma (= C_p/C_v)$  is  $\left(\gamma = 1 + \frac{2}{f}\right)$**

**Choose the correct answer from the options given below**

### JEE Main 2026 (Online) 5th April Morning Shift

**Options:**

A.

Both **A** and **R** are true and **R** is the correct explanation of **A**

B.

Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

C.

A is true but **R** is false

D.

A is false but **R** is true

**Answer: B**

## Solution:

Internal energy (  $U$  ) of an ideal gas depends only on temperature. The change in internal energy for any process is given by:

$$\Delta U = nC_v \Delta T = nC_v (T_f - T_i)$$

According to Mayer's relation,  $C_p - C_v = R$ .

$$\frac{C_p}{C_v} - 1 = \frac{R}{C_v}$$

$$\text{Since } \gamma = \frac{C_p}{C_v}, \text{ then } \gamma - 1 = \frac{R}{C_v} \Rightarrow C_v = \frac{R}{\gamma - 1}.$$

Substituting  $C_v$  back into the  $\Delta U$  equation:

$$\Delta U = \frac{nR}{\gamma - 1} (T_f - T_i)$$

Thus, Statement I is true.

From the law of equipartition of energy, the molar specific heat at constant volume is  $C_v = \frac{f}{2} R$ , where  $f$  is the degree of freedom.

$$\text{Using } C_p = C_v + R = \frac{f}{2} R + R = R \left( 1 + \frac{f}{2} \right).$$

$$\text{The ratio } \gamma = \frac{C_p}{C_v} = \frac{R(1 + \frac{f}{2})}{(\frac{f}{2})R} = \frac{1 + f/2}{f/2} = \frac{2}{f} + 1.$$

$$\text{So, } \gamma = 1 + \frac{2}{f}.$$

Thus, Statement II is true.

Statement II is the definition of  $\gamma$  in terms of degrees of freedom, but Statement I is based on Mayer's relation which is the definition of  $C_v$ , i.e., not directly from the degrees of freedom.

Therefore, both A and R are true but R is the correct explanation of A .

Thus, the correct option is (B).

-----

## Question32

**An ideal gas at pressure  $P$  and temperature  $T$  is expanding such that  $PT^3 = \text{constant}$ . The coefficient of volume expansion of the gas is \_\_\_\_\_ .**

## JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

$$\frac{2}{T}$$

B.

$$\frac{1}{T}$$

C.

$$\frac{4}{T}$$

D.

$$\frac{3}{T}$$

**Answer: C**

**Solution:**

The coefficient of volume expansion ( $\gamma$ ) is defined as the fractional change in volume per unit change in temperature at constant pressure :

$$\gamma = \frac{1}{V} \left( \frac{dV}{dT} \right)$$

The relation between pressure and temperature is,

$$PT^3 = C \text{ (where } C \text{ is a constant)}$$

$$\text{For an ideal gas, } PV = nRT \Rightarrow P = \frac{nRT}{V}$$

Substitute P into the given process equation :

$$\left( \frac{nRT}{V} \right) T^3 = C$$

$$\Rightarrow \frac{nRT^4}{V} = C$$

$$\Rightarrow V = \left( \frac{nR}{C} \right) T^4$$

Let  $k = \frac{nR}{C}$  (another constant), then:

$$V = kT^4$$

On differentiating  $V$  with respect to  $T$  :

$$\frac{dV}{dT} = \frac{d}{dT} (kT^4) = 4kT^3$$

Substituting the expressions for V and  $\frac{dV}{dT}$  into the definition of  $\gamma$  :

$$\gamma = \frac{1}{V} \left( \frac{dV}{dT} \right)$$

$$\Rightarrow \gamma = \frac{1}{kT^4} (4kT^3)$$

$$\Rightarrow \gamma = \frac{4kT^3}{kT^4}$$

$$\Rightarrow \gamma = \frac{4}{T}$$

Therefore, the coefficient of volume expansion of the gas is  $\frac{4}{T}$ .

Hence, the correct option is (C).

-----

## Question33

**Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K . Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K . The final pressure in the vessels is \_\_\_\_\_ kPa .**

### JEE Main 2026 (Online) 6th April Morning Shift

**Options:**

A.

100

B.

120

C.

90

D.

105

**Answer: A**

**Solution:**

Using Ideal Gas Law:

$$PV = nRT$$

Where P is pressure, V is volume, n is number of moles, R is the gas constant, and T is absolute temperature. Since the system is closed, the total number of moles ( $n_{\text{total}}$ ) remains constant before and after heating.

Initial State:

Both vessels (each of volume V) are at  $P_0 = 90\text{kPa}$  and  $T_0 = 400\text{ K}$ .

$$n_{\text{total}} = \frac{P_0 V}{RT_0} + \frac{P_0 V}{RT_0} = \frac{2P_0 V}{RT_0} \dots (i)$$

Final State: Pressure becomes  $P_f$  in both vessels. Vessel 1 stays at  $T_1 = 400\text{ K}$ , and vessel 2 is at  $T_2 = 500\text{ K}$ .

$$n_{\text{total}} = \frac{P_f V}{RT_1} + \frac{P_f V}{RT_2} = \frac{P_f V}{R} \left( \frac{1}{T_1} + \frac{1}{T_2} \right) \dots (ii)$$

From both the equations,

$$\frac{2P_0 V}{RT_0} = \frac{P_f V}{R} \left( \frac{1}{T_1} + \frac{1}{T_2} \right) \Rightarrow \frac{2P_0}{T_0} = P_f \left( \frac{1}{T_1} + \frac{1}{T_2} \right)$$

Substituting the values ( $P_0 = 90$ ,  $T_0 = 400$ ,  $T_1 = 400$ ,  $T_2 = 500$ ):

$$2 \times \frac{90}{400} = P_f \left( \frac{1}{400} + \frac{1}{500} \right)$$

$$\Rightarrow \frac{180}{400} = P_f \left( \frac{5+4}{2000} \right)$$

$$\Rightarrow 0.45 = P_f \left( \frac{9}{2000} \right)$$

$$\Rightarrow P_f = \frac{0.45 \times 2000}{9} = \frac{900}{9} = 100\text{kPa}$$

Therefore, the final pressure of the vessel is 100 kPa . Thus, the correct option is (A).

-----

## Question34

**If 2 mole of an ideal monoatomic gas at temperature  $T$ , is mixed with 6 mole of another ideal monoatomic gas at temperature  $2T$  then the temperature of mixture is:**

**JEE Main 2026 (Online) 6th April Evening Shift**

**Options:**

A.

$$\frac{5}{2}T$$

B.

$$\frac{5}{4}T$$

C.

$$\frac{7}{2}T$$

D.

$$\frac{7}{4}T$$

**Answer: D**

### Solution:

The internal energy ( $U$ ) of  $n$  moles of an ideal gas at temperature  $T$  is given by  $U = nC_vT$ , where  $C_v$  is the molar specific heat at constant volume.

For the mixture, total initial internal energy  $U_{\text{initial}} = U_1 + U_2$

Total final internal energy  $U_{\text{final}} = (n_1 + n_2)C_{v, \text{mix}} T_{\text{mix}}$

Since both gases are monoatomic, their molar specific heats are the same,

$$C_{v1} = C_{v2} = C_{v, \text{mix}} = \frac{3}{2}R$$

Therefore, using conservation of energy (total initial internal energy is equal to final internal energy).

$$n_1 C_v T_1 + n_2 C_v T_2 = (n_1 + n_2) C_v T_{\text{mix}}$$

$$\Rightarrow T_{\text{mix}} = \frac{n_1 T_1 + n_2 T_2}{(n_1 + n_2)}$$

Moles of first gas is  $n_1 = 2$

Temperature of first gas is  $T_1 = T$

Moles of second gas  $n_2 = 6$

Temperature of second gas  $T_2 = 2T$

On substituting the values into expression for the temperature of mixture,

$$T_{\text{mix}} = \frac{(2 \times T) + (6 \times 2T)}{2 + 6}$$

$$\Rightarrow T_{\text{mix}} = \frac{2T + 12T}{8}$$

$$\Rightarrow T_{\text{mix}} = \frac{14T}{8}$$

$$\Rightarrow T_{\text{mix}} = \frac{7}{4}T$$

Therefore, the correct option is (D).

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## Question35

A cylinder with adiabatic walls is closed at both ends and is divided into two compartments by a frictionless adiabatic piston. Ideal gas is filled in both (left and right) the compartments at same  $P, V,$

**T. Heating is started from left side until pressure changes to  $27P/8$ . If initial volume of each compartment was 9 litres then the final volume in right-hand side compartment is \_\_\_\_\_ litres. (for this ideal gas  $C_P/C_V = 1.5$ )**

**JEE Main 2026 (Online) 6th April Evening Shift**

**Options:**

A.

3

B.

4

C.

14

D.

9

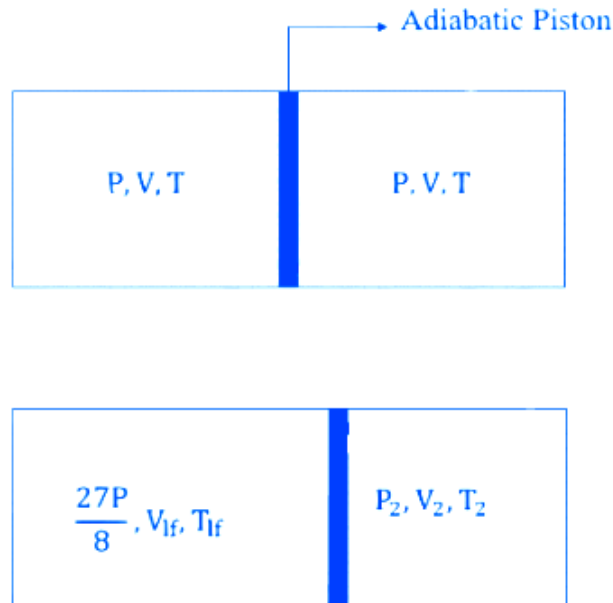
**Answer: B**

**Solution:**

In left compartment heat is added. The gas expands, increasing its pressure and volume.

In right compartment heat is not added because the wall is adiabatic, but the piston moves toward it because the pressure on the left increases. Since the walls and the piston are adiabatic, the gas in the right compartment undergoes an adiabatic compression.

The piston stops moving when the pressure in both compartments is equal. Therefore, the final pressure in the right compartment ( $P_R$ ) will also be  $\frac{27P}{8}$ .



For an adiabatic process, the relationship between pressure (P) and volume (V) is:

$$P V^\gamma = \text{constant}$$

Where  $\gamma = \frac{C_p}{C_v} = 1.5$ .

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow \frac{V_2}{V_1} = \left( \frac{P_1}{P_2} \right)^{1/\gamma}$$

The initial pressure in right compartment is  $P_1 = P$  and its initial volume is  $V_1 = 9$  litres

The final pressure in right compartment is  $P_2 = \frac{27P}{8}$

Substituting these values:

$$\frac{V_{\text{final}}}{9} = \left( \frac{P}{27P/8} \right)^{1/1.5}$$

$$\Rightarrow \frac{V_{\text{final}}}{9} = \left( \frac{8}{27} \right)^{2/3}$$

$$\Rightarrow \frac{V_{\text{final}}}{9} = \frac{4}{9}$$

$$\Rightarrow V_{\text{final}} = 4 \text{ litres}$$

Therefore, the final volume in the right-hand side compartment is 4 litres.

Hence, the correct option is (B).

## Question36



Initial pressure and volume of a monoatomic ideal gas are  $P$  and  $V$ . The change in internal energy of this gas in adiabatic expansion to volume  $V_{\text{final}} = 27 V$  is \_\_\_\_\_ J.

### JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

$$-2PV(3\sqrt{3} - 1)$$

B.

$$\frac{4}{3}PV$$

C.

$$-\frac{4}{3}PV$$

D.

$$\frac{3}{4}PV$$

**Answer: C**

**Solution:**

In an adiabatic process, no heat is exchanged with the surroundings ( $Q = 0$ ).

For an ideal gas undergoing an adiabatic change, the relationship between pressure and volume is :

$$PV^\gamma = \text{constant}$$

Where  $\gamma$  is the ratio of specific heats  $\left(\frac{C_p}{C_v}\right)$ .

For a monoatomic gas,  $\gamma = \frac{5}{3}$ .

The initial pressure of gas is  $P_1 = P$ .

The initial volume of gas is  $V_1 = V$

The final volume of gas is  $V_2 = 27 V$

Using the adiabatic relation  $P_1 V_1^\gamma = P_2 V_2^\gamma$  :

$$P \cdot V^{5/3} = P_{\text{final}} \cdot (27 V)^{5/3}$$

$$\Rightarrow P_{\text{final}} = P \left( \frac{V}{27V} \right)^{5/3} = P \left( \frac{1}{27} \right)^{5/3}$$

$$P_{\text{final}} = P \left( \frac{1}{3^3} \right)^{5/3} = P \left( \frac{1}{3} \right)^5 = \frac{P}{243}$$

From the 1<sup>st</sup> law of thermodynamics ( $Q = \Delta U + W$ ), since  $Q = 0$ , we have:

$$\Delta U = -W$$

The work done (  $W$  ) in an adiabatic process is:

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

Therefore:

$$\Delta U = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

Substitute the given values,  $P_1 V_1 = PV$ ,  $P_2 V_2 = \left( \frac{P}{243} \right) \times (27V) = \frac{PV}{9}$

$$\Delta U = \frac{\frac{PV}{9} - PV}{\frac{5}{3} - 1}$$

$$\Rightarrow \Delta U = \frac{-\frac{8}{9}PV}{2/3}$$

$$\Rightarrow \Delta U = -\frac{8}{9}PV \times \frac{3}{2}$$

$$\Rightarrow \Delta U = -\frac{4}{3}PV$$

Therefore, the change in internal energy is  $-\frac{4}{3}PV$ .

Hence, the correct option is (C).

## Question37

**One mole of diatomic gas having rotational modes only is kept in a cylinder with a piston system. The cross-section area of the cylinder is  $4 \text{ cm}^2$ . The gas is heated slowly to raise the temperature by  $1.2^\circ\text{C}$  during which the piston moves by  $25 \text{ mm}$ . The amount of heat supplied to the gas is \_\_\_\_\_ J.**

**(Atmospheric pressure =  $100 \text{ kPa}$ ,  $R = 8.3 \text{ J/mol} \cdot \text{K}$ ) (Neglect mass of the piston)**

**JEE Main 2026 (Online) 8th April Evening Shift**

**Options:**

A.

24.8

B.

10.96

C.

15.04

D.

29.98

**Answer: B**

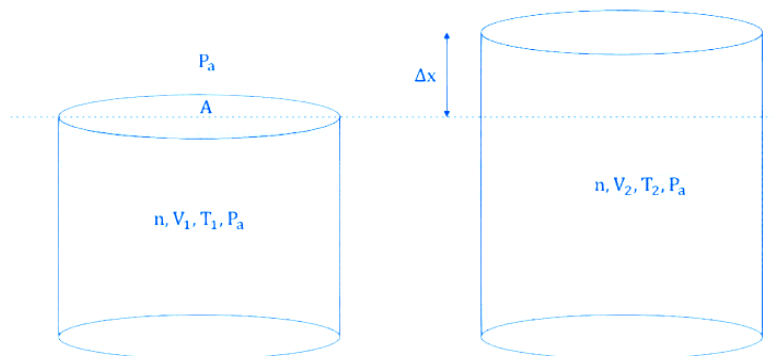
### Solution:

To find the total heat supplied (  $Q$  ), we use the 1<sup>st</sup> law of thermodynamics :

$$Q = \Delta U + W$$

Where,  $\Delta U$  is the change in internal energy and  $W$  is the work done by the gas.

The internal energy of an ideal gas depends on its degrees of freedom ( $f$ ). For a diatomic gas, there are 3 translational and 2 rotational degrees of freedom (  $f = 5$  ). However, it is given that the gas has rotational modes only, which means we only consider  $f = 2$ .



The formula for internal energy change is :

$$\Delta U = \frac{nfR\Delta T}{2}$$

The amount of gas is 1 mole,  $n = 1$  mole, the degree of freedom is  $f = 2$ , the universal gas constant is  $R = \frac{8.3 \text{ J}}{\text{mol} \cdot \text{K}}$  and the change in temperature of gas is  $\Delta T = 1.2^\circ\text{C} = 1.2 \text{ K}$

$$\Delta U = \frac{1 \times 2 \times 8.3 \times 1.2}{2}$$

$$\Rightarrow \Delta U = 8.3 \times 1.2 = 9.96 \text{ J}$$

Work done by the gas in a piston system is given by:

$$W = P \Delta V$$

Since the mass of the piston is neglected and it moves slowly, the internal pressure  $P$  equals the atmospheric pressure ( $P_{\text{atm}}$ ).

The change in volume  $\Delta V$  is  $\Delta V = A \times \Delta x$ , where the cross-sectional area is  $A$  and displacement of the piston is  $\Delta x$ .

The atmospheric pressure is  $P = 100\text{kPa} = 10^5 \text{ Pa}$

The area of cross-section is  $A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$

The displacement of piston  $d = 25 \text{ mm} = 25 \times 10^{-3} \text{ m}$

$$\Delta V = A \times \Delta x = (4 \times 10^{-4}) \times (25 \times 10^{-3}) = 100 \times 10^{-7} = 10^{-5} \text{ m}^3$$

So, the work done by the gas is;

$$W = 10^5 \times 10^{-5} = 1 \text{ J}$$

Using 1<sup>st</sup> law of thermodynamics,

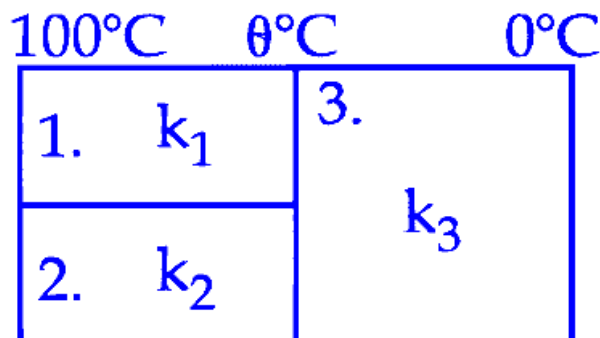
$$Q = \Delta U + W$$

$$\Rightarrow Q = 9.96 \text{ J} + 1 \text{ J} = 10.96 \text{ J}$$

Therefore, the amount of heat supplied by the gas is 10.96 J.

## Question38

**Three conductors of same length having thermal conductivity  $k_1, k_2$  and  $k_3$  are connected as shown in figure.**



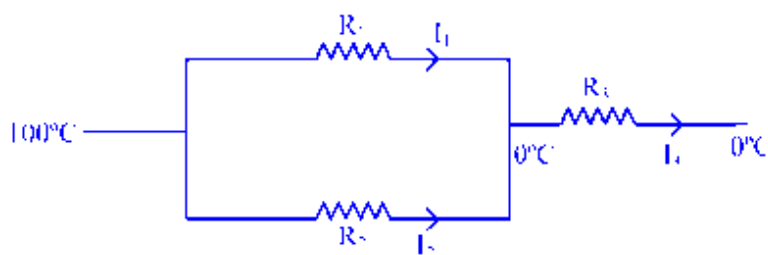
**Area of cross sections of 1<sup>st</sup> and 2<sup>nd</sup> conductor are same and for 3<sup>rd</sup> conductor it is double of the 1<sup>st</sup> conductor. The temperatures are given in the figure. In steady state condition, the value of  $\theta$  is \_\_\_\_\_ °C. (Given :**

$$k_1 = 60 \text{Js}^{-1} \text{ m}^{-1} \text{ K}^{-1}, k_2 = 120 \text{Js}^{-1} \text{ m}^{-1} \text{ K}^{-1}, k_3 = 135 \text{Js}^{-1} \text{ m}^{-1} \text{ K}^{-1}$$

## JEE Main 2025 (Online) 22nd January Morning Shift

**Answer: 40**

**Solution:**



$$R_1 = \frac{2L}{K_1 A}$$

$$(\text{As } l_1 = l_2 = l_3 = l$$

$$A_1 = A_2 = A/2$$

$$A_3 = A)$$

$$R_2 = \frac{2L}{K_2 A}$$

$$R_3 = \frac{L}{K_3 A}$$

$$I_1 + I_2 = I_3$$

$$\Rightarrow \frac{100-\theta}{\frac{2L}{K_1 A}} + \frac{100-\theta}{\frac{2L}{K_2 A}} = \frac{\theta-0}{\frac{L}{K_3 A}}$$

$$\Rightarrow \frac{100-\theta}{2} (K_1 + K_2) = K_3 \theta$$

$$\Rightarrow 50(K_1 + K_2) = \theta \left[ K_3 + \frac{K_1 + K_2}{2} \right]$$

$$\Rightarrow 50(60 + 120) = \theta \left[ 135 + \frac{60+120}{2} \right]$$

$$\Rightarrow 50 \times 80 = \theta(135 + 90)$$

$$\Rightarrow \theta = \frac{50 \times 180}{225}$$

$$\Rightarrow \theta = 40^\circ\text{C}$$

---

## Question39

**An ideal gas initially at  $0^\circ\text{C}$  temperature, is compressed suddenly to one fourth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is  $3/2$ , the change in temperature due to the thermodynamic process is \_\_\_\_\_ K.**

## **JEE Main 2025 (Online) 23rd January Morning Shift**

**Answer: 273**

### **Solution:**

We're given an ideal gas initially at  $T_i = 0^\circ\text{C} = 273\text{ K}$ . The gas is suddenly compressed to one-fourth of its initial volume, and we are told that

$$\gamma = \frac{c_p}{c_v} = \frac{3}{2}.$$

For an adiabatic process (one in which there is no heat exchange), if the process were reversible, the relation between the temperature and volume would be

$$TV^{\gamma-1} = \text{constant}.$$

Even though the compression is "sudden" (and hence irreversible), the final equilibrium state of the gas is uniquely determined by its internal energy (which is a state function). That allows us to use the adiabatic relation between the initial and final states.

Here are the steps:

Write the adiabatic relation between the initial and final states:

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}.$$

Since the gas is compressed to one-fourth of its volume, we have

$$V_f = \frac{1}{4} V_i.$$

Substitute this into the adiabatic relation:

$$273 V_i^{\gamma-1} = T_f \left(\frac{1}{4} V_i\right)^{\gamma-1}.$$

$$\text{With } \gamma = \frac{3}{2},$$

$$\gamma - 1 = \frac{3}{2} - 1 = \frac{1}{2}.$$

So the relation becomes:

$$273 V_i^{1/2} = T_f \left(\frac{1}{4}\right)^{1/2} V_i^{1/2}.$$

Cancel the common factor  $V_i^{1/2}$  (provided it is nonzero) and simplify:

$$273 = T_f \left( \frac{1}{2} \right).$$

Now solve for the final temperature  $T_f$ :

$$T_f = 273 \times 2 = 546 \text{ K}.$$

The change in temperature is then:

$$\Delta T = T_f - T_i = 546 - 273 = 273 \text{ K}.$$

So, the change in temperature due to the thermodynamic process is 273 K.

## Question40

**The temperature of 1 mole of an ideal monoatomic gas is increased by  $50^\circ\text{C}$  at constant pressure. The total heat added and change in internal energy are  $E_1$  and  $E_2$ , respectively. If  $\frac{E_1}{E_2} = \frac{x}{9}$  then the value of  $x$  is \_\_\_\_\_.**

**JEE Main 2025 (Online) 24th January Morning Shift**

**Answer: 15**

**Solution:**

$$Q = n C_P \Delta T \quad \text{and} \quad \Delta U = n C_V \Delta T$$

For a monoatomic ideal gas, the molar specific heats are given by:

$$C_V = \frac{3}{2}R \quad \text{and} \quad C_P = C_V + R = \frac{5}{2}R$$

Thus, the ratio of the heat added to the change in internal energy is:

$$\frac{Q}{\Delta U} = \frac{C_P}{C_V} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3}$$

Given that:

$$\frac{E_1}{E_2} = \frac{x}{9}$$

We equate the two ratios:

$$\frac{5}{3} = \frac{x}{9}$$

Multiplying both sides by 9:

$$x = \frac{5}{3} \times 9 = 15$$

Thus, the value of  $x$  is 15.

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## Question41

**A container of fixed volume contains a gas at 27°C. To double the pressure of the gas, the temperature of gas should be raised to \_\_\_\_\_ °C.**

**JEE Main 2025 (Online) 29th January Morning Shift**

**Answer: 327**

### Solution:

To solve this problem, we can use the combined gas law in terms of pressure ( $P$ ) and temperature ( $T$ ). For a gas with a fixed volume, the relationship is given as:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

where:

$P_1$  is the initial pressure,

$T_1$  is the initial temperature in Kelvin,

$P_2$  is the final pressure,

$T_2$  is the final temperature in Kelvin.

Given that the initial temperature  $T_1 = 27^\circ\text{C}$ , we need to convert it to Kelvin:

$$T_1 = 27^\circ\text{C} + 273.15 = 300.15\text{ K}$$

The final pressure  $P_2$  is twice the initial pressure ( $P_2 = 2P_1$ ). Substituting these values into the gas law equation gives:

$$\frac{P_1}{300.15} = \frac{2P_1}{T_2}$$

Solving for  $T_2$ :

Cancel  $P_1$  from both sides:

$$\frac{1}{300.15} = \frac{2}{T_2}$$

Rearrange to solve for  $T_2$ :



$$T_2 = 2 \times 300.15 = 600.3 \text{ K}$$

Convert the final temperature back to Celsius:

$$T_2 = 600.3 \text{ K} - 273.15 = 327.15^\circ \text{C}$$

Therefore, to double the pressure of the gas, the temperature should be raised to approximately **327°C**.

---

## Question 42

$\gamma_A$  is the specific heat ratio of monoatomic gas A having 3 translational degrees of freedom.  $\gamma_B$  is the specific heat ratio of polyatomic gas B having 3 translational, 3 rotational degrees of freedom and 1 vibrational mode. If  $\frac{\gamma_A}{\gamma_B} = \left(1 + \frac{1}{n}\right)$ , then the value of  $n$  is \_\_\_\_\_.

### JEE Main 2025 (Online) 2nd April Morning Shift

**Answer: 3**

#### Solution:

To determine the value of  $n$ , we start by considering the specific heat ratios for gases A and B.

For gas A, which is monoatomic with 3 translational degrees of freedom:

The degrees of freedom,  $f_A$ , is 3.

$$\text{Therefore, } \gamma_A = \frac{f_A + 2}{f_A} = \frac{3 + 2}{3} = \frac{5}{3}.$$

For gas B, which is polyatomic with 3 translational, 3 rotational degrees of freedom, and 1 vibrational mode:

The total degrees of freedom,  $f_B$ , is  $3 + 3 + 2 \times 1 = 8$  (since one vibrational mode contributes 2 degrees of freedom: 1 kinetic + 1 potential).

$$\text{Therefore, } \gamma_B = \frac{f_B + 2}{f_B} = \frac{8 + 2}{8} = \frac{10}{8} = \frac{5}{4}.$$

Now, the relation given is:

$$\frac{\gamma_A}{\gamma_B} = \left(1 + \frac{1}{n}\right)$$

Substitute the values of  $\gamma_A$  and  $\gamma_B$ :

$$\frac{\frac{5}{3}}{\frac{5}{4}} = \frac{5}{3} \times \frac{4}{5} = \frac{20}{15} = \frac{4}{3}$$

Equating to the given form:

$$\frac{4}{3} = 1 + \frac{1}{n}$$

Thus, we have:

$$\frac{4}{3} - 1 = \frac{1}{n}$$

$$\frac{1}{3} = \frac{1}{n}$$

Solving for  $n$ :

$$n = 3$$

Hence, the value of  $n$  is 3.

---

## Question43

**The internal energy of air in  $4\text{ m} \times 4\text{ m} \times 3\text{ m}$  sized room at 1 atmospheric pressure will be \_\_\_\_\_  $\times 10^6\text{ J}$**

**(Consider air as diatomic molecule)**

**JEE Main 2025 (Online) 2nd April Evening Shift**

**Answer: 12**

**Solution:**

To determine the internal energy of the air in a room that measures  $4\text{ m} \times 4\text{ m} \times 3\text{ m}$  at 1 atmospheric pressure, we can follow these steps using the properties of a diatomic gas:

The volume of the room,  $V$ , is calculated as:

$$V = 4\text{ m} \times 4\text{ m} \times 3\text{ m} = 48\text{ m}^3$$

For a diatomic gas like air, the internal energy  $U$  is given by the formula:

$$U = nC_vT = n\frac{5RT}{2}$$

Where:

$n$  is the number of moles,

$C_v$  is the molar heat capacity at constant volume,

$R$  is the ideal gas constant, and

$T$  is the temperature in Kelvin.

Using the ideal gas law,  $PV = nRT$ , we can substitute for  $nRT$ :

$$nRT = PV$$

Therefore, the internal energy  $U$  becomes:

$$U = \frac{5}{2}PV$$

Substituting the given values for atmospheric pressure  $P = 10^5 \text{ Pa}$  (1 atm) and volume  $V = 48 \text{ m}^3$ :

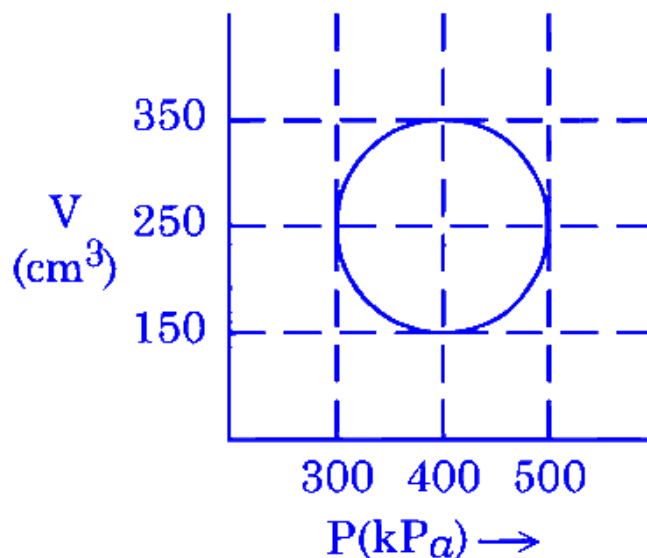
$$U = \frac{5}{2} \times 10^5 \text{ Pa} \times 48 \text{ m}^3 = 12 \times 10^6 \text{ J}$$

Thus, the internal energy of the air in the room is  $12 \times 10^6 \text{ J}$ .

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## Question44

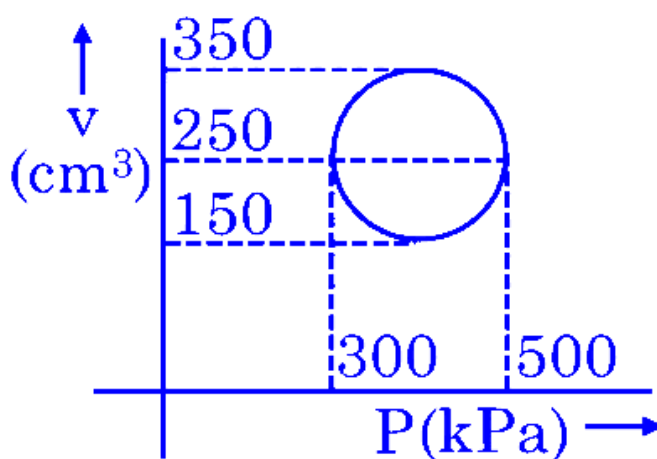
An ideal gas has undergone through the cyclic process as shown in the figure. Work done by the gas in the entire cycle is \_\_\_\_\_  $\times 10^{-1} \text{ J}$ . (Take  $\pi = 3.14$ )



**JEE Main 2025 (Online) 7th April Morning Shift**

**Answer: 314**

**Solution:**



Area of circle,  $W = \frac{\pi}{4} d_1 d_2$

$$W = \frac{\pi}{4} (500 - 300) \times 10^3 (350 - 150) \times 10^{-6}$$

$$W = 31.4 \text{ Joule}$$

$$W = 314 \times 10^{-1} \text{ Joule}$$


---

## Question 45

A wire of length 10 cm and diameter 0.5 mm is used in a bulb. The temperature of the wire is  $1727^\circ\text{C}$  and power radiated by the wire is 94.2 W. Its emissivity is  $\frac{x}{8}$  where  $x = \underline{\hspace{2cm}}$ .

(Given  $\sigma = 6.0 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ ,  $\pi = 3.14$  and assume that the emissivity of wire material is same at all wavelength.)

**JEE Main 2025 (Online) 7th April Morning Shift**

**Answer: 5**

**Solution:**

To determine the emissivity ( $\epsilon$ ) of the wire used in the bulb, we start with the given parameters and the formula for power radiated by a black body:

Length of the wire ( $L$ ): 10 cm = 0.1 m

Diameter of the wire ( $d$ ): 0.5 mm = 0.0005 m

Temperature of the wire ( $T$ ):  $1727^{\circ}\text{C} = 2000\text{K}$

Power radiated ( $P$ ):  $94.2\text{ W}$

The power radiated by the wire can be expressed as:

$$P = \epsilon \sigma A T^4$$

Where:

$\sigma$  is the Stefan-Boltzmann constant ( $6.0 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ )

$A$  is the surface area of the wire, calculated using the formula for the surface area of a cylinder:  $A = \pi dL$

Substituting the given values into the formula:

$$94.2 = \epsilon \times (6 \times 10^{-8}) \times (3.14 \times 0.0005 \times 0.1) \times (2000)^4$$

Simplifying the expression inside the parentheses:

$$94.2 = \epsilon \times (6 \times 10^{-8}) \times 3.14 \times 0.0005 \times 0.1 \times (2000)^4$$

On solving for  $\epsilon$ :

$$\epsilon = \frac{94.2}{[(6 \times 10^{-8}) \times 3.14 \times 0.0005 \times 0.1 \times (2000)^4]}$$

Given that  $\epsilon$  is  $\frac{x}{8}$ , evaluating this expression provides the value of  $x = 5$ . Thus, the emissivity  $\epsilon$  is  $\frac{5}{8}$ .

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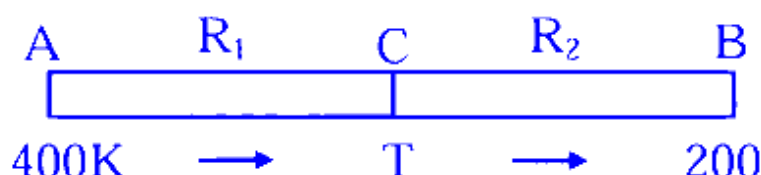
## Question46

Two cylindrical rods A and B made of different materials, are joined in a straight line. The ratios of lengths, radii and thermal conductivities of these rods are:  $\frac{L_A}{L_B} = \frac{1}{2}$ ,  $\frac{r_A}{r_B} = 2$  and  $\frac{K_A}{K_B} = \frac{1}{2}$ . The free ends of rods A and B are maintained at  $400\text{ K}$ ,  $200\text{ K}$ , respectively. The temperature of rods interface is \_\_\_\_\_  $K$ , when equilibrium is established.

**JEE Main 2025 (Online) 7th April Evening Shift**

**Answer: 360**

**Solution:**



$$\begin{aligned}
R_1 &= \frac{\ell_1}{K_1 A_1}, R_2 = \frac{\ell_2}{K_2 A_2} \\
\frac{dQ}{dt} &= \frac{\Delta T}{R} \\
\left( \frac{dQ}{dt} \right)_1 &= \left( \frac{dQ}{dt} \right)_2 \\
\frac{400 - T}{R_1} &= \frac{T - 200}{R_2} \\
\frac{400 - T}{T - 200} &= \frac{R_1}{R_2} = \left( \frac{\ell_1}{\ell_2} \right) \left( \frac{r_2}{r_1} \right)^2 \times \frac{K_2}{K_1} \\
&= \frac{1}{2} \times \left( \frac{1}{2} \right)^2 \times 2 \\
&= \left( \frac{1}{4} \right) \\
\frac{400 - T}{T - 200} &= \frac{1}{4} \\
1600 - 4T &= T - 200 \\
5T &= 1800 \\
T &= 360 \text{ K}
\end{aligned}$$


---

## Question47

**An amount of ice of mass  $10^{-3}$  kg and temperature  $-10^\circ\text{C}$  is transformed to vapour of temperature  $110^\circ\text{C}$  by applying heat. The total amount of work required for this conversion is, (Take, specific heat of ice =  $2100\text{Jkg}^{-1} \text{ K}^{-1}$ , specific heat of water =  $4180\text{Jkg}^{-1} \text{ K}^{-1}$ , specific heat of steam =  $1920\text{Jkg}^{-1} \text{ K}^{-1}$ , Latent heat of ice =  $3.35 \times 10^5\text{Jkg}^{-1}$  and Latent heat of steam =  $2.25 \times 10^6\text{Jkg}^{-1}$ )**

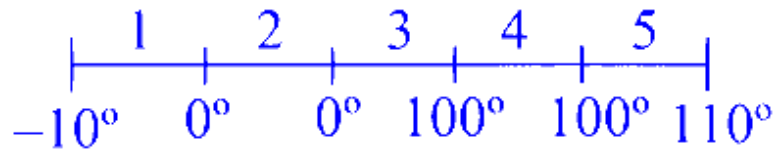
**JEE Main 2025 (Online) 22nd January Morning Shift**

**Options:**

- A. 3022 J
- B. 3043 J
- C. 3003 J
- D. 3024 J

**Answer: B**

**Solution:**



$$\Delta Q_1 = ms_i \Delta T_2, \Delta Q_2 = mL_i$$

$$\Delta Q_3 = mS_w \Delta T_3, \Delta Q_4 = mL_s, \Delta Q_5 = mS_s \Delta T_5$$

$$\Delta Q_{net} = m(S_i \Delta T_1 + L_i + S_w \Delta T_3 + L_s + S_s \Delta T_5)$$

$$= 10^{-3}(2100 \times 10 + 335000 + 4180 \times 100 + 2250000 + 1920 \times 10)$$

$$= 10^{-3}(3043200) = 3043.2 \text{ J}$$

$$\text{So, } \Delta Q_{net} \approx 3043 \text{ J}$$

---

## Question48

Two spherical bodies of same materials having radii 0.2 m and 0.8 m are placed in same atmosphere. The temperature of the smaller body is 800 K and temperature of the bigger body is 400 K . If the energy radiated from the smaller body is E, the energy radiated from the bigger body is (assume, effect of the surrounding temperature to be negligible),

**JEE Main 2025 (Online) 22nd January Morning Shift**

**Options:**

A. 64 E

B. 16 E

C. E

D. 256 E

**Answer: C**

**Solution:**

To determine the energy radiated by two spherical bodies, both made of the same material but with different radii and temperatures, we can use the Stefan-Boltzmann law. This law states that the energy radiated by a black body per unit time is proportional to the product of the surface area  $A$  and the fourth power of its temperature  $T$ .

The formula for power  $P$  is given by:

$$P = \sigma e A T^4$$

Where:

$\sigma$  is the Stefan-Boltzmann constant

$e$  is the emissivity of the material

$A$  is the surface area of the sphere

$T$  is the temperature in Kelvin

Expressing this relationship as a proportionality:

$$P \propto A T^4$$

For a sphere, the surface area  $A$  is:

$$A = 4\pi R^2$$

Hence, the power radiated is proportional to:

$$P \propto R^2 T^4$$

For two bodies with radii  $R_1$  and  $R_2$ , and temperatures  $T_1$  and  $T_2$ , the ratio of their powers is:

$$\frac{P_1}{P_2} = \left(\frac{R_1}{R_2}\right)^2 \left(\frac{T_1}{T_2}\right)^4$$

Given:

$$R_1 = 0.2 \text{ m}, T_1 = 800 \text{ K}$$

$$R_2 = 0.8 \text{ m}, T_2 = 400 \text{ K}$$

Substituting into the formula:

$$\frac{E}{E'} = \left(\frac{0.2}{0.8}\right)^2 \left(\frac{800}{400}\right)^4$$

Calculating each part:

$$\left(\frac{0.2}{0.8}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

$$\left(\frac{800}{400}\right)^4 = 2^4 = 16$$

Combining these calculations:

$$\frac{E}{E'} = \frac{1}{16} \times 16 = 1$$

This implies that the energy radiated by the bigger body  $E'$  equals the energy radiated by the smaller body  $E$ . Thus,  $E' = E$ .

-----



## Question49

For a diatomic gas, if  $\gamma_1 = \left(\frac{C_p}{C_v}\right)$  for rigid molecules and  $\gamma_2 = \left(\frac{C_p}{C_v}\right)$  for another diatomic molecules, but also having vibrational modes. Then, which one of the following options is correct? ( $C_p$  and  $C_v$  are specific heats of the gas at constant pressure and volume)

**JEE Main 2025 (Online) 22nd January Evening Shift**

**Options:**

A.  $\gamma_2 < \gamma_1$

B.  $\gamma_2 > \gamma_1$

C.  $\gamma_2 = \gamma_1$

D.  $2\gamma_2 = \gamma_1$

**Answer: A**

**Solution:**

Let's analyze the problem step by step:

For a rigid diatomic molecule (without vibrational modes):

Degrees of freedom:

Translation: 3

Rotation: 2

Total: 5

The molar specific heat at constant volume is given by:

$$C_v = \frac{5}{2}R$$

At constant pressure, it is:

$$C_p = C_v + R = \frac{5}{2}R + R = \frac{7}{2}R$$

Therefore, the ratio is:

$$\gamma_1 = \frac{C_p}{C_v} = \frac{\frac{7}{2}R}{\frac{5}{2}R} = \frac{7}{5} \approx 1.4$$

For the diatomic molecule that also has vibrational modes (assuming the vibrational mode is fully excited):

Additional vibrational mode contributions:

Each vibrational mode contributes 2 degrees of freedom (one kinetic and one potential).

Therefore, the degrees of freedom become:

$$3 \text{ (translation)} + 2 \text{ (rotation)} + 2 \text{ (vibration)} = 7$$

The molar specific heat at constant volume now is:

$$C_v = \frac{7}{2}R$$

And at constant pressure:

$$C_p = C_v + R = \frac{7}{2}R + R = \frac{9}{2}R$$

Thus, the ratio is:

$$\gamma_2 = \frac{C_p}{C_v} = \frac{\frac{9}{2}R}{\frac{7}{2}R} = \frac{9}{7} \approx 1.2857$$

Comparison:

For the rigid molecule,  $\gamma_1 \approx 1.4$ .

For the molecule with vibrational modes,  $\gamma_2 \approx 1.2857$ .

Clearly,  $\gamma_2 < \gamma_1$ .

Therefore, the correct option is:

Option A:  $\gamma_2 < \gamma_1$

-----

## Question50

**Given are statements for certain thermodynamic variables,**

**(A) Internal energy, volume (V) and mass (M) are extensive variables.**

**(B) Pressure (P), temperature ( T ) and density (  $\rho$  ) are intensive variables.**

**(C) Volume (V), temperature (T) and density (  $\rho$  ) are intensive variables.**

**(D) Mass (M), temperature (T) and internal energy are extensive variables.**

**Choose the correct answer from the options given below :**

**JEE Main 2025 (Online) 22nd January Evening Shift**

**Options:**

A. (C) and (D) Only

B. (A) and (B) Only

C. (D) and (A) Only

D. (B) and (C) Only

**Answer: B**

**Solution:**

**Check Statement (A):**

“Internal energy (U), volume (V), and mass (M) are extensive variables.”

**Internal energy** is extensive.

**Volume** is extensive.

**Mass** is extensive.

Hence, statement (A) is **correct**.

**Check Statement (B):**

“Pressure (P), temperature (T), and density ( $\rho$ ) are intensive variables.”

**Pressure** is intensive.

**Temperature** is intensive.

**Density** is intensive.

Hence, statement (B) is **correct**.

**Check Statement (C):**

“Volume (V), temperature (T), and density ( $\rho$ ) are intensive variables.”

**Volume** is **not** intensive; it is an extensive property.

(T) and ( $\rho$ ) are indeed intensive.

Since volume is miscategorized here, (C) is **incorrect**.

**Check Statement (D):**

“Mass (M), temperature (T), and internal energy are extensive variables.”

**Mass** is extensive.

**Temperature** is **not** extensive (it is intensive).

**Internal energy** is extensive.

Because temperature is miscategorized, (D) is **incorrect**.

Only (A) and (B) are correct statements.

Therefore, the correct choice is:

**Option B: (A) and (B) Only.**

---

## Question 51

A gun fires a lead bullet of temperature 300 K into a wooden block. The bullet having melting temperature of 600 K penetrates into the block and melts down. If the total heat required for the process is 625 J, then the mass of the bullet is \_\_\_\_\_ grams. (Latent heat of fusion of lead =  $2.5 \times 10^4 \text{ J Kg}^{-1}$  and specific heat capacity of lead =  $125 \text{ J Kg}^{-1} \text{ K}^{-1}$ )

**JEE Main 2025 (Online) 23rd January Morning Shift**

**Options:**

A. 20

B. 15

C. 5

D. 10

**Answer: D**

**Solution:**

Let's determine the mass step by step:

The bullet is heated from 300 K to 600 K, so the increase in temperature is:

$$\Delta T = 600 \text{ K} - 300 \text{ K} = 300 \text{ K}.$$

The heat required to raise the temperature of the bullet (using the specific heat capacity of lead) is:

$$Q_1 = m \times c \times \Delta T = m \times 125 \text{ J/kg K} \times 300 \text{ K} = 37500 m \text{ J},$$

where  $m$  is the mass in kilograms.

After reaching 600 K, the bullet melts. The heat required for the phase change (melting) is given by:

$$Q_2 = m \times L = m \times 2.5 \times 10^4 \text{ J/kg} = 25000 m \text{ J}.$$

The total heat required for the process is:

$$Q = Q_1 + Q_2 = 37500 m + 25000 m = 62500 m \text{ J}.$$

We are given that the total heat is 625 J, so:

$$62500 m = 625.$$

Solving for  $m$ :

$$m = \frac{625}{62500} = 0.01 \text{ kg}.$$

Finally, converting the mass from kilograms to grams:

$$0.01 \text{ kg} = 0.01 \times 1000 \text{ g} = 10 \text{ g}.$$

Thus, the mass of the bullet is 10 grams. This corresponds to Option D.

-----

## Question52

### Match the List - I with List - II

|     | List - I                                                                     |       | List - II          |
|-----|------------------------------------------------------------------------------|-------|--------------------|
| (A) | Pressure varies inversely with volume of an ideal gas.                       | (I)   | Adiabatic process  |
| (B) | Heat absorbed goes partly to increase internal energy and partly to do work. | (II)  | Isochoric process  |
| (C) | Heat is neither absorbed nor released by a system.                           | (III) | Isothermal process |
| (D) | No work is done on or by a gas.                                              | (IV)  | Isobaric process   |

**Choose the correct answer from the options given below:**

**JEE Main 2025 (Online) 23rd January Morning Shift**

**Options:**

A. A-I, B-III, C-II, D-IV

B. A-I, B-IV, C-II, D-III

C. A-III, B-I, C-IV, D-II

D. A-III, B-IV, C-I, D-II

**Answer: D**

## **Solution:**

Let's analyze each statement step by step:

For (A): "Pressure varies inversely with volume of an ideal gas."

At constant temperature (isothermal process), Boyle's law applies:

$$PV = \text{constant}.$$

Thus, pressure and volume have an inverse relationship.

Hence, (A) corresponds to the Isothermal process, which is (III).

For (B): "Heat absorbed goes partly to increase internal energy and partly to do work."

In a constant pressure (isobaric) process, when heat is supplied, part of it is used to do work (expansion, for example) and the rest increases the internal energy.

Therefore, (B) matches with the Isobaric process, which is (IV).

For (C): "Heat is neither absorbed nor released by a system."

This implies that the process is adiabatic (no heat exchange).

Thus, (C) matches with the Adiabatic process, which is (I).

For (D): "No work is done on or by a gas."

Work done by a gas is given by  $W = P\Delta V$ .

If the volume does not change (constant volume), then no work is done.

So, (D) identifies with the Isochoric process, which is (II).

Putting it all together:

(A) → (III) Isothermal process

(B) → (IV) Isobaric process

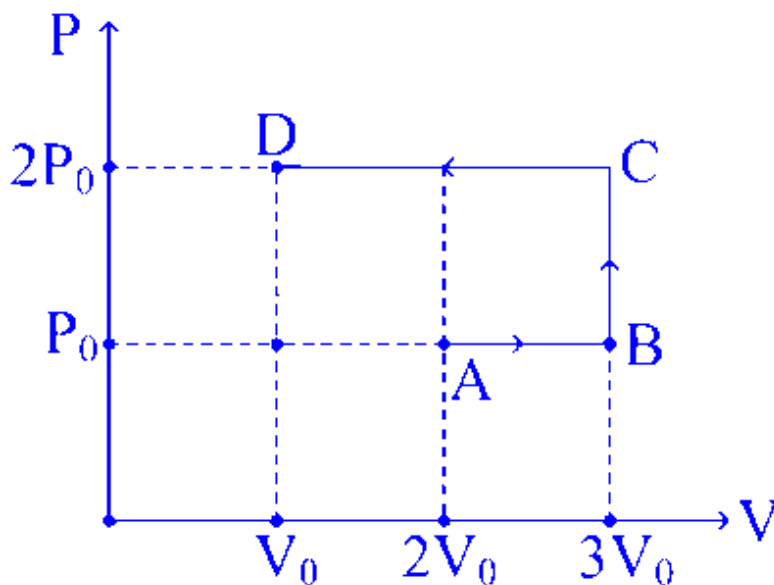
(C) → (I) Adiabatic process

(D) → (II) Isochoric process

Thus, the correct answer is:

Option D

## Question 53



Using the given  $P - V$  diagram, the work done by an ideal gas along the path  $ABCD$  is :

**JEE Main 2025 (Online) 23rd January Evening Shift**

**Options:**

- A.  $3P_0 V_0$
- B.  $-3P_0 V_0$
- C.  $-4P_0 V_0$
- D.  $4P_0 V_0$

**Answer: B**

**Solution:**

To find the work done by an ideal gas along the path  $ABCD$  on the given  $P - V$  diagram, we calculate the work done on each segment individually and then sum them up.

**Segment AB:** The work is given by  $w_{AB} = P_0 \times V_0$ , contributing  $P_0 V_0$ .

**Segment BC:** No volume change occurs along this path, resulting in zero work:  $w_{BC} = 0$ .

**Segment  $CD$ :** The gas compresses, performing negative work:  $w_{CD} = -(2P_0 \times 2V_0) = -4P_0V_0$ .

Adding the work done in each segment:

$$\begin{aligned}w_{ABCD} &= w_{AB} + w_{BC} + w_{CD} \\&= P_0V_0 + 0 - 4P_0V_0 \\&= -3P_0V_0\end{aligned}$$

Thus, the total work done by the gas along the path  $ABCD$  is  $-3P_0V_0$ .

-----

## Question 54

**Water of mass  $m$  gram is slowly heated to increase the temperature from  $T_1$  to  $T_2$ . The change in entropy of the water, given specific heat of water is  $1\text{Jkg}^{-1}\text{K}^{-1}$ , is :**

**JEE Main 2025 (Online) 23rd January Evening Shift**

**Options:**

A.  $m(T_2 - T_1)$

B. zero

C.  $m \ln \left( \frac{T_1}{T_2} \right)$

D.  $m \ln \left( \frac{T_2}{T_1} \right)$

**Answer: D**

**Solution:**

To determine the change in entropy when water is heated from temperature  $T_1$  to  $T_2$  (here,  $T_2$  is equivalent to  $T_1$ ), we use the definition of entropy change for a reversible process:

$$\Delta S = \int_{T_1}^{T_2} \frac{dQ}{T}$$

Since the water is being heated slowly, the process is reversible, and the heat added is given by:

$$dQ = mc dT$$

where:

- $m$  is the mass of the water,



- $c$  is the specific heat capacity (given as  $1 \text{ J kg}^{-1} \text{ K}^{-1}$ ).

Substitute  $dQ$  into the integral:

$$\Delta S = \int_{T_1}^{T_2} \frac{mc dT}{T} = mc \int_{T_1}^{T_2} \frac{dT}{T}.$$

Since  $c = 1$ , the equation simplifies to:

$$\Delta S = m \int_{T_1}^{T_2} \frac{dT}{T}.$$

Evaluating the integral, we have:

$$\int_{T_1}^{T_2} \frac{dT}{T} = \ln T \Big|_{T_1}^{T_2} = \ln \left( \frac{T_2}{T_1} \right).$$

Thus, the change in entropy is:

$$\Delta S = m \ln \left( \frac{T_2}{T_1} \right).$$

Comparing with the options provided, the correct answer is:

Option D:  $m \ln \left( \frac{T_2}{T_1} \right).$

---

## Question 55

**An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.**

- A. The work done by gas during the process is zero.**
- B. The heat added to gas is different from change in its internal energy.**
- C. The volume of the gas is increased.**
- D. The internal energy of the gas is increased.**
- E. The process is isochoric (constant volume process)**

**Choose the correct answer from the options given below:**

**Options:**

A. A, C Only

B. A, D, E Only

C. E Only

D. A, B, C, D Only

**Answer: B**

**Solution:**

For an ideal gas, the equation of state is given by

$$pV = nRT.$$

If during a process the pressure increases linearly with temperature (i.e.,  $p \propto T$ ), then from the ideal gas law (with a fixed amount of gas,  $n$ , and with the gas constant,  $R$ ), it follows that

$$\frac{p}{T} = \frac{nR}{V}.$$

Since  $nR$  is constant, a linear relationship between  $p$  and  $T$  implies that  $\frac{p}{T}$  remains constant throughout the process. This can only be true if the volume  $V$  is constant. Hence, the process is isochoric.

For an isochoric (constant volume) process:

The work done by the gas is given by

$$W = \int p dV,$$

and since  $dV = 0$ , we have  $W = 0$ . (Statement A is true.)

The internal energy change for an ideal gas depends only on temperature:

$$\Delta U = nC_V \Delta T.$$

If the temperature increases, then  $\Delta U > 0$ . (Statement D is true.)

For an isochoric process, no expansion work is done, so any heat added goes entirely into increasing the internal energy. This means the heat added is not different from the change in internal energy (in fact,  $Q = \Delta U$ ). (Statement B is false.)

Since the process is isochoric, the volume does not change (and certainly is not increased). (Statement C is false, and Statement E stating it is isochoric is true.)

Thus, the correct true statements are A, D, and E.

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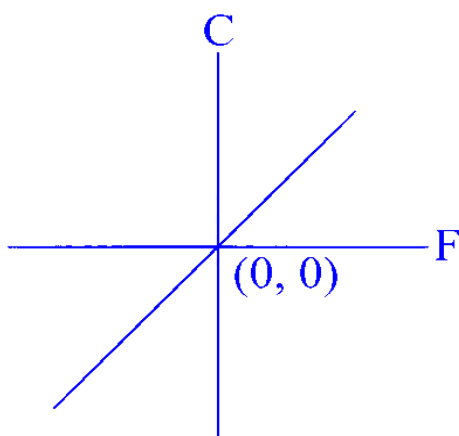
## Question 56

Which of the following figure represents the relation between Celsius and Fahrenheit temperatures?

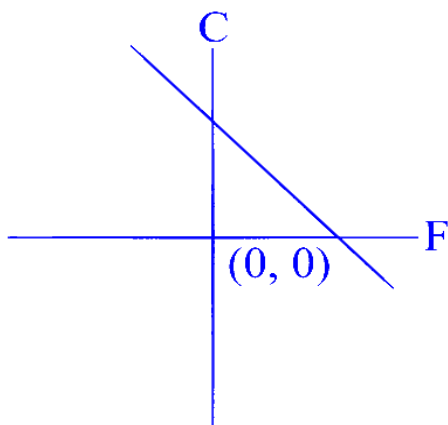
**JEE Main 2025 (Online) 24th January Evening Shift**

**Options:**

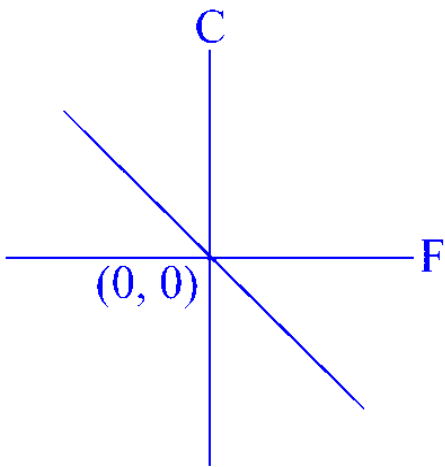
A.



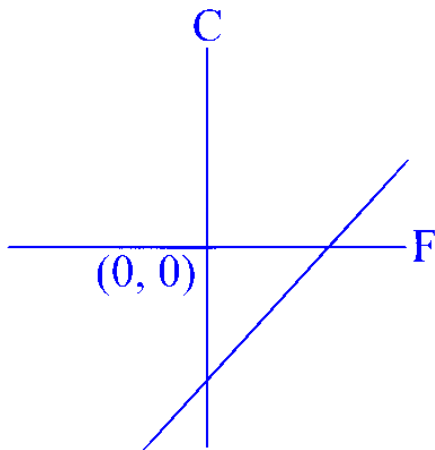
B.



C.



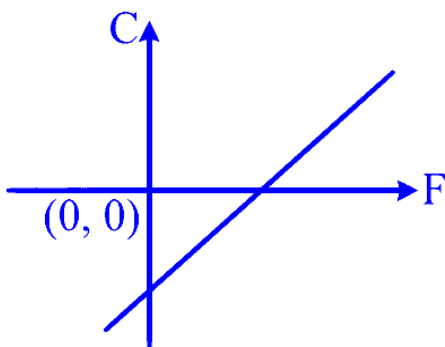
D.



**Answer: D**

**Solution:**

$$\frac{C}{5} = \frac{F-32}{9} \Rightarrow C = \frac{5F}{9} - \frac{160}{9}$$



---

## Question 57

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A) :** In an insulated container, a gas is adiabatically shrunk to half of its initial volume. The temperature of the gas decreases.

**Reason (R) :** Free expansion of an ideal gas is an irreversible and an adiabatic process.

In the light of the above statements, choose the correct answer from the options given below :

**JEE Main 2025 (Online) 24th January Evening Shift**

**Options:**

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
- B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- C. (A) is false but (R) is true
- D. (A) is true but (R) is false

**Answer: C**

**Solution:**

Let's analyze both statements step by step.

Assertion (A): "In an insulated container, a gas is adiabatically shrunk to half of its initial volume. The temperature of the gas decreases."

- In an adiabatic process, no heat is exchanged with the surroundings.
- For a reversible adiabatic (compression) process, the relationship between temperature and volume is given by

$$TV^{\gamma-1} = \text{constant},$$

where  $\gamma$  is the heat capacity ratio (greater than 1).

- If the volume is reduced from  $V$  to  $V/2$ , then

$$T_{\text{final}} = T_{\text{initial}} \left( \frac{V}{V/2} \right)^{\gamma-1} = T_{\text{initial}} (2)^{\gamma-1}.$$

- Since  $(2)^{\gamma-1} > 1$ , the final temperature is higher than the initial temperature.
- Therefore, the assertion that the temperature decreases is incorrect.

Reason (R): "Free expansion of an ideal gas is an irreversible and an adiabatic process."

- In a free expansion, the gas expands into a vacuum without doing any work on the surroundings.
- Since the container is insulated, no heat is exchanged, making it adiabatic.
- However, the process is inherently irreversible due to the spontaneous expansion and mixing of states.
- Thus, this statement is true.

Since Assertion (A) is false and Reason (R) is true (but the reason does not explain the false assertion about temperature change during compression), the correct answer is:

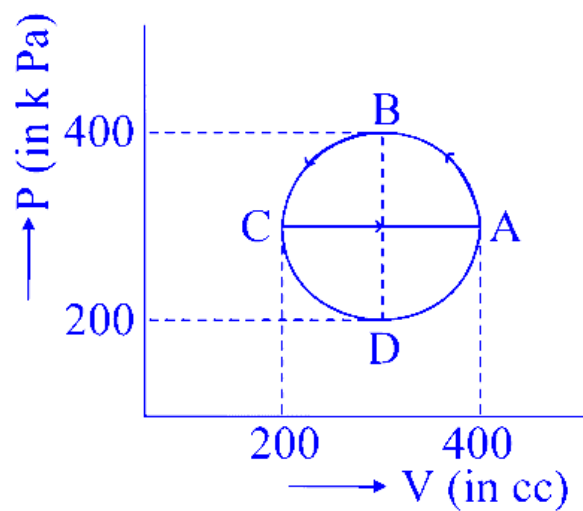
Option C

(A) is false but (R) is true.

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## Question58

**The magnitude of heat exchanged by a system for the given cyclic process ABCA (as shown in figure) is (in SI unit) :**



**JEE Main 2025 (Online) 24th January Evening Shift**

**Options:**

A. zero

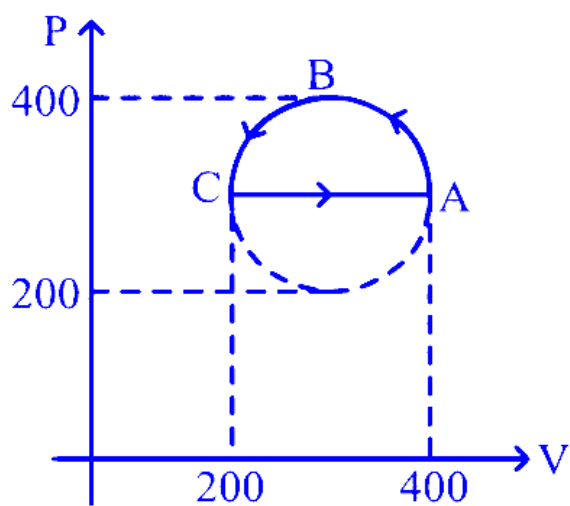
B.  $40\pi$

C.  $5\pi$

D.  $10\pi$

**Answer: C**

**Solution:**



$$\begin{aligned}
 W &= \frac{1}{2} \pi R^2 \\
 &= \frac{1}{2} \times \pi \times \left( \frac{200}{2} \times 10^3 \right) \times \frac{200}{2} \times 10^{-6} \\
 &= \frac{10\pi}{2} = 5\pi \text{ J}
 \end{aligned}$$


---

## Question 59

The temperature of a body in air falls from  $40^\circ\text{C}$  to  $24^\circ\text{C}$  in 4 minutes. The temperature of the air is  $16^\circ\text{C}$ . The temperature of the body in the next 4 minutes will be :

**JEE Main 2025 (Online) 24th January Evening Shift**

**Options:**

- A.  $\frac{28}{3}^\circ\text{C}$
- B.  $\frac{56}{3}^\circ\text{C}$
- C.  $\frac{42}{3}^\circ\text{C}$
- D.  $\frac{14}{3}^\circ\text{C}$

**Answer: B**

**Solution:**

$$\begin{aligned}
 \frac{T_2 - T_1}{t} &= K [T_{\text{avg}} - T_s] \\
 T_1 &= 24^\circ\text{C}; T_2 = 40^\circ\text{C}, t = 4, T_s = 16^\circ\text{C} \\
 \frac{40 - 24}{4} &= K[32 - 16]
 \end{aligned}$$



$$K = \frac{4}{16} = \frac{1}{4}$$

$$\text{Now } \frac{24 - T}{4} = K \left[ \frac{T + 24}{2} - 16 \right]$$

$$24 - T = \frac{T - 16}{2} + 16$$

$$\frac{3T}{2} = 28$$

$$T = \frac{56}{3}^\circ\text{C}$$

## Question60

A Carnot engine ( $E$ ) is working between two temperatures  $473\text{ K}$  and  $273\text{ K}$ . In a new system two engines - engine  $E_1$  works between  $473\text{ K}$  to  $373\text{ K}$  and engine  $E_2$  works between  $373\text{ K}$  to  $273\text{ K}$ . If  $\eta_{12}$ ,  $\eta_1$  and  $\eta_2$  are the efficiencies of the engines  $E$ ,  $E_1$  and  $E_2$ , respectively, then

**JEE Main 2025 (Online) 28th January Morning Shift**

**Options:**

A.  $\eta_{12} = \eta_1 \eta_2$

B.  $\eta_{12} = \eta_1 + \eta_2$

C.  $\eta_{12} \geq \eta_1 + \eta_2$

D.  $\eta_{12} < \eta_1 + \eta_2$

**Answer: D**

**Solution:**

We know, efficiency of a carnot engine,  $\eta$  or  $e = 1 - \frac{T_C}{T_H}$

where,  $T_C$  = temperature of cold sink

$T_H$  = temperature of hot source

$$\text{So, } \eta_{12} = 1 - \frac{273}{473} = \frac{200}{473} = 0.423$$

$$\eta_1 = 1 - \frac{373}{473} = \frac{100}{473} = 0.211$$

$$\eta_2 = 1 - \frac{273}{373} = \frac{100}{373} = 0.268$$

$$\text{Here, } \eta_1 + \eta_2 = 0.211 + 0.268 = 0.479 > 0.423$$

$$\text{So, } \eta_{12} < \eta_1 + \eta_2$$

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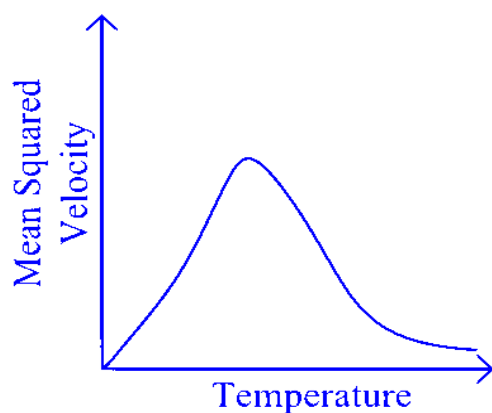
## Question61

**For a particular ideal gas which of the following graphs represents the variation of mean squared velocity of the gas molecules with temperature?**

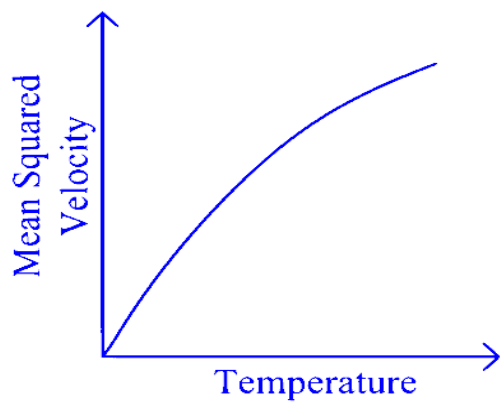
**JEE Main 2025 (Online) 28th January Morning Shift**

**Options:**

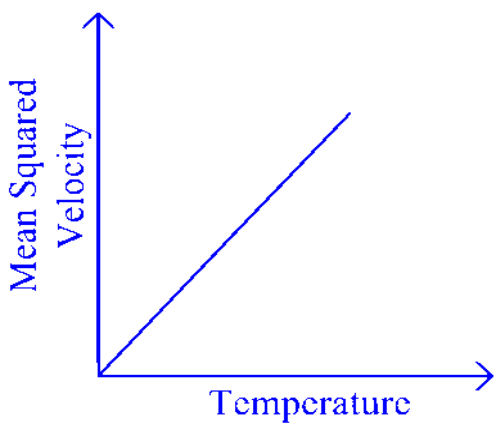
A.



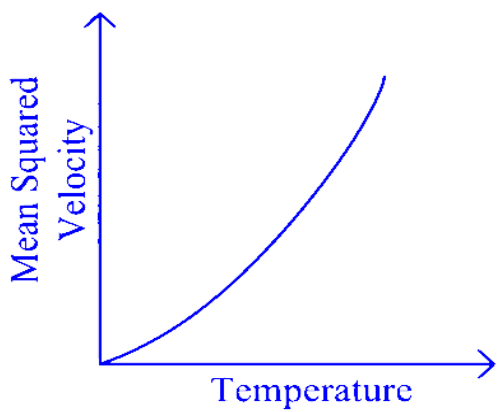
B.



C.



D.



**Answer: C**

## Solution:

We know, for an ideal gas,

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\Rightarrow V_{ms} = V_{rms}^2 = \frac{3RT}{M}$$

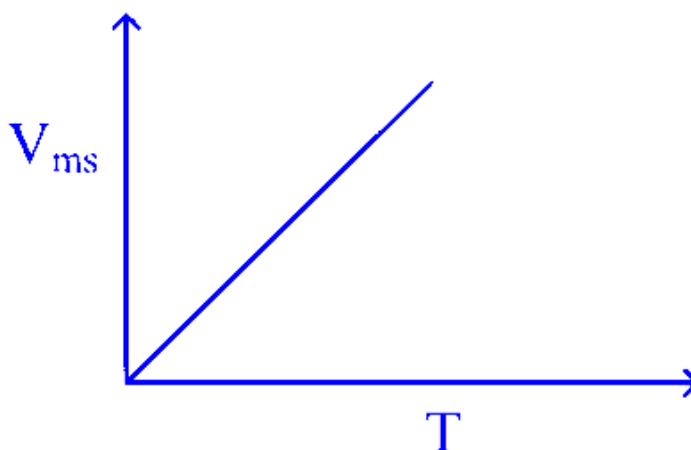
$$\Rightarrow V_{ms} \propto T$$

$$\Rightarrow V_{ms} \propto T$$

i.e. mean squared velocity is directly proportional to temperature.

$$\Rightarrow V_{ms} = \left(\frac{3R}{M}\right)T$$

by comparing it with  $y = mx$  (straight line)



Hence, option 3 is correct.

---

## Question62

**The kinetic energy of translation of the molecules in 50 g of CO<sub>2</sub> gas at 17°C is :**

## JEE Main 2025 (Online) 28th January Evening Shift

**Options:**

A.

4205.5 J

B.

3582.7 J

C.

3986.3 J

D.

4102.8 J

**Answer: D**

**Solution:**

$$(\text{KE})_{\text{Transsational}} = \left[ \frac{3}{2}KT \right] \times \text{no. of molecule}$$

$$\text{No. of molecule} = \left[ \frac{50}{44} \times 6.023 \times 10^{23} \right]$$

$$(\text{KE})_{\text{Transsational}} = 4108.644 \text{ J}$$

---

## Question63

The ratio of vapour densities of two gases at the same temperature is  $\frac{4}{25}$ , then the ratio of r.m.s. velocities will be :

## JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

$$\frac{5}{2}$$

B.

$$\frac{25}{4}$$

C.

$$\frac{4}{25}$$

D.

$$\frac{2}{5}$$

**Answer: A**

**Solution:**

We are given that the ratio of the vapour densities of the two gases is

$$\frac{4}{25}.$$

Since vapour density is proportional to the molecular mass, we can write

$$\frac{M_1}{M_2} = \frac{4}{25},$$

where  $M_1$  and  $M_2$  are the molecular masses of the gases.

The root mean square (r.m.s.) velocity of a gas is given by

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}},$$

where:

$R$  is the universal gas constant,

$T$  is the temperature,

$M$  is the molecular mass of the gas.

Since both gases are at the same temperature, the ratio of their r.m.s. velocities is

$$\frac{v_{\text{rms},1}}{v_{\text{rms},2}} = \sqrt{\frac{M_2}{M_1}}.$$

Substitute the ratio of molecular masses:

$$\frac{v_{\text{rms},1}}{v_{\text{rms},2}} = \sqrt{\frac{25}{4}} = \frac{5}{2}.$$

Thus, the ratio of their r.m.s. velocities is

$$\frac{5}{2}.$$

The correct answer is Option A:  $\frac{5}{2}$ .

---

## Question64

**The work done in an adiabatic change in an ideal gas depends upon only :**

**JEE Main 2025 (Online) 29th January Morning Shift**

**Options:**

A.

change in its pressure

B.

change in its temperature

C.

change in its specific heat

D.

change in its volume

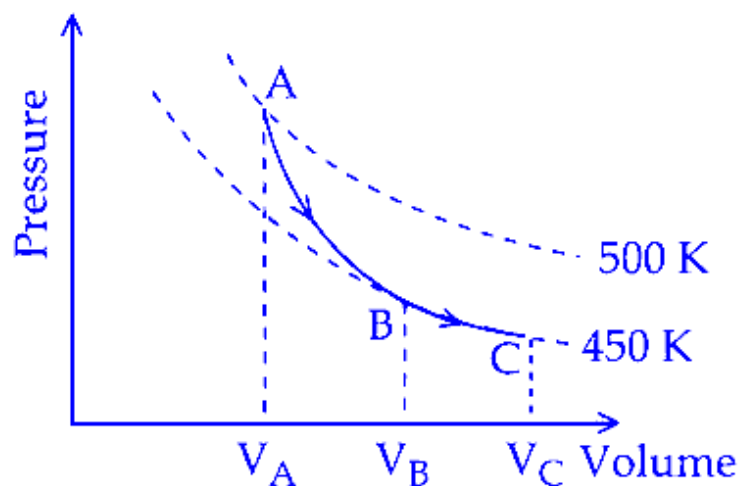
**Answer: B**

**Solution:**

$$\Delta W = -\Delta U = -nC_v\Delta T$$

---

## Question65



A poly-atomic molecule ( $C_V = 3R$ ,  $C_P = 4R$ , where  $R$  is gas constant) goes from phase space point A ( $P_A = 10^5 \text{ Pa}$ ,  $V_A = 4 \times 10^{-6} \text{ m}^3$ ) to point B ( $P_B = 5 \times 10^4 \text{ Pa}$ ,  $V_B = 6 \times 10^{-6} \text{ m}^3$ ) to point C ( $P_C = 10^4 \text{ Pa}$ ,  $V_C = 8 \times 10^{-6} \text{ m}^3$ ). A to B is an adiabatic path and B to C is an isothermal path.

The net heat absorbed per unit mole by the system is :

**JEE Main 2025 (Online) 29th January Evening Shift**

**Options:**

A.  $500R(\ln 3 + \ln 4)$

B.  $450R(\ln 4 - \ln 3)$

C.  $500R \ln 2$

D.  $400R \ln 4$

**Answer: B**

**Solution:**

As A to B is an adiabatic path,

So,  $\Delta Q_{A \rightarrow B} = 0 \dots (1)$

So, now we need to find the heat absorbed for the isothermal path (B to C)



We know, change in internal energy

$$\Delta u = mc_v \Delta T$$

$$\Rightarrow \Delta u = 0 \text{ (for isothermal as } \Delta T = 0 \text{)}$$

Now, using 1st law of thermodynamics

$$\Delta Q = \Delta u + w$$

$$\Rightarrow \Delta Q = 0 + w \Rightarrow \Delta Q = w$$

$$\text{So, } \Delta Q_{B \rightarrow C} = W_{B \rightarrow C}$$

$$= nRT \ln \left( \frac{v_C}{v_B} \right)$$

$$\Rightarrow \Delta Q_{B \rightarrow C} = nR(450) \ln \left( \frac{4}{3} \right)$$

$$\Rightarrow \Delta Q_{B \rightarrow C} = 450nR \ln \left( \frac{4}{3} \right) \dots (2)$$

So net heat absorbed,

$$Q = \Delta Q_{A \rightarrow B} + \Delta Q_{B \rightarrow C}$$

$$\Rightarrow Q = 450nR \ln \left( \frac{4}{3} \right) \dots \text{(from 1 and 2)}$$

Net heat absorbed per mole,

$$\Rightarrow \frac{Q}{n} = 450R(\ln 4 - \ln 3).$$

Option 2 is correct.

---

## Question 66

**A cup of coffee cools from  $90^\circ\text{C}$  to  $80^\circ\text{C}$  in  $t$  minutes when the room temperature is  $20^\circ\text{C}$ . The time taken by the similar cup of coffee to cool from  $80^\circ\text{C}$  to  $60^\circ\text{C}$  at the same room temperature is:**

**JEE Main 2025 (Online) 29th January Evening Shift**

**Options:**

A.

$$\frac{13}{5}t$$

B.

$$\frac{10}{13}t$$

C.

$$\frac{5}{13}t$$

D.

$$\frac{13}{10}t$$

**Answer: A**

## **Solution:**

Here, we will use the Newton's law of cooling.

$$\text{We know, } \frac{T_1 - T_2}{\Delta t} = k \left( \frac{T_1 + T_2}{2} - T_s \right) \dots (1)$$

where,  $T_1$  and  $T_2$  are initial and final temperatures respectively.

$T_s$  = surrounding temperature

$\Delta t$  = time taken

$k$  = positive constant

Given,

Ist case :

$$T_1 = 90^\circ C$$

$$T_2 = 80^\circ C$$

$$\Delta t = t, T_s = 20^\circ C$$

IInd case :

$$T_1 = 80^\circ C$$

$$T_2 = 60^\circ C$$

$$T_s = 20^\circ C, \Delta t = ?$$

Now, for Ist case,

$$\frac{90-80}{t} = k \left( \frac{90+80}{2} - 20 \right) \dots\dots (\text{from (1)})$$

$$\Rightarrow \frac{10}{t} = k(85 - 20)$$

$$\Rightarrow \frac{10}{t} = 65k \dots (2)$$

For IInd case,

$$\frac{80-60}{\Delta t} = k \left( \frac{80+60}{2} - 20 \right) \dots (\text{From (1)})$$

$$\Rightarrow \frac{20}{\Delta t} = k(70 - 20)$$

$$\Rightarrow \frac{20}{\Delta t} = 50k \dots (3)$$

Now, by (2)  $\div$  (3),

$$\frac{\frac{10}{t}}{\frac{20}{\Delta t}} = \frac{13}{10}$$

$$\Rightarrow \frac{\Delta t}{t} \left( \frac{1}{2} \right) = \frac{13}{5}$$

$$\Rightarrow \Delta t = \frac{13}{5}t$$

---

## Question 67

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A) :** With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process.

**Reason (R) :** In isothermal process,  $PV = \text{constant}$ , while in adiabatic process  $PV^\gamma = \text{constant}$ . Here  $\gamma$  is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas.

In the light of the above statements, choose the correct answer from the options given below:

**JEE Main 2025 (Online) 29th January Evening Shift**

**Options:**

A.

(A) is true but (R) is false

B.

Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

C.

(A) is false but (R) is true

D.

Both (A) and (R) are true and (R) is the correct explanation of (A)

**Answer: D**

### Solution:

We know, ideal gas equation,

$$PV = nRT$$

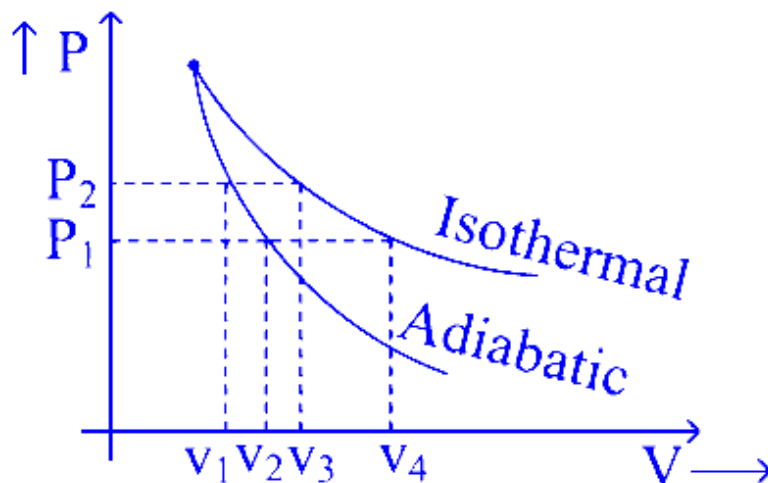
For isothermal process,  $T = \text{constant}$

$$\Rightarrow PV = \text{const.}$$

$$\Rightarrow P \propto \frac{1}{V}$$

For an adiabatic process,

$$PV^\gamma = \text{constant}$$



We can see, for the same pressure increment,  $P_2 - P_1$ ,  $(v_4 - v_3) > (v_2 - v_1)$

i.e. the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process. Hence, option 4 is correct.

---

## Question 68

**The difference of temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have**

**JEE Main 2025 (Online) 29th January Evening Shift**

**Options:**

A.

low thermal conductivity and high electrical conductivity

B.

low thermal conductivity and low electrical conductivity

C.

high thermal conductivity and high electrical conductivity

D.

high thermal conductivity and low electrical conductivity

**Answer: A**

### **Solution:**

To convert a temperature difference into electrical energy effectively, the material in question should have low thermal conductivity and high electrical conductivity. Such materials are classified as thermoelectric materials.

A low thermal conductivity ensures that the material does not easily transfer heat, which is crucial for maintaining a significant temperature difference between its hot and cold sides. This helps in generating a greater voltage. Meanwhile, high electrical conductivity is essential for facilitating the efficient movement of electrons across the material when there is a temperature difference, leading to maximal energy conversion.

Therefore, a material that optimally harvests heat energy into electrical energy should embody these characteristics.

-----

## **Question69**

**The equation for real gas is given by  $(P + \frac{a}{V^2})(V - b) = RT$ , where P, V, T and R are the pressure, volume, temperature and gas constant, respectively. The dimension of  $ab^{-2}$  is equivalent to that of :**

### **JEE Main 2025 (Online) 2nd April Morning Shift**

**Options:**

A. Compressibility

B. Planck's constant

C. Energy density

D. Strain

**Answer: C**

### **Solution:**

The equation for a real gas is given by:

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

Here,  $P$ ,  $V$ ,  $T$ , and  $R$  are the pressure, volume, temperature, and gas constant, respectively.

To find the dimension of  $ab^{-2}$ , we proceed with the following steps:

Rearrange the equation for dimensions:

$$[a] = [P][V^2]$$

Since  $[P] = ML^{-1}T^{-2}$  and  $[V] = L^3$ , we have:

$$[a] = ML^{-1}T^{-2}(L^6) = ML^5T^{-2}$$

As derived from the equation:

$$[b] = [V] = L^3$$

Now, to find  $[ab^{-2}]$ :

$$[ab^{-2}] = \frac{[a]}{[b]^2} = ML^5T^{-2} \cdot L^{-6} = ML^{-1}T^{-2}$$

This dimension,  $ML^{-1}T^{-2}$ , is equivalent to the dimension of energy density.

-----

## **Question70**

**In an adiabatic process, which of the following statements is true?**

### **JEE Main 2025 (Online) 2nd April Morning Shift**

**Options:**

A. The internal energy of the gas decreases as the temperature increases

B. The molar heat capacity is zero

C. Work done by the gas equals the increase in internal energy

D. The molar heat capacity is infinite

**Answer: B**

**Solution:**

Option B is correct.

- In an adiabatic process no heat is exchanged, so

$$\delta Q = 0.$$

- The (molar) heat capacity for any process is defined by

$$C = \frac{\delta Q}{n dT}.$$

Since  $\delta Q = 0$  even when  $dT \neq 0$ , it follows that

$$C_{\text{adiabatic}} = 0.$$

All other statements are therefore false.

-----

## Question 71

**Identify the characteristics of an adiabatic process in a monoatomic gas. (A) Internal energy is constant. (B) Work done in the process is equal to the change in internal energy. (C) The product of temperature and volume is a constant. (D) The product of pressure and volume is a constant. (E) The work done to change the temperature from  $T_1$  to  $T_2$  is proportional to  $(T_2 - T_1)$ . Choose the correct answer from the options given below :**

**JEE Main 2025 (Online) 2nd April Evening Shift**

**Options:**

A. (B), (D) only

B. (B), (E) only

C. (A), (C), (E) only

D. (A), (C), (D) only

**Answer: B**

**Solution:**

In an adiabatic process, the key characteristics for a monoatomic gas are:

**No Heat Exchange:** The adiabatic process occurs without heat transfer, meaning  $Q = 0$ .

**Work Done and Internal Energy Change:** The work done on or by the system is equal to the change in internal energy. Mathematically, this is expressed as:

$$\Delta U = -W$$

where  $\Delta U$  is the change in internal energy and  $W$  is the work done.

**Work and Temperature Relationship:** The work done when changing the temperature from  $T_1$  to  $T_2$  is proportional to the difference between these temperatures. This relationship can be expressed as:

$$|\text{Work Done}| = nC_v\Delta T \propto (T_2 - T_1)$$

Here,  $n$  is the number of moles,  $C_v$  is the molar heat capacity at constant volume, and  $\Delta T$  is the change in temperature.

From these characteristics, the correct features of an adiabatic process in a monoatomic gas relate to the relationships involving work done and temperature change, specifically options (B) and (E).

-----

## Question 72

### Match List - I with List - II

List - I

List - II

- |                                    |                                                    |
|------------------------------------|----------------------------------------------------|
| (A) Heat capacity of body          | (I) $\text{J kg}^{-1}$                             |
| (B) Specific heat capacity of body | (II) $\text{J K}^{-1}$                             |
| (C) Latent heat                    | (III) $\text{J kg}^{-1} \text{K}^{-1}$             |
| (D) Thermal conductivity           | (IV) $\text{J m}^{-1} \text{K}^{-1} \text{s}^{-1}$ |

Choose the correct answer from the options given below :

Options:

- A. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
- B. (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
- C. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
- D. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

**Answer: D**



## Solution:

To match the entries in List-I with their corresponding units in List-II, we can use the definitions and formulas related to each concept in thermal physics:

**Heat Capacity of a Body (C')** is defined as:

$$C' = \frac{\Delta Q}{\Delta T}$$

The unit for heat capacity (C') is Joules per Kelvin ( $\text{J K}^{-1}$ ).

**Specific Heat Capacity of a Body (S)** indicates how much heat energy is required to change the temperature of a unit mass of a substance by one degree Kelvin:

$$S = \frac{\Delta Q}{m\Delta T}$$

The unit for specific heat capacity (S) is Joules per kilogram per Kelvin ( $\text{J kg}^{-1} \text{K}^{-1}$ ).

**Latent Heat (L)** is the heat energy absorbed or released during a phase change of a substance at a constant temperature:

$$L = \frac{\Delta Q}{m}$$

The unit for latent heat (L) is Joules per kilogram ( $\text{J kg}^{-1}$ ).

**Thermal Conductivity (K)** measures the ability of a material to conduct heat:

$$\Delta Q = \frac{KA\Delta T}{L} \Rightarrow K = \frac{\Delta Q \cdot L}{A\Delta T}$$

The unit for thermal conductivity (K) is Joules per meter per Kelvin per second ( $\text{J m}^{-1} \text{K}^{-1} \text{s}^{-1}$ ).

Using these formulas and their corresponding units, the correct matches are:

(A) Heat capacity of body  $\rightarrow$  (II)  $\text{J K}^{-1}$

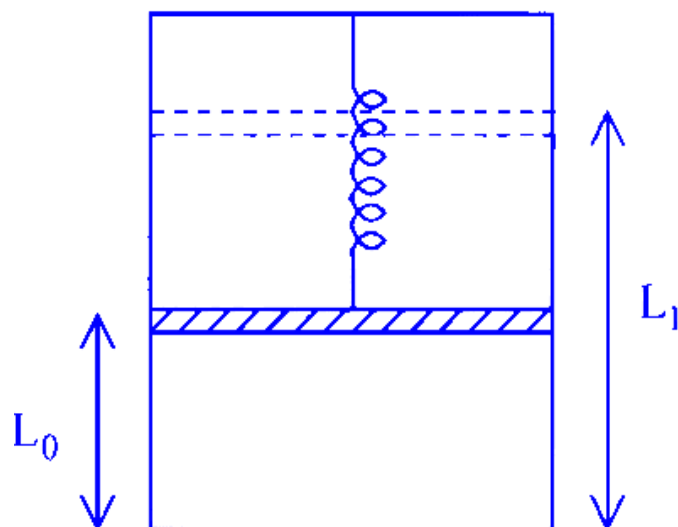
(B) Specific heat capacity of body  $\rightarrow$  (III)  $\text{J kg}^{-1} \text{K}^{-1}$

(C) Latent heat  $\rightarrow$  (I)  $\text{J kg}^{-1}$

(D) Thermal conductivity  $\rightarrow$  (IV)  $\text{J m}^{-1} \text{K}^{-1} \text{s}^{-1}$

---

## Question73



A piston of mass  $M$  is hung from a massless spring whose restoring force law goes as  $F = -kx^3$ , where  $k$  is the spring constant of appropriate dimension. The piston separates the vertical chamber into two parts, where the bottom part is filled with ' $n$ ' moles of an ideal gas. An external work is done on the gas isothermally (at a constant temperature  $T$ ) with the help of a heating filament (with negligible volume) mounted in lower part of the chamber, so that the piston goes up from a height  $L_0$  to  $L_1$ , the total energy delivered by the filament is: (Assume spring to be in its natural length before heating)

**JEE Main 2025 (Online) 3rd April Morning Shift**

**Options:**

- A.  $nRT \ln \left( \frac{L_1}{L_0} \right) + Mg(L_1 - L_0) + \frac{3k}{4}(L_1^4 - L_0^4)$
- B.  $nRT \ln \left( \frac{L_1}{L_0} \right) + Mg(L_1 - L_0) + \frac{k}{4}(L_1^4 - L_0^4)$
- C.  $nRT \ln \left( \frac{L_1^2}{L_0^2} \right) + \frac{Mg}{2}(L_1 - L_0) + \frac{k}{4}(L_1^4 - L_0^4)$
- D.  $3nRT \ln \left( \frac{L_1}{L_0} \right) + 2Mg(L_1 - L_0) + \frac{k}{3}(L_1^3 - L_0^3)$

**Answer: B**

## Solution:

Using WET

Total energy supplied = gravitational potential energy + spring potential energy + work done by gas

$$Mg(L_1 - L_0) + \int_{L_0}^{L_1} kx^3 dx + nRT \ln$$

$$\left[ \frac{L_1}{L_0} \frac{A}{A} \right] + W_{\text{ext}} = 0$$

$$\frac{K}{4} [x^4]_{L_0}^{L_1} + Mg(L_1 - L_0) + \int_{L_0}^{L_1} kx^3 dx + nRT \ln$$

$$\left[ \frac{L_1}{L_0} \right] + W_{\text{ext}} = 0$$

$$\frac{k}{4} (L_1^4 - L_0^4) + Mg(L_1 - L_0) + nRT \ln$$

$$\left[ \frac{L_1}{L_0} \right] + W_{\text{ext}} = 0$$

$$W_{\text{ext}} = \frac{k}{4} (L_1^4 - L_0^4) + Mg(L_1 - L_0) + nRT \ln \left[ \frac{L_1}{L_0} \right]$$

---

## Question 74

During the melting of a slab of ice at 273 K at atmospheric pressure :

**JEE Main 2025 (Online) 3rd April Morning Shift**

**Options:**

- A. Internal energy of ice-water system remains unchanged.
- B. Positive work is done by the ice-water system on the atmosphere.
- C. Positive work is done on the ice-water system by the atmosphere.
- D. Internal energy of the ice-water system decreases.

**Answer: C**

## Solution:

During the melting of a slab of ice at 273 K under atmospheric pressure, the following occurs:

As the ice melts, its volume decreases. Because of this volume reduction, the atmosphere exerts a positive work on the ice-water system. When heat is absorbed by the ice-water system, we denote the absorbed heat as  $\Delta Q$ , which is positive. The work done by the ice-water system is negative because it is being compressed by the external pressure.

According to the first law of thermodynamics:

$$\Delta U = \Delta Q + \Delta W$$

In this equation,  $\Delta U$  represents the change in internal energy,  $\Delta Q$  is the heat absorbed by the system, and  $\Delta W$  is the work done by the system. Since the work done is negative and the absorbed heat is positive, the overall internal energy ( $\Delta U$ ) increases.

-----

## Question 75

**A gas is kept in a container having walls which are thermally non-conducting. Initially the gas has a volume of  $800 \text{ cm}^3$  and temperature  $27^\circ \text{C}$ . The change in temperature when the gas is adiabatically compressed to  $200 \text{ cm}^3$  is:**

**(Take  $\gamma = 1.5$ ;  $\gamma$  is the ratio of specific heats at constant pressure and at constant volume)**

**JEE Main 2025 (Online) 3rd April Morning Shift**

**Options:**

- A. 300 K
- B. 600 K
- C. 327 K
- D. 522 K

**Answer: A**

**Solution:**

To determine the change in temperature when a gas is adiabatically compressed, we are given the following initial conditions:

Initial volume,  $V_1 = 800 \text{ cm}^3$

Final volume,  $V_2 = 200 \text{ cm}^3$

Initial temperature,  $T_1 = 27^\circ \text{C} = 300 \text{ K}$

Adiabatic index (ratio of specific heats)  $\gamma = 1.5$

In an adiabatic process, the relationship between temperature and volume is given by the equation:

$$TV^{\gamma-1} = \text{constant}$$

Applying this to the initial and final states:

$$300 \times (800)^{0.5} = T_2 \times (200)^{0.5}$$

Solving for the final temperature  $T_2$ :

$$T_2 = 300 \times \left(\frac{800}{200}\right)^{0.5} = 300 \times (4)^{0.5}$$

$$T_2 = 300 \times 2 = 600 \text{ K}$$

Thus, the change in temperature is:

$$\Delta T = T_2 - T_1 = 600 \text{ K} - 300 \text{ K} = 300 \text{ K}$$

Therefore, when the gas is adiabatically compressed, the temperature increases by 300 K.

-----

## Question 76

**Pressure of an ideal gas, contained in a closed vessel, is increased by 0.4% when heated by 1°C. Its initial temperature must be:**

**JEE Main 2025 (Online) 3rd April Evening Shift**

**Options:**

A. 2500 K

B. 25°C

C. 250°C

D. 250 K

**Answer: D**

**Solution:**

In this scenario, we're dealing with an isochoric process, meaning the volume of the gas remains constant. For an ideal gas in an isochoric process, the pressure ( $P$ ) is directly proportional to the temperature ( $T$ ) in Kelvin.

The relationship can be expressed as:

$$P \propto T$$

Given the percentage increase in pressure is 0.4% when the temperature is increased by 1°C, we can use the relationship:

$$\frac{\Delta P}{P} = \frac{\Delta T}{T}$$

Substituting the given values into the equation, we have:

$$\frac{0.4}{100} = \frac{1}{T}$$

Solving for  $T$ , we find:

$$T = 250 \text{ K}$$

Therefore, the initial temperature of the gas in the vessel must be 250 K.

---

## Question 77

**An ideal gas exists in a state with pressure  $P_0$ , volume  $V_0$ . It is isothermally expanded to 4 times of its initial volume ( $V_0$ ), then isobarically compressed to its original volume. Finally the system is heated isochorically to bring it to its initial state. The amount of heat exchanged in this process is**

**JEE Main 2025 (Online) 3rd April Evening Shift**

**Options:**

A.  $P_0 V_0 (\ln 2 - 0.75)$

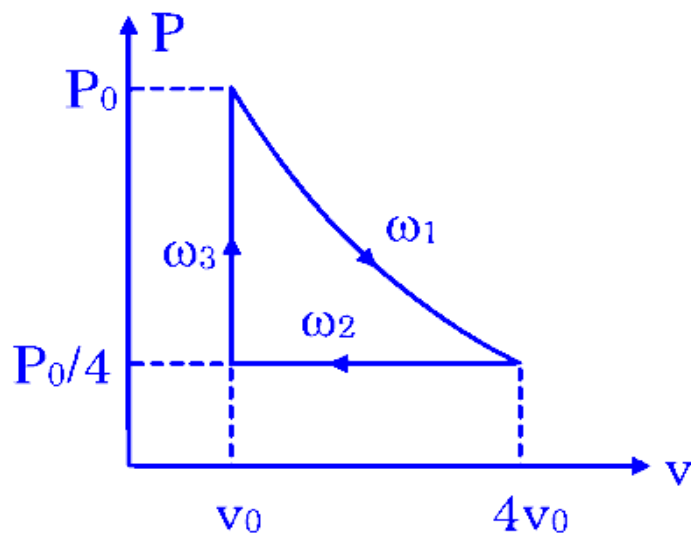
B.  $P_0 V_0 (2 \ln 2 - 0.75)$

C.  $P_0 V_0 (2 \ln 2 - 0.25)$

D.  $P_0 V_0 (\ln 2 - 0.25)$

**Answer: B**

**Solution:**



$$\omega_1 = P_0 v_0 \ln 4$$

$$\omega_2 = \frac{P_0}{4}(-3v_0) = -\frac{3P_0 v_0}{4}$$

$$\omega_3 = 0$$

$$Q_T = \Delta U_{\text{cyclic}} + \omega$$

$$Q_T = \omega \quad (\Delta U_{\text{cyclic}} = 0)$$

$$Q_T = P_0 v_0 \left( \ln 4 - \frac{3}{4} \right)$$

$$= P_0 v_0 (2 \ln 2 - 0.75)$$

## Question 78

The mean free path and the average speed of oxygen molecules at 300 K and 1 atm are  $3 \times 10^{-7}$  m and 600 m/s, respectively. Find the frequency of its collisions.

**JEE Main 2025 (Online) 4th April Morning Shift**

**Options:**

A.  $5 \times 10^8/\text{s}$

B.  $9 \times 10^5/\text{s}$

C.  $2 \times 10^{10}/\text{s}$

D.  $2 \times 10^9/\text{s}$

**Answer: D**

## Solution:

To determine the frequency of collisions for oxygen molecules, we use the formula for frequency:

$$\text{Frequency} = \frac{1}{T} = \frac{V_{\text{avg}}}{\lambda}$$

where  $V_{\text{avg}}$  is the average speed of the molecules and  $\lambda$  is the mean free path.

Given:

Average speed ( $V_{\text{avg}}$ ) = 600 m/s

Mean free path ( $\lambda$ ) =  $3 \times 10^{-7}$  m

Substitute these values into the formula:

$$\text{Frequency} = \frac{600}{3 \times 10^{-7}} = 2 \times 10^9 \text{ s}^{-1}$$

Therefore, the frequency of collisions is  $2 \times 10^9$  collisions per second.

-----

## Question 79

**Consider the sound wave travelling in ideal gases of He, CH<sub>4</sub>, and CO<sub>2</sub>. All the gases have the same ratio  $\frac{P}{\rho}$ , where  $P$  is the pressure and  $\rho$  is the density. The ratio of the speed of sound through the gases  $V_{\text{He}} : V_{\text{CH}_4} : V_{\text{CO}_2}$  is given by**

**JEE Main 2025 (Online) 4th April Morning Shift**

**Options:**

A.  $\sqrt{\frac{7}{5}} : \sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}}$

B.  $\sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}} : \sqrt{\frac{4}{3}}$

C.  $\sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}} : \sqrt{\frac{7}{5}}$

D.  $\sqrt{\frac{4}{3}} : \sqrt{\frac{5}{3}} : \sqrt{\frac{7}{5}}$

**Answer: C**



## Solution:

The speed of sound in an ideal gas is given by

$$v = \sqrt{\gamma \frac{P}{\rho}},$$

where:

$\gamma$  is the ratio of specific heats,

$P$  is the pressure, and

$\rho$  is the density.

Since the problem states that  $\frac{P}{\rho}$  is the same for all the gases, the speed of sound in each gas is determined solely by the factor  $\sqrt{\gamma}$ .

Let's determine the appropriate  $\gamma$  for each gas:

### Helium (He):

Helium is a monatomic gas.

For a monatomic gas,  $\gamma = \frac{5}{3}$ .

Therefore,  $v_{\text{He}} \propto \sqrt{\frac{5}{3}}$ .

### Methane (CH<sub>4</sub>):

Methane is a polyatomic gas (a tetrahedral molecule with three rotational degrees of freedom).

For nonlinear polyatomic gases,  $\gamma$  is typically taken as  $\frac{4}{3}$ .

Thus,  $v_{\text{CH}_4} \propto \sqrt{\frac{4}{3}}$ .

### Carbon Dioxide (CO<sub>2</sub>):

Carbon dioxide is a linear molecule.

For linear molecules,  $\gamma = \frac{7}{5}$ .

Hence,  $v_{\text{CO}_2} \propto \sqrt{\frac{7}{5}}$ .

Therefore, the ratio of the speeds of sound in the gases is:

$$v_{\text{He}} : v_{\text{CH}_4} : v_{\text{CO}_2} = \sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}} : \sqrt{\frac{7}{5}}.$$

Comparing this expression with the given options, we see that it matches **Option C**.

-----

## Question80

### Match List - I with List - II.

|     | List - I   |       | List - II                         |
|-----|------------|-------|-----------------------------------|
| (A) | Isobaric   | (I)   | $\Delta Q = \Delta W$             |
| (B) | Isochoric  | (II)  | $\Delta Q = \Delta U$             |
| (C) | Adiabatic  | (III) | $\Delta Q = \text{zero}$          |
| (D) | Isothermal | (IV)  | $\Delta Q = \Delta U + P\Delta V$ |

$\Delta Q$  = Heat supplied

$\Delta W$  = Work done by the system

$\Delta U$  = Change in internal energy

P = Pressure of the system

$\Delta V$  = Change in volume of the system

Choose the correct answer from the options given below :

**JEE Main 2025 (Online) 4th April Evening Shift**

**Options:**

A. (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

B. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

C. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

D. (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

**Answer: B**

**Solution:**

(A) Isobaric ( $P = C$ )

$$\Delta Q = \Delta U + P\Delta V$$

(B) Isochoric ( $V = C$ )

$$\Delta Q = \Delta U$$

(C) Adiabatic ( $\Delta Q = 0$ )

$$\Delta Q = 0$$

(D) Isothermal ( $\Delta U = 0$ )

$$\Delta Q = \Delta W$$

---

## Question81

**There are two vessels filled with an ideal gas where volume of one is double the volume of other. The large vessel contains the gas at 8 kPa at 1000 K while the smaller vessel contains the gas at 7 kPa at 500 K . If the vessels are connected to each other by a thin tube allowing the gas to flow and the temperature of both vessels is maintained at 600 K , at steady state the pressure in the vessels will be (in kPa ).**

**JEE Main 2025 (Online) 4th April Evening Shift**

**Options:**

A. 24

B. 4.4

C. 18

D. 6

**Answer: D**

**Solution:**



C.  $3.0 \times 10^{-7} \text{ m}^2$

D.  $4.0 \times 10^{-7} \text{ m}^2$

**Answer: A**

## **Solution:**

### **Calculate the Change in Temperature:**

Use the formula:

$$\Delta Q = mC_v\Delta T$$

Given:

$$\Delta Q = 8.1 \times 10^2 \text{ J}$$

$$m = 0.1 \text{ kg}$$

$$C_v = 900 \text{ J kg}^{-1} \text{ K}^{-1}$$

Substitute the values to find  $\Delta T$ :

$$8.1 \times 10^2 = 0.1 \times 900 \times \Delta T$$

Solving for  $\Delta T$ :

$$\Delta T = \frac{8.1 \times 10^2}{0.1 \times 900} = 9 \text{ K}$$

### **Calculate the Change in Area:**

The change in area  $\Delta A$  is calculated using:

$$\Delta A = A_0 \times 2\alpha\Delta T$$

Where:

$A_0$  is the original area of the sheet.

$$\alpha = 3.1 \times 10^{-5} \text{ K}^{-1}.$$

$$\Delta T = 9 \text{ K}.$$

First, calculate the original area  $A_0$ :

$$A_0 = l \times d = 9 \text{ cm} \times 4 \text{ cm} = 36 \text{ cm}^2 = 0.0036 \text{ m}^2$$

Now, substitute into the formula for  $\Delta A$ :

$$\Delta A = 0.0036 \times 2 \times 3.1 \times 10^{-5} \times 9$$

Simplifying:

$$\Delta A = 2.0 \times 10^{-6} \text{ m}^2$$

Therefore, the change in area of the rectangular sheet is  $\boxed{2.0 \times 10^{-6} \text{ m}^2}$ .

## Question83

### Match the List I with List II

|     | List - I               |       | List - II                       |
|-----|------------------------|-------|---------------------------------|
| (A) | Triatomic rigid gas    | (I)   | $\frac{C_p}{C_v} = \frac{5}{3}$ |
| (B) | Diatomic non-rigid gas | (II)  | $\frac{C_p}{C_v} = \frac{7}{5}$ |
| (C) | Monoatomic gas         | (III) | $\frac{C_p}{C_v} = \frac{4}{3}$ |
| (D) | Diatomic rigid gas     | (IV)  | $\frac{C_p}{C_v} = \frac{9}{7}$ |

Choose the correct answer from the options given below:

### JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. A-III, B-IV, C-I, D-II

B. A-II, B-IV, C-I, D-III

C. A-IV, B-II, C-III, D-I

D. A-III, B-II, C-IV, D-I

**Answer: A**

**Solution:**

The explanation uses the equation for the heat capacity ratio,  $\gamma = 1 + \frac{2}{f}$ , where f is the degrees of freedom of the gas. Here's how it applies to different types of gases:

**Triatomic Rigid Gas:**

Degrees of freedom (f) = 6

$$\gamma = 1 + \frac{2}{6} = \frac{4}{3}$$

**Diatomic Non-Rigid Gas:**

Degrees of freedom (f) = 7

$$\gamma = 1 + \frac{2}{7} = \frac{9}{7}$$

**Diatomic Rigid Gas:**

Degrees of freedom (f) = 5

$$\gamma = 1 + \frac{2}{5} = \frac{7}{5}$$

**Monoatomic Gas:**

Degrees of freedom (f) = 3

$$\gamma = 1 + \frac{2}{3} = \frac{5}{3}$$

Therefore, when matching List I with List II, the correct pairing is:

**A** (Triatomic rigid gas) - **III** ( $\frac{4}{3}$ )

**B** (Diatomic non-rigid gas) - **IV** ( $\frac{9}{7}$ )

**C** (Monoatomic gas) - **I** ( $\frac{5}{3}$ )

**D** (Diatomic rigid gas) - **II** ( $\frac{7}{5}$ )

Thus, the correct answer from the options given is:

**A-III, B-IV, C-I, D-II**

-----

## Question84

**Match List - I with List - II.**

| List - I       | List - II                                             |
|----------------|-------------------------------------------------------|
| (A) Isothermal | (I) $\Delta W$ (work done) = 0                        |
| (B) Adiabatic  | (II) $\Delta Q$ (supplied heat) = 0                   |
| (C) Isobaric   | (III) $\Delta U$ (change in internal energy) $\neq 0$ |
| (D) Isochoric  | (IV) $\Delta U = 0$                                   |

**Choose the correct answer from the options given below :**

**JEE Main 2025 (Online) 7th April Evening Shift**

**Options:**

A.

(A)-(III), (B)-(II), (C)-(I), (D)-(IV)

B.

(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

C.

(A)-(IV), (B)-(II), (C)-(III), (D)-(I)

D.

(A)-(IV), (B)-(I), (C)-(III), (D)-(II)

**Answer: C**

## Solution:

To accurately match the terms from List-I with the corresponding statements in List-II, consider the properties of each thermodynamic process:

### Isothermal Process (A):

During an isothermal process, the temperature remains constant ( $\Delta T = 0$ ), which means that the change in internal energy ( $\Delta U$ ) is zero. Therefore, it corresponds to (IV).

### Adiabatic Process (B):

In an adiabatic process, no heat is exchanged with the surroundings ( $\Delta Q = 0$ ). This matches with (II).

### Isobaric Process (C):

An isobaric process involves constant pressure conditions ( $\Delta P = 0$ ), but internal energy ( $\Delta U$ ) can change. Thus, it's associated with (III).

### Isochoric Process (D):

For an isochoric process, volume remains constant ( $\Delta V = 0$ ), which means that no work is done ( $\Delta W = 0$ ). This aligns with (I).

Based on these descriptions, the correct matching is:

(A) Isothermal  $\rightarrow$  (IV)

(B) Adiabatic  $\rightarrow$  (II)

(C) Isobaric  $\rightarrow$  (III)

(D) Isochoric  $\rightarrow$  (I)

---

## Question85



**The helium and argon are put in the flask at the same room temperature (300 K). The ratio of average kinetic energies (per molecule) of helium and argon is:**

**(Give: Molar mass of helium = 4 g/mol, Molar mass of argon = 40 g/mol)**

**JEE Main 2025 (Online) 7th April Evening Shift**

**Options:**

A.

$$1 : \sqrt{10}$$

B.

$$10 : 1$$

C.

$$1 : 10$$

D.

$$1 : 1$$

**Answer: D**

**Solution:**

Given the same room temperature of 300 K for helium and argon, we are tasked with finding the ratio of their average kinetic energies per molecule.

**Formula for Kinetic Energy per Molecule:**

$$\text{K. E.} = \frac{f}{2}kT$$

In this equation:

$f$  is the degrees of freedom, which is 3 for both helium (He) and argon (Ar) since they are monatomic gases.

$k$  is the Boltzmann constant.

$T$  is the temperature in Kelvin.

**Kinetic Energy Comparison:**

For both helium and argon, since  $f = 3$ , the expression simplifies:

$$\frac{K \cdot E_{\text{He}}}{K \cdot E_{\text{Ar}}} = \frac{1}{1}$$

Thus, the average kinetic energy per molecule for both gases is the same at the same temperature, leading to a ratio of:

1 : 1

This result implies that despite their differences in molar masses (helium being 4 g/mol and argon 40 g/mol), the average kinetic energy per molecule remains equal at the same temperature.

---

## Question 86

**Water falls from a height of 200 m into a pool. Calculate the rise in temperature of the water assuming no heat dissipation from the water in the pool.**

**(Take  $g = 10 \text{ m/s}^2$ , specific heat of water =  $4200 \text{ J/(kg K)}$ )**

**JEE Main 2025 (Online) 8th April Evening Shift**

**Options:**

A.

0.36 K

B.

0.23 K

C.

0.48 K

D.

0.14 K

**Answer: C**

**Solution:**

To find the rise in temperature of the water as it falls from a height, we start by equating the potential energy loss to the heat gained, assuming no heat is lost from the water in the pool.

The potential energy lost by the water can be expressed as:

$$mgh$$

where:

$m$  is the mass of water,

$g$  is the acceleration due to gravity ( $10 \text{ m/s}^2$ ),

$h$  is the height (200 m).

The heat gained by the water when temperature changes is given by:

$$ms\Delta T$$

where:

$s$  is the specific heat capacity of water ( $4200 \text{ J/(kg K)}$ ),

$\Delta T$  is the change in temperature.

Setting the potential energy loss equal to the heat gained:

$$mgh = ms\Delta T$$

We can simplify this equation by canceling out the mass  $m$ :

$$gh = s\Delta T$$

Now, solve for  $\Delta T$ :

$$\Delta T = \frac{gh}{s}$$

Substitute the known values:

$$\Delta T = \frac{10 \times 200}{4200}$$

Calculate  $\Delta T$ :

$$\Delta T = \frac{2000}{4200}$$

Simplify the fraction:

$$\Delta T = \frac{10}{21}$$

Therefore, the rise in temperature is  $\Delta T \approx 0.48 \text{ K}$ .

## Question 87

**A monoatomic gas having  $\gamma = \frac{5}{3}$  is stored in a thermally insulated container and the gas is suddenly compressed to  $\left(\frac{1}{8}\right)^{\text{th}}$  of its initial volume. The ratio of final pressure and initial pressure is:**

**( $\gamma$  is the ratio of specific heats of the gas at constant pressure and at constant volume)**

## JEE Main 2025 (Online) 8th April Evening Shift

**Options:**

A.

16

B.

32

C.

28

D.

40

**Answer: B**

**Solution:**

To find the ratio of final pressure ( $P_f$ ) to initial pressure ( $P_i$ ), we use the adiabatic condition:

$$P_i V_i^\gamma = P_f V_f^\gamma$$

We aim to express the pressure ratio:

$$\frac{P_f}{P_i} = \left( \frac{V_i}{V_f} \right)^\gamma$$

Given that the final volume  $V_f$  is  $\frac{1}{8}$  of the initial volume  $V_i$ , this simplifies to:

$$\left( \frac{V_i}{V_f} \right) = 8$$

Thus, the pressure ratio becomes:

$$\frac{P_f}{P_i} = 8^{\frac{5}{3}}$$

Calculating  $8^{\frac{5}{3}}$  gives:

$$\frac{P_f}{P_i} = 32$$

---

## Question 88

0.08kg air is heated at constant volume through  $5^{\circ}\text{C}$ . The specific heat of air at constant volume is  $0.17\text{kcal/kg}^{\circ}\text{C}$  and  $J = 4.18 \text{ joule/cal}$ . The change in its internal energy is approximately.

[27-Jan-2024 Shift 1]

Options:

A.

318J

B.

298J

C.

284J

D.

142J

**Answer: C**

**Solution:**

$$Q = \Delta U \text{ as work done is zero [constant volume]}$$

$$\Delta U = ms \Delta T$$

$$= 0.08 \times (170 \times 4.18) \times 5$$

$$\simeq 284\text{J}$$

---

## Question89

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio of  $C_p/C_v$  for the gas is :

[27-Jan-2024 Shift 2]

**Options:**

A.

$$5/3$$

B.

$$3/2$$

C.

$$7/5$$

D.

$$9/7$$

**Answer: B****Solution:**

$$P \propto T^3 \Rightarrow PT^{-3} = \text{constant}$$

$$PV^\gamma = \text{const}$$

$$P \left( \frac{nRT}{P} \right)^\gamma = \text{const}$$

$$P^{1-\gamma} T^\gamma = \text{const}$$

$$PT^{\frac{\gamma}{1-\gamma}} = \text{const}$$

$$\frac{\gamma}{1-\gamma} = -3$$

$$\gamma = -3 + 3\gamma$$

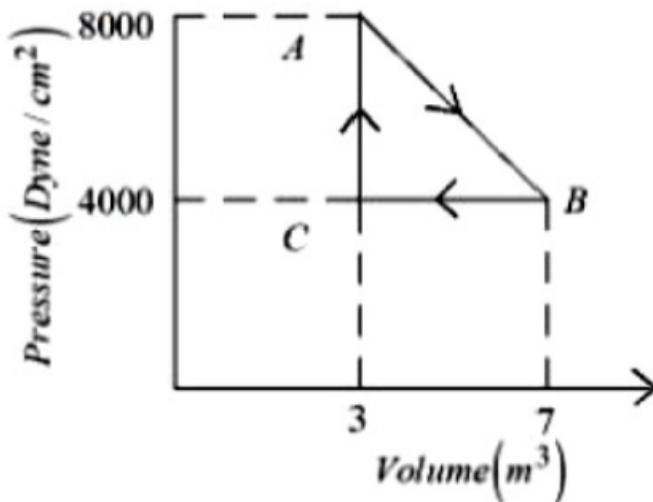
$$3 = 2\gamma$$

$$\gamma = \frac{3}{2}$$

---

**Question90**

A thermodynamic system is taken from an original state A to an intermediate state B by a linear process as shown in the figure. Its volume is then reduced to the original value from B to C by an isobaric process. The total work done by the gas from A to B and B to C would be :



[29-Jan-2024 Shift 1]

Options:

A.

33800J

B.

2200J

C.

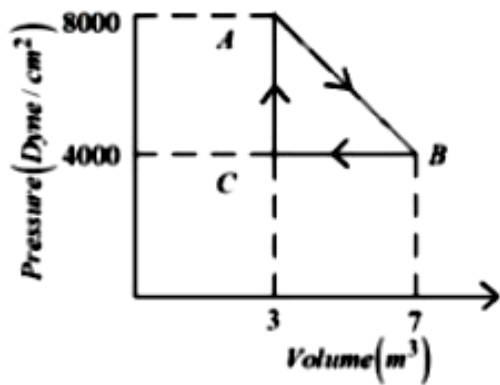
800J

D.

1200J

**Answer: C**

**Solution:**



$$\text{Work done } AB = \frac{1}{2}(8000 + 4000) \text{ Dyne / cm}^2 \times 4\text{m}^3 = (6000 \text{ Dyne / cm}^2) \times 4\text{m}^3$$

$$\text{Work done } BC = -(4000 \text{ Dyne / cm}^2) \times 4\text{m}^3$$

$$\text{Total work done} = 2000 \text{ Dyne / cm}^2 \times 4\text{m}^3$$

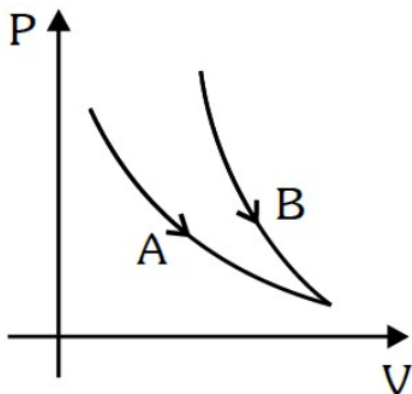
$$= 2 \times 10^3 \times \frac{1}{10^5} \frac{\text{N}}{\text{cm}^2} \times 4\text{m}^3$$

$$= 2 \times 10^{-2} \times \frac{\text{N}}{10^{-4}\text{m}^2} \times 4\text{m}^3$$

$$= 2 \times 10^2 \times 4 \text{ Nm} = 800\text{J}$$

## Question91

Choose the correct statement for processes A & B shown in figure.



[30-Jan-2024 Shift 2]



**Options:**

A.

$PV^\gamma = k$  for process B and  $PV = k$  for process A.

B.

$PV = k$  for process B and A.

C.

$\frac{P^{\gamma-1}}{T^\gamma} = k$  for process B and  $T = k$  for process A.

D.

$\frac{T^\gamma}{P^{\gamma-1}} = k$  for process A and  $PV = k$  for process B.

**Answer: 0**

**Solution:**

Steeper curve (B) is adiabatic

Adiabatic  $\Rightarrow PV^\gamma = \text{const.}$

Or  $P \left( \frac{T}{P} \right)^\gamma = \text{const.}$

$\frac{T^\gamma}{P^{\gamma-1}} = \text{const.}$

Curve (A) is isothermal

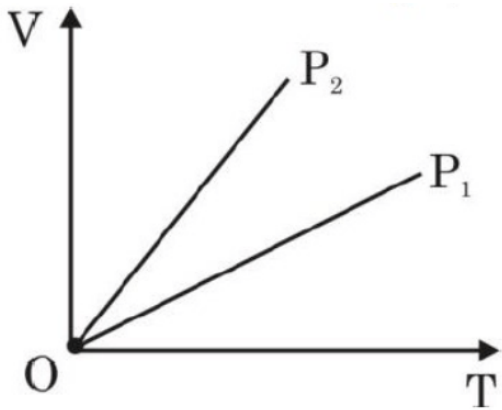
$T = \text{const.}$

$PV = \text{const.}$

---

## Question92

The given figure represents two isobaric processes for the same mass of an ideal gas, then



[31-Jan-2024 Shift 1]

Options:

A.

$$P_2 \geq P_1$$

B.

$$P_2 > P_1$$

C.

$$P_1 = P_2$$

D.

$$P_1 > P_2$$

**Answer: D**

**Solution:**

$$PV = nRT$$

$$V = \left( \frac{nR}{P} \right) T$$

$$\text{Slope} = \frac{nR}{P}$$

$$\text{Slope} \propto \frac{1}{P}$$

$$(\text{Slope})_2 > (\text{Slope})_1$$

$$P_2 < P_1$$

## Question93

The pressure and volume of an ideal gas are related as  $PV^{3/2} = K$  (Constant). The work done when the gas is taken from state A( $P_1, V_1, T_1$ ) to state B( $P_2, V_2, T_2$ ) is :

[1-Feb-2024 Shift 1]

Options:

A.

$$2(P_1V_1 - P_2V_2)$$

B.

$$2(P_2V_2 - P_1V_1)$$

C.

$$2(\sqrt{P_1}V_1 - \sqrt{P_2}V_2)$$

D.

$$2(P_2\sqrt{V_2} - P_1\sqrt{V_1})$$

**Answer: B**

## Solution:

For  $PV^x = \text{constant}$

If work done by gas is asked then

$$W = \frac{nR \Delta T}{1-x}$$

$$\text{Here } x = \frac{3}{2}$$

$$\therefore W = \frac{P_2 V_2 - P_1 V_1}{-\frac{1}{2}}$$

$$= 2(P_1 V_1 - P_2 V_2) \dots \text{Option (1) is correct}$$

If work done by external is asked then  $W = -2(P_1 V_1 - P_2 V_2) \dots \text{Option (2) is correct}$

---

## Question94

**A diatomic gas ( $\gamma = 1.4$ ) does 200J of work when it is expanded isobarically. The heat given to the gas in the process is :**

**[1-Feb-2024 Shift 2]**

**Options:**

A.

850J

B.

800J

C.

600J

D.

700J

**Answer: D**

**Solution:**

$$\gamma = 1 + \frac{2}{f} = 1.4 \Rightarrow \frac{2}{f} = 0.4$$

$$\Rightarrow f = 5$$

$$W = nR \Delta T = 200J$$

$$Q = \left( \frac{f+2}{2} \right) nR \Delta T$$

$$= \frac{7}{2} \times 200 = 700J$$

---

## Question95

**1g of a liquid is converted to vapour at  $3 \times 10^5$  Pa pressure. If 10% of the heat supplied is used for increasing the volume by  $1600\text{cm}^3$  during this phase change, then the increase in internal energy in the process will be :**

**[24-Jan-2023 Shift 1]**

**Options:**

A. 4320J

B. 432000J

C. 4800J

D.  $4.32 \times 10^8$ J

**Answer: A**

**Solution:**

$$\text{Work done} = P \Delta V$$

$$= 3 \times 10^5 \times 1600 \times 10^{-6}$$

$$= 480J$$

Only 10% of heat is used in work done.

Hence  $\Delta Q = 4800\text{J}$

The rest goes in internal energy, which is 90% of heat.

Change in internal energy  $= 0.9 \times 4800 = 4320\text{J}$

---

## Question96

Let  $\gamma_1$  be the ratio of molar specific heat at constant pressure and molar specific heat at constant volume of a monoatomic gas and  $\gamma_2$  be the similar ratio of diatomic gas. Considering the diatomic gas molecule as a rigid rotator, the ratio,  $\frac{\gamma_1}{\gamma_2}$  is

[24-Jan-2023 Shift 2]

Options:

A.  $\frac{27}{35}$

B.  $\frac{35}{27}$

C.  $\frac{25}{21}$

D.  $\frac{21}{25}$

**Answer: C**

**Solution:**

For monoatomic gas  $\gamma_1 = \frac{5}{3}$

For diatomic gas at low temperatures

$$\gamma_2 = \frac{7}{5}$$

$$\therefore \frac{\gamma_1}{\gamma_2} = \frac{\frac{5}{3}}{\frac{7}{5}} = \frac{25}{21}$$

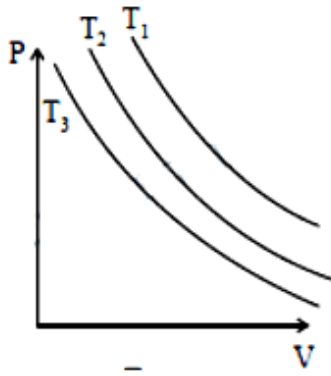
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## Question97

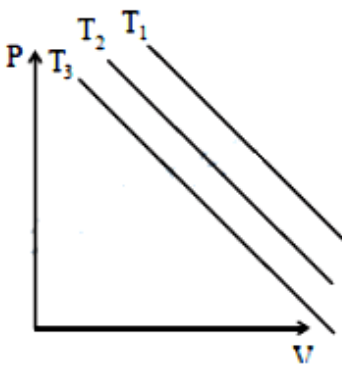
In an Isothermal change, the change in pressure and volume of a gas can be represented for three different temperature;  $T_3 > T_2 > T_1$  as :  
[24-Jan-2023 Shift 2]

Options:

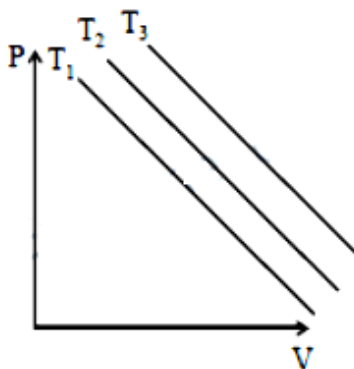
A.



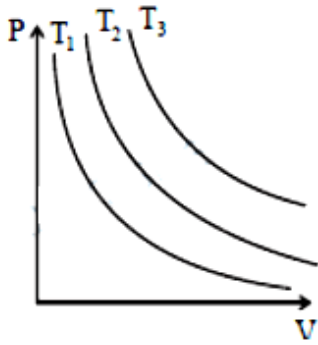
B.



C.



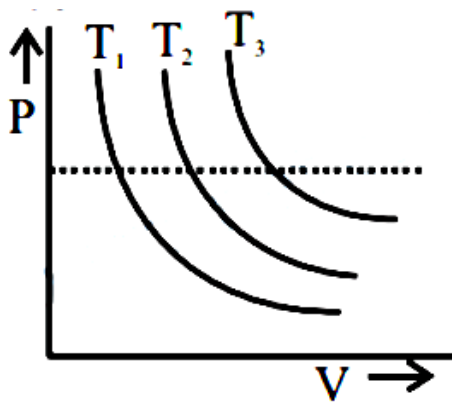
D.



**Answer: D**

**Solution:**

For isothermal process  $P - V$  graph is rectangular hyperbola



As dotted line is isobaric line which implies  $T_3 > T_2 > T_1$  as volume is increasing.

## Question98

**A Carnot engine with efficiency 50% takes heat from a source at 600K. In order to increase the efficiency to 70%, keeping the temperature of sink same, the new temperature of the source will be : [25-Jan-2023 Shift 1]**

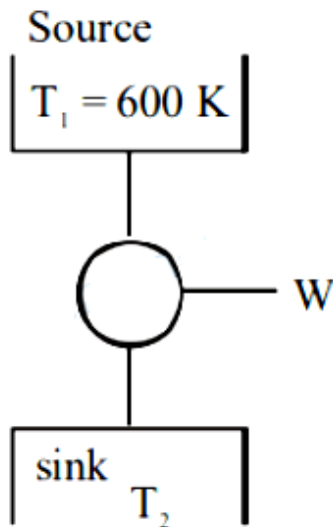
**Options:**

- A. 360K
- B. 1000K
- C. 900K
- D. 300K



**Answer: B**

**Solution:**



Initially  $\eta = \frac{1}{2}$

But  $\eta = 1 - \frac{T_2}{T_1}$

$$\therefore \frac{1}{2} = 1 - \frac{T_2}{600}$$

$$\Rightarrow \frac{T_2}{600} = \frac{1}{2}$$

Now efficiency is increased to 70% and  $T_2 = 300$

K, Let temp of source  $T_1 = T$

$$\Rightarrow \frac{7}{10} = 1 - \frac{300}{T}$$

$$\Rightarrow \frac{300}{T} = 1 - \frac{7}{10}$$

$$\Rightarrow \frac{300}{T} = \frac{3}{10} \therefore T_2 = 300\text{K}$$

---

## Question99

**A bowl filled with very hot soup cools from  $98^\circ\text{C}$  to  $86^\circ\text{C}$  in 2 minutes when the room temperature is  $22^\circ\text{C}$ . How long it will take to cool from  $75^\circ\text{C}$  to  $69^\circ\text{C}$  ?**  
**[25-Jan-2023 Shift 1]**

**Options:**

A. 2 minutes

B. 1.4 minutes

C. 0.5 minute

D. 1 minute

**Answer: B**

**Solution:**

$$\frac{\Delta Q}{\Delta t} = -K(T - T_0)$$

$$\frac{\Delta Q}{\Delta t} = -K(T_{\text{avg}} - T_0)$$

$$(i) \frac{ms \times 12}{2} = -K \left( \frac{98 + 86}{2} - 22 \right)$$

$$6 = - \frac{K}{ms} \left[ \frac{98 + 86}{2} - 22 \right]$$

$$6 = - \frac{K}{ms} [70] \dots \dots \dots (i)$$

$$(ii) \frac{ms \times 6}{\Delta t} = -K \left( \frac{75 + 69}{2} - 22 \right)$$

$$\frac{6}{\Delta t} = - \frac{K}{ms} (50) \dots \dots \dots (II)$$

$$(ii) \div (i)$$

$$\frac{6}{\Delta t(6)} = \frac{50}{70}$$

$$\Delta t = \frac{7}{5} = 1.4 \text{ min}$$

## Question100

**Match List I with List II :**

| List I                | List II                                                                              |
|-----------------------|--------------------------------------------------------------------------------------|
| A. Isothermal Process | I. Work done by the gas decreases internal energy                                    |
| B. Adiabatic Process  | II. No change in internal energy                                                     |
| C. Isochoric Process  | III. The heat absorbed goes partly to increase internal energy and partly to do work |
| D. Isobaric Process   | IV. No work is done on or by the gas                                                 |

**Choose the correct answer from the options given below :**

**[25-Jan-2023 Shift 2]**

**Options:**

A. A-II, B-I, C-III, D-IV

B. A-II, B-I, C-IV, D-III

C. A-I, B-II, C-IV, D-III

D. A-I, B-II, C-III, D-IV

**Answer: A**

**Solution:**

$$\Delta U = nC_v \Delta T$$

For isothermal process T is constant

$$\text{So } \Delta U = 0$$

$$A \rightarrow II$$

Adiabatic process

$$\Delta Q = 0$$

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta U = -\Delta W$$

Work done by gas is positive

So  $\Delta U$  is negative

$$B \rightarrow I$$

For Isochoric process  $\Delta W = 0$

$$C \rightarrow IV$$

For Isobaric process

$$\Delta W = P \Delta V \neq 0$$

$$\Delta U = nC_v \Delta T \neq 0$$

Heat absorbed goes partly to increase internal energy and partly do work.

---

## Question101

**According to law of equipartition of energy the molar specific heat of a diatomic gas at constant volume where the molecule has one additional vibrational mode is :-**

**[25-Jan-2023 Shift 2]**

**Options:**

A.  $\frac{9}{2}R$

B.  $\frac{5}{2}R$

C.  $\frac{3}{2}R$

D.  $\frac{7}{2}R$

**Answer: D**

### Solution:

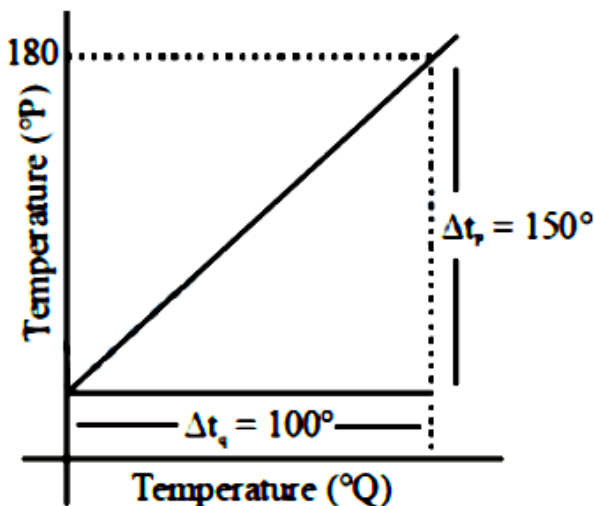
Diatomic gas molecules have three translational degree of freedom, two rotational degree of freedom & it is given that it has one vibrational mode so there are two additional degree of freedom corresponding to one vibrational mode, so total degree of freedom = 7

$$C_v = \frac{fR}{2} = \frac{7R}{2}$$

---

## Question102

The graph between two temperature scales P and Q is shown in the figure. Between upper fixed point and lower fixed point there are 150 equal divisions of scale P and 100 divisions on scale Q. The relationship for conversion between the two scales is given by :



[25-Jan-2023 Shift 2]

Options:

A.  $\frac{t_Q}{150} = \frac{t_P - 180}{100}$

B.  $\frac{t_Q}{100} = \frac{t_P - 30}{150}$

C.  $\frac{t_P}{180} = \frac{t_Q - 40}{100}$

D.  $\frac{t_P}{100} = \frac{t_Q - 180}{150}$

**Answer: B**

**Solution:**

$$\frac{\text{reading on scale} - \text{Lower fixed point}}{\text{upper fixed point} - \text{lower fixed point}} = \text{constant}$$

$$\frac{t_p - 30}{180 - 30} = \frac{t_Q - 0}{100 - 0}$$

$$\frac{t_p - 30}{150} = \frac{t_Q}{100}$$

---

## Question103

**A bicycle tyre is filled with air having pressure of 270 kPa at 27°C. The approximate pressure of the air in the tyre when the temperature increases to 36°C is**  
**[29-Jan-2023 Shift 1]**

**Options:**

A. 270 kPa

B. 262 KPa

C. 278 kPa

D. 360 kPa

**Answer: C**

**Solution:**

$$\text{Taking volume constant : } \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\Rightarrow P_2 = \frac{P_1}{T_1} \times T_2 = \frac{270 \times (309)}{300}$$
$$= 278 \text{ kPa}$$

---

## Question104

**Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.**

**Assertion A :** If  $dQ$  and  $dW$  represent the heat supplied to the system and the work done on the system respectively. Then according to the first law of thermodynamics  $dQ = dU - dW$ .

**Reason R :** First law of thermodynamics is based on law of conservation of energy.

**In the light of the above statements, choose the correct answer from the option given below :**

**[29-Jan-2023 Shift 1]**

**Options:**

- A. A is correct but R is not correct
- B. A is not correct but R is correct
- C. Both A and R are correct and R is the correct explanation of A
- D. Both A and R are correct but R is not the correct explanation of A

**Answer: C**

**Solution:**

First law of thermodynamics is based on law of conservation of energy and it can be written as  $dQ = dU - dW$ .  
where  $dW$  is work done on the system

-----

## **Question105**

**A body cools from  $60^{\circ}\text{C}$  to  $40^{\circ}\text{C}$  in 6 minutes. If, temperature of surroundings is  $10^{\circ}\text{C}$ . Then, after the next 6 minutes, its temperature will be  $^{\circ}\text{C}$ .**

**[29-Jan-2023 Shift 1]**

**Answer: 28**

## Solution:

By average form of Newton's law of cooling

$$\frac{20}{6} = k(50 - 10) \dots (i)$$

$$\frac{40 - T}{6} = K \left( \frac{40 + T}{2} - 10 \right) \dots (ii)$$

From equation (i) and (ii)

$$\frac{20}{40 - T} = \frac{40}{10 + T/2}$$

$$10 + \frac{T}{2} = 80 - 2T$$

$$\frac{5T}{2} = 70 \Rightarrow T = 28^\circ\text{C}$$

---

## Question106

Heat energy of 184 kJ is given to ice of mass 600g at  $-12^\circ\text{C}$ , Specific heat of ice is  $2222.3\text{Jkg}^{-1}^\circ\text{C}^{-1}$  and latent heat of ice is  $336\text{kJ/kg}^{-1}$

(A) Final temperature of system will be  $0^\circ\text{C}$ .

(B) Final temperature of the system will be greater than  $0^\circ\text{C}$ .

(C) The final system will have a mixture of ice and water in the ratio of 5 : 1.

(D) The final system will have a mixture of ice and water in the ratio of 1 : 5.

(E) The final system will have water only.

Choose the correct answer from the options given below:

[29-Jan-2023 Shift 2]

Options:

A. A and D only

B. B and D only

C. A and E only

D. A and C only

**Answer: A**

**Solution:**

$$\Delta Q = 184 \times 10^3$$

$$m = 0.600 \text{ kg at } -12^\circ\text{C}$$

$$S = 222.3 \text{ J/kg/}^\circ\text{C}$$

$$L = 336 \times 10^3 \text{ J/kg}$$

$$Q_1 = 0.600 \times 222.3 \times 12 = 16000.56 \text{ J}$$

$$\text{Remaining heat } \Delta Q_1 = 184000 - 16000.56$$

$$= 167999.44 \text{ J}$$

For melting at  $0^\circ\text{C}$

$$\Delta Q_2 = 0.600 \times 336000 = 201600 \text{ J needed}$$

$\therefore$  100% ice is not melted

Amount of ice melted

$$167999.44 = m \times 336000 = 0.4999 \text{ kg}$$

$\therefore$  mass of water = 0.4999 kg

Mass of ice = 0.1001

$$\therefore \text{Ratio} = \frac{0.1001}{0.4999} \approx 1 : 5$$

## Question107

Heat is given to an ideal gas in an isothermal process.

- A. Internal energy of the gas will decrease.
- B. Internal energy of the gas will increase.
- C. Internal energy of the gas will not change.
- D. The gas will do positive work.
- E. The gas will do negative work.

Choose the correct answer from the options given below:

[30-Jan-2023 Shift 1]

Options:

- A. A and E only
- B. B and D only
- C. C and E only
- D. C and D only

**Answer: D**

**Solution:**



Sol.  $dQ = dU + dW \Rightarrow dU = nC_V dT$   
 $dU = 0$  (for isothermal)  
 $\therefore U = \text{constant}$   
Also  $dQ > 0$  (supplied)  
Hence  $dW > 0$

---

## Question108

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A :** Efficiency of a reversible heat engine will be highest at  $-273^\circ\text{C}$  temperature of cold reservoir.

**Reason R :** The efficiency of Carnot's engine depends not only on temperature of cold reservoir but it depends on the temperature of

hot reservoir too and is given as  $\eta = \left(1 - \frac{T_2}{T_1}\right)$ .

In the light of the above statements, choose the correct answer from the options given below :

[30-Jan-2023 Shift 2]

**Options:**

- A. A is true but R is false
- B. Both A and R are true but R is NOT the correct explanation of A
- C. A is false but R is true
- D. Both A and R are true and R is the correct explanation of A

**Answer: D**

**Solution:**

Both A and R are true and R is the correct explanation of A

---

## Question109

**The correct relation between  $\gamma = \frac{C_p}{C_v}$  and temperature T is:**  
**[31-Jan-2023 Shift 1]**

**Options:**

A.  $\gamma \propto \frac{1}{\sqrt{T}}$

B.  $\gamma \propto T^\circ$

C.  $\gamma \propto \frac{1}{T}$

D.  $\gamma \propto T$

**Answer: B**

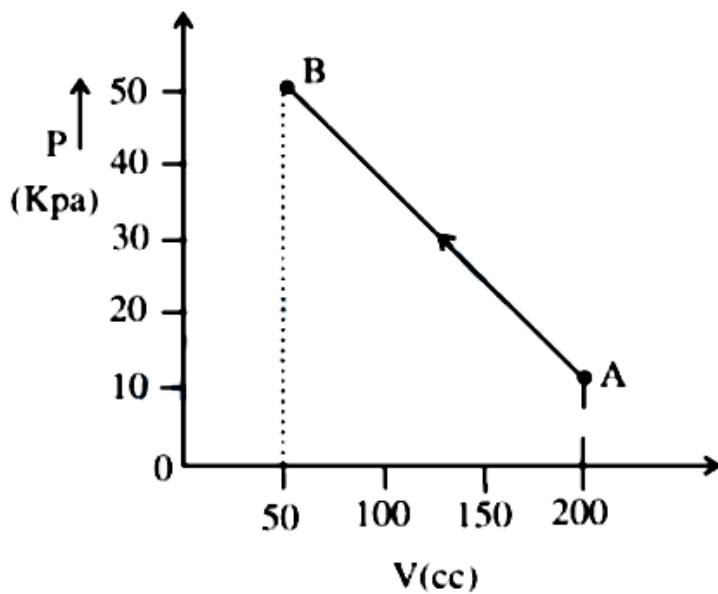
**Solution:**

$\gamma$  is independent of temperature  
Option 2

-----

## **Question110**

**The pressure of a gas changes linearly with volume from A to B as shown in figure. If no heat is supplied to or extracted from the gas then change in the internal energy of the gas will be**



**[31-Jan-2023 Shift 1]**

**Options:**

- A. 6J
- B. Zero
- C.  $-4.5\text{J}$
- D.  $4.5\text{J}$

**Answer: D**

**Solution:**

As  $\Delta q = 0$

$\Delta u = -W$

$W = \int P dV$

$\Delta u = -W = 30 \times 10^3 \times 150 \times 10^{-6}$

$= 4500 \times 10^{-3}$

$= 4.5\text{J}$

## Question111

A hypothetical gas expands adiabatically such that its volume changes from 08 litres to 27 litres. If the ratio of final pressure of the gas to

initial pressure of the gas is  $\frac{16}{81}$ . Then the ratio of  $\frac{C_p}{C_v}$  will be.

[31-Jan-2023 Shift 1]

Options:

A. 3/1

B. 4/3

C. 1/2

D. 3/2

**Answer: A**

**Solution:**

Let  $\gamma$  be the ratio of  $\frac{C_p}{C_v}$

Then for adiabatic process

$$PV^\gamma = \text{Constant}$$

$$\Rightarrow \frac{P_i}{P_f} = \left( \frac{V_f}{V_i} \right)^\gamma$$

$$\Rightarrow \frac{81}{16} = \left( \frac{27}{8} \right)^\gamma$$

$$\Rightarrow \gamma = \frac{4}{3}$$

---

## Question112

Heat energy of 735J is given to a diatomic gas allowing the gas to expand at constant pressure. Each gas molecule rotates around an internal axis but do not oscillate. The increase in the internal energy of the gas will be:

[31-Jan-2023 Shift 2]

Options:

A. 525J

B. 441J

C. 572J

D. 735J

**Answer: A**

**Solution:**

$$\Delta Q = nC_p \Delta T = 735J$$

$$\Rightarrow \frac{5nR \Delta T}{2} = 735J$$

$$\Delta U = nC_v \Delta T = \frac{3}{2}(nR \Delta T) = \frac{3}{2} \times \frac{2}{5} \times 735 \\ = 441J$$

---

## Question113

**A sample of gas at temperature T is adiabatically expanded to double its volume. The work done by the gas in the process is ( . given,  $\gamma = \frac{3}{2}$  )**

**:**

**[1-Feb-2023 Shift 1]**

**Options:**

A.  $W = TR[\sqrt{2} - 2]$

B.  $W = \frac{T}{R}[\sqrt{2} - 2]$

C.  $W = \frac{R}{T}[2 - \sqrt{2}]$

D.  $W = RT[2 - \sqrt{2}]$

**Answer: D**

**Solution:**

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T V^{1/2} = T_2 (2V)^{1/2}$$

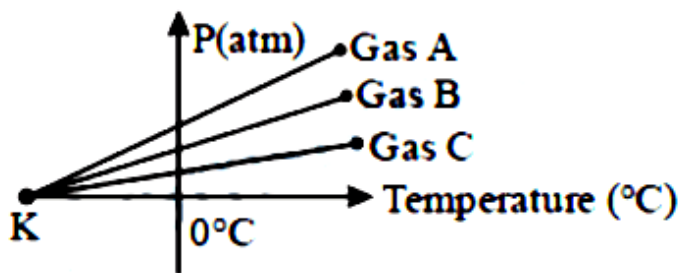
$$T_2 = \frac{T}{\sqrt{2}}$$

$$W = \frac{R(T_1 - T_2)}{\gamma - 1} = \frac{R\left(T - \frac{T}{\sqrt{2}}\right)}{\frac{1}{2}} = RT(2 - \sqrt{2})$$


---

## Question114

For three low density gases A, B, C pressure versus temperature graphs are plotted while keeping them at constant volume, as shown in the figure.



The temperature corresponding to the point ' K ' is:  
[1-Feb-2023 Shift 2]

Options:

A.  $-273^{\circ}\text{C}$

B.  $-100^{\circ}\text{C}$

C.  $-373^{\circ}\text{C}$

D.  $-40^{\circ}\text{C}$

**Answer: A**

**Solution:**

For isochoric process

$$\frac{P}{T} = n \frac{R}{V} = \text{constant}$$

$$P = \frac{nR}{V}(t + 273)$$

$$\text{If } P = 0 \Rightarrow t = -273^{\circ}\text{C}$$


---

## Question115

A Carnot engine operating between two reservoirs has efficiency  $\frac{1}{3}$ . When the temperature of cold reservoir raised by  $x$ , its efficiency decreases to  $\frac{1}{6}$ . The value of  $x$ , if the temperature of hot reservoir is  $99^\circ\text{C}$ , will be:  
[1-Feb-2023 Shift 2]

Options:

- A. 16.5K
- B. 33K
- C. 66K
- D. 62K

**Answer: D**

**Solution:**

$$\begin{aligned}T_H &= 99^\circ\text{C} = 99 + 273 \\&= 372\text{K} \\1 - \frac{T_C}{T_H} &= \frac{1}{3} \\1 - \frac{T_C}{372} &= \frac{1}{3} \quad (1) \Rightarrow T_C = \frac{2}{3} \times 372 \\&= 2 \times 124 = 248\text{K} \\1 - \frac{T_C + X}{T_H} &= \frac{1}{6} \\\frac{5}{6} &= \frac{T_C + X}{T_H} \\\frac{5}{6} &= \frac{248 + X}{372} \\248 + X &= 5 \times 62 \\X &= 310 - 248 = 62\text{K}\end{aligned}$$

---

## Question116

A source supplies heat to a system at the rate of 1000W. If the system performs work at a rate of 200W. The rate at which internal energy

**of the system increase is  
[6-Apr-2023 shift 1]**

**Options:**

- A. 500W
- B. 600W
- C. 800W
- D. 1200W

**Answer: C**

**Solution:**

From 1st law of thermodynamics,

$$dQ = dU + dW$$

Also, we can write this as,  $\frac{dQ}{dt} = \frac{dU}{dt} + \frac{dW}{dt}$

$$\Rightarrow 1000W = \frac{dU}{dt} + 200W$$

$$\Rightarrow \frac{dU}{dt} = 800W$$

---

## Question117

**Given below are two statements:**

**Statement I : If heat is added to a system, its temperature must increase.**

**Statement II : If positive work is done by a system in a thermodynamic process, its volume must increase. In the light of the above statements, choose the correct answer from the options given below**

**[8-Apr-2023 shift 1]**



**Options:**

- A. Both Statement I and Statement II are true
- B. Statement I is true but Statement II is false
- C. Both Statement I and Statement II are false
- D. Statement I is false but Statement II is true

**Answer: D**

**Solution:**

St I False

Ex. in isothermal process temp. is constant but heat can be added. ST II True

$$w = \int P dV$$

If volume increases the  $w = +ve$

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## Question118

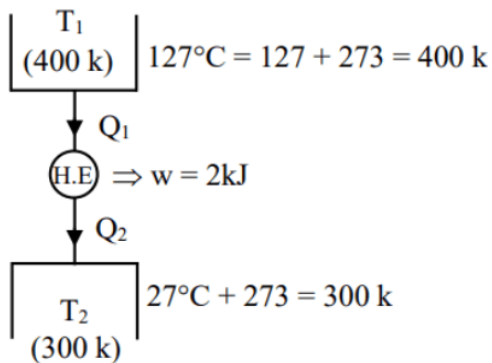
**Work done by a Carnot engine operating between temperatures  $127^{\circ}\text{C}$  and  $27^{\circ}\text{C}$  is 2 kJ. The amount of heat transferred to the engine by the reservoir is:**  
**[8-Apr-2023 shift 2]**

**Options:**

- A. 2 kJ
- B. 4 kJ
- C. 2.67 kJ
- D. 8 kJ

**Answer: D**

**Solution:**



$$n = 1 - \frac{300}{400} = \frac{1}{4}$$

$$n = \frac{w}{Q_1} = \frac{1}{4} \Rightarrow Q_1 = 8 \text{ kJ}$$

---

## Question119

Consider two containers A and B containing monoatomic gases at the same Pressure (P), Volume (V) and Temperature (T). The gas in A is compressed isothermally to  $\frac{1}{8}$  of its original volume while the gas in B is compressed adiabatically to  $\frac{1}{8}$  of its original volume. The ratio of final pressure of gas in B to that of gas in A is  
[10-Apr-2023 shift 1]

**Options:**

A. 8

B. 4

C.  $\frac{1}{8}$

D.  $8^{\frac{3}{2}}$

**Answer: B**

**Solution:**

By Isothermal Process for (A)

$$P_1 V_1 = P_2 V_2$$

$$PV = P_2 \frac{V}{8}$$

$$P_2 = 8P$$

For B adiabatically  $\gamma_{\text{mono}} = \frac{5}{3}$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$PV^{5/3} = P_2 \left( \frac{V}{8} \right)^{5/3}$$

$$P_2 = (8)^{5/3} P$$

$$\frac{P_2}{P_1} = \frac{8^{5/3}}{8P} = (8)^{\frac{2}{3}} = 4$$

---

## Question 120

**A gas is compressed adiabatically, which one of the following statement is NOT true.**

**[10-Apr-2023 shift 2]**

**Options:**

- A. There is no change in the internal energy
- B. The temperature of the gas increases.
- C. The change in the internal energy is equal to the work done on the gas
- D. There is no heat supplied to the system

**Answer: A**

**Solution:**

In Adiabatic process,  $\Delta Q = 0$

If gas is compressed, then  $w$  (by gas)  $\neq 0$

Therefore by 1<sup>st</sup> law

$$\Delta Q = \Delta u + w$$

$$0 = \Delta u + w$$

$$\Delta u = -w \neq 0$$

It implies in adiabatic compression, internal energy of gas changes.

---

## Question121

**The Thermodynamic process, in which internal energy of the system remains constant is**

**[11-Apr-2023 shift 2]**

**Options:**

- A. Isobaric
- B. Isochoric
- C. Adiabatic
- D. Isothermal

**Answer: D**

**Solution:**

$\Delta U = nC_v \Delta T$ , for all process

For isothermal process,  $\Delta T = 0$

So,  $\Delta U = 0$

That means internal energy of system remains constant.

-----

## Question122

**An engine operating between the boiling and freezing points of water will have**

- A. efficiency more than 27%**
- B. efficiency less than the efficiency of a Carnot engine operating between the same two temperatures.**
- C. efficiency equal to 27%**
- D. efficiency less than 27%**

**Choose the correct answer from the options given below :**

**[12-Apr-2023 shift 1]**

**Options:**

- A. B and C only
- B. B and D only

C. B, C and D only

D. A and b only

**Answer: B**

### **Solution:**

Engine is operating between FP and BP of water.

We asked to find efficiency :

$$T_1 = 100^\circ\text{C} \Rightarrow 100 + 273 = 373\text{K}$$

$$T_2 = 0^\circ\text{C} \Rightarrow 0 + 273 = 273\text{K}$$

We know

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\eta = 1 - \frac{273}{373}$$

$$\eta = \frac{373 - 273}{373} = \frac{100}{373}$$

$$\eta = 0.268$$

and

$$\eta\% = 0.268 \times 100$$

$$\eta\% = 26.8\%$$

---

## **Question123**

**The initial pressure and volume of an ideal gas are  $P_0$  and  $V_0$ . The final pressure of the gas when the gas is suddenly compressed to volume  $\frac{V_0}{4}$  will be :**

**(Given  $\gamma$  = ratio of specific heats at constant pressure and at constant volume)**

**[13-Apr-2023 shift 2]**

**Options:**

A.  $P_0(4)^{\frac{1}{\gamma}}$

B.  $4P_0$

C.  $P_0$

D.  $P_0(4)^\gamma$

**Answer: D**

**Solution:**

As, gas is suddenly compressed, the process is adiabatic

So equation of gas for adiabatic process is:

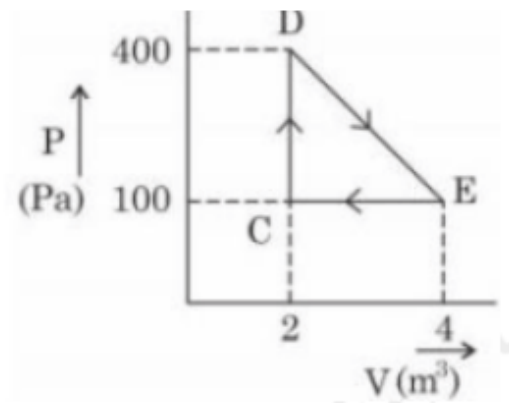
$$PV^\gamma = \text{constant}$$

$$P_0 V_0^\gamma = P_2 \left( \frac{V_0}{4} \right)^\gamma$$

$$P_2 = P_0(4)^\gamma$$

## Question124

**A thermodynamic system is taken through cyclic process. The total work done in the process is :**



**[15-Apr-2023 shift 1]**

**Options:**

A. 100J

B. 300J

C. 200J

D. Zero

**Answer: B**

**Solution:**

Work done = Area of graph

$$W = \frac{1}{2}(400 - 100)(4 - 2)$$

$$W = 300\text{J}$$

---

## Question125

**A Carnot engine whose heat sinks at  $27^\circ\text{C}$ , has an efficiency of 25%. By how many degrees should the temperature of the source be changed to increase the efficiency by 100% of the original efficiency? [24-Jun-2022-Shift-1]**

**Options:**

- A. Increases by  $18^\circ\text{C}$
- B. Increases by  $200^\circ\text{C}$
- C. Increases by  $120^\circ\text{C}$
- D. Increases by  $73^\circ\text{C}$

**Answer: B**

**Solution:**

$$\text{Initially : } \frac{1}{4} = 1 - \frac{300}{T_H}$$

$$\Rightarrow T_H = 400\text{K}$$

$$\text{Finally : Efficiency becomes } \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} = 1 - \frac{300}{T_H'}$$

$$\Rightarrow T_H' = 600\text{K}$$

$$\Rightarrow \text{Temperature of the source increases by } 200^\circ\text{C}.$$

---

## Question126

**A Carnot engine takes 5000 kcal of heat from a reservoir at  $727^\circ\text{C}$  and gives heat to a sink at  $127^\circ\text{C}$ . The work done by the engine is**

**[24-Jun-2022-Shift-2]**

**Options:**

A.  $3 \times 10^6 \text{J}$

B. Zero

C.  $12.6 \times 10^6 \text{J}$

D.  $8.4 \times 10^6 \text{J}$

**Answer: C**

**Solution:**

$$\text{Efficiency } \eta = 1 - \frac{T_L}{T_H}$$

$$= 1 - \frac{400}{1000} = 0.6$$

$$\Rightarrow 0.6 = \frac{W}{Q}$$

$$\Rightarrow W = 0.6Q = 3000 \text{ kcal} = 12.6 \times 10^6 \text{J}$$

---

## Question127

A monoatomic gas performs a work of  $\frac{Q}{4}$  where Q is the heat supplied to it. The molar heat capacity of the gas will be \_\_\_\_\_ R during this transformation. Where R is the gas constant.

**[24-Jun-2022-Shift-2]**

**Answer: 2**

**Solution:**

By 1<sup>st</sup> law,

$$\Delta U = \Delta Q - \frac{\Delta Q}{4} = \frac{3}{4} \Delta Q$$



$$\Rightarrow nC_v \Delta T = \frac{3}{4}nC \Delta T$$

$$\Rightarrow C = \frac{4C_v}{3} = 2R$$


---

## Question128

The efficiency of a Carnot's engine, working between steam point and ice point, will be :

[26-Jun-2022-Shift-1]

Options:

A. 26.81%

B. 37.81%

C. 47.81%

D. 57.81%

**Answer: A**

**Solution:**

$$\begin{aligned}\eta &= 1 - \frac{T_2}{T_1} = 1 - \frac{273}{373} \\ &= 0.26809 \\ &\approx 26.81\%\end{aligned}$$


---

## Question129

A thermally insulated vessel contains an ideal gas of molecular mass  $M$  and ratio of specific heats 1.4. Vessel is moving with speed  $v$  and is suddenly brought to rest. Assuming no heat is lost to the surrounding and vessel temperature of the gas increases by :

( $R$  = universal gas constant)

[26-Jun-2022-Shift-1]

Options:

A.  $\frac{M v^2}{7R}$

B.  $\frac{M v^2}{5R}$

C.  $2 \frac{M v^2}{7R}$

D.  $7 \frac{M v^2}{5R}$

**Answer: B**

**Solution:**

Let there be  $n$  moles of gas

$$E_{\text{loss}} = E_{\text{gain}}$$

$$\frac{1}{2}(nM)v^2 = nC_v \Delta T$$

$$\frac{1}{2}M v^2 = C_v \Delta T$$

here,  $\gamma = 1.4 = \frac{7}{5}$  i.e. diatomic gas

$$\therefore C_v = \frac{5R}{2}$$

$$\text{Now, } \frac{1}{2}M v^2 = \frac{5R}{2} \Delta T$$

$$\Delta T = \frac{M v^2}{5R}$$

## Question130

**A heat engine operates with the cold reservoir at temperature 324K. The minimum temperature of the hot reservoir, if the heat engine takes 300J heat from the hot reservoir and delivers 180J heat to the cold reservoir per cycle, is \_\_\_\_\_K.**

**[26-Jun-2022-Shift-2]**

**Answer: 540**

**Solution:**

$$\left(1 - \frac{324}{T_H}\right) = \frac{300 - 180}{300}$$

$$1 - \frac{2}{5} = \frac{324}{T_H}$$

$$T_H = \frac{324 \times 5}{3} = 540$$


---

## Question131

In a carnot engine, the temperature of reservoir is  $527^\circ\text{C}$  and that of sink is  $200\text{K}$ . If the work done by the engine when it transfers heat from reservoir to sink is  $12000\text{ kJ}$ , the quantity of heat absorbed by the engine from reservoir is  $\text{_____} \times 10^6\text{J}$ .

[27-Jun-2022-Shift-1]

**Answer: 16**

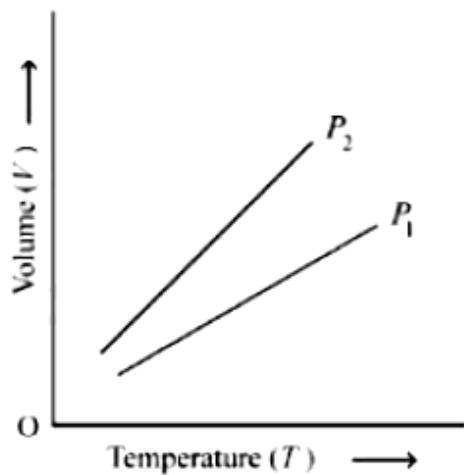
**Solution:**

$$\begin{aligned}\eta &= 1 - \frac{T_2}{T_1} \\ &= 1 - \frac{200}{800} = \frac{3}{4} \\ \therefore \eta &= \frac{W}{Q_1} \\ \Rightarrow \frac{3}{4} &= \frac{12000 \times 10^3}{Q_1} \\ \Rightarrow Q_1 &= 16 \times 10^6\text{J}\end{aligned}$$


---

## Question132

For a perfect gas, two pressures  $P_1$  and  $P_2$  are shown in figure. The graph shows:



**[27-Jun-2022-Shift-2]**

**Options:**

- A.  $P_1 > P_2$
- B.  $P_1 < P_2$
- C.  $P_1 = P_2$
- D. Insufficient data to draw any conclusion

**Answer: A**

**Solution:**

As per ideal gas equation,  $V = \frac{nR}{P}T$

$\Rightarrow$  Slope of V-T graph is inversely proportional to P.

As  $m_2 > m_1 \Rightarrow P_1 > P_2$

## Question133

**A diatomic gas ( $\gamma = 1.4$ ) does 400J of work when it is expanded isobarically. The heat given to the gas in the process is\_\_\_\_\_ J.**

**[27-Jun-2022-Shift-2]**

**Answer: 1400**

### Solution:

$$W = nR \Delta T = 400\text{J}$$

$$\therefore \Delta Q = nC_p \Delta T$$

$$= n \times \frac{7}{2} R \times \Delta T = \frac{7}{2} \times (400) = 1400$$

---

## Question134

The total internal energy of two mole monoatomic ideal gas at temperature  $T = 300\text{K}$  will be \_\_\_\_J. ( Given  $R = 8.31 \text{ J/mol} \cdot \text{K}$ )  
[28-Jun-2022-Shift-1]

**Answer: 7479**

### Solution:

$$U = 2 \left( \frac{3}{2} R \right) 300$$

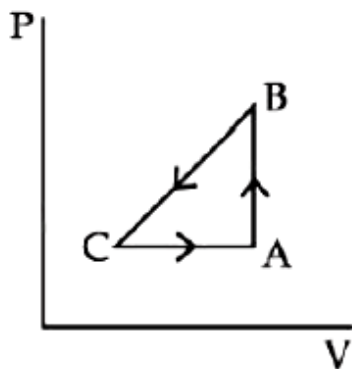
$$= 3 \times 8.31 \times 300$$

$$= 7479\text{J}$$

---

## Question135

A sample of an ideal gas is taken through the cyclic process ABCA as shown in figure. It absorbs, 40J of heat during the part AB, no heat during BC and rejects 60J of heat during CA. A work of 50J is done on the gas during the part BC. The internal energy of the gas at A is 1560J. The work done by the gas during the part CA is :



**[28-Jun-2022-Shift-2]**

**Options:**

- A. 20J
- B. 30J
- C. -30J
- D. -60J

**Answer: B**

**Solution:**

$\Delta U_{AB} = 40\text{J}$  as process is isochoric.

$\Delta U_{BC} = +50 (W_{BC} = -50\text{J})$

$U_C = U_A + \Delta U_{AB} + \Delta U_{BC} = 1650$

For CA process,

$Q_{CA} = -60\text{J}$

$\Delta U_{CA} + W_{CA} = -60$

$-90 + W_{CA} = -60$

$\Rightarrow W_{CA} = +30\text{J}$

The graph given is inconsistent with the statement BC may be adiabatic and CA cannot be like isobaric as shown, as increasing volume while rejecting heat at same time.

## Question136

**A cylinder of fixed capacity of 44.8 litres contains helium gas at standard temperature and pressure. The amount of heat needed to raise the temperature of gas in the cylinder by  $20.0^\circ\text{C}$  will be :**

**(Given gas constant  $R = 8.3\text{JK}^{-1} - \text{mol}^{-1}$  )**

**[29-Jun-2022-Shift-1]**

**Options:**

A. 249J

B. 415J

C. 498J

D. 830J

**Answer: C**

**Solution:**

$$\text{No of moles} = \frac{44.8}{22.4} = 2$$

Gas is mono atomic so  $C_v = \frac{3}{2}R$

$$\Delta Q = nC_v \Delta T$$

$$= 2 \times \frac{3}{2}R(20)$$

$$= 60R$$

$$= 60 \times 8.3$$

$$= 498J$$

---

## Question137

300 cal. of heat is given to a heat engine and it rejects 225 cal. If source temperature is  $227^\circ\text{C}$ , then the temperature of sink will be \_\_\_\_\_ $^\circ\text{C}$ .

[29-Jun-2022-Shift-1]

**Answer: 102**

**Solution:**

$$\eta = \frac{W}{Q} = \frac{300 - 225}{300}$$

$$\Rightarrow \frac{75}{300} = 1 - \frac{T_L}{T_H}$$

$$\Rightarrow T_L = \frac{3}{4}T_H = \frac{3}{4}(500) = 375\text{K}$$

$$\Rightarrow T_L = 102^\circ\text{C}$$

## Question 138

Starting with the same initial conditions, an ideal gas expands from volume  $V_1$  to  $V_2$  in three different ways. The work done by the gas is  $W_1$  if the process is purely isothermal,  $W_2$  if the process is purely adiabatic and  $W_3$  if the process is purely isobaric. Then, choose the correct option  
[29-Jun-2022-Shift-2]

Options:

A.  $W_1 < W_2 < W_3$

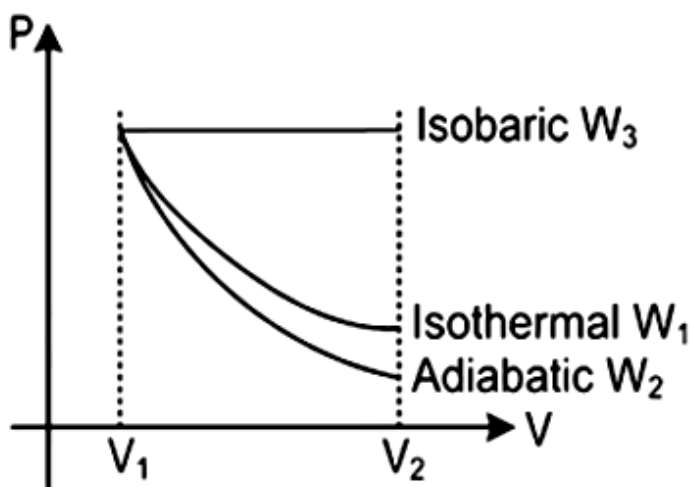
B.  $W_2 < W_3 < W_1$

C.  $W_3 < W_1 < W_2$

D.  $W_2 < W_1 < W_3$

**Answer: D**

**Solution:**



Comparing the area under the PV graph  $A_3 > A_1 > A_2$

$$\Rightarrow W_3 > W_1 > W_2$$



---

## Question139

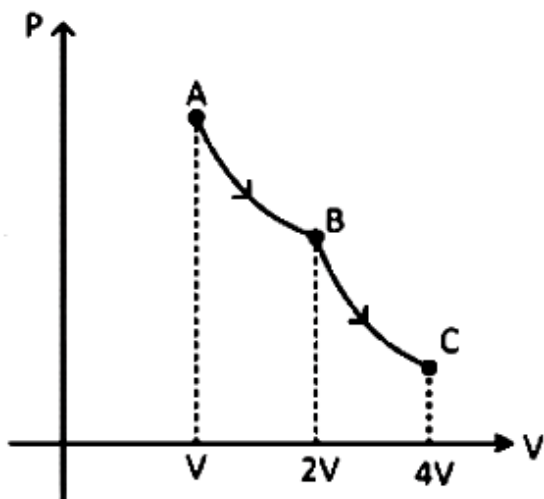
A certain amount of gas of volume  $V$  at  $27^\circ\text{C}$  temperature and pressure  $2 \times 10^7 \text{ N m}^{-2}$  expands isothermally until its volume gets doubled. Later it expands adiabatically until its volume gets redoubled. The final pressure of the gas will be (Use  $\gamma = 1.5$ ) :  
[25-Jul-2022-Shift-1]

Options:

- A.  $3.536 \times 10^5 \text{ Pa}$
- B.  $3.536 \times 10^6 \text{ Pa}$
- C.  $1.25 \times 10^6 \text{ Pa}$
- D.  $1.25 \times 10^5 \text{ Pa}$

Answer: B

Solution:



Let AB is isothermal process and BC is adiabatic process then for AB process

$$P_A V_A = P_B V_B$$
$$\Rightarrow P_B = 10^7 \text{ N m}^{-2}$$

For process BC

$$P_B V_B^\gamma = P_C V_C^\gamma$$

$$P_C = 3.536 \times 10^6 \text{ Pa}$$

---

## Question140

Let  $\eta_1$  is the efficiency of an engine at  $T_1 = 447^\circ\text{C}$  and  $T_2 = 147^\circ\text{C}$  while  $\eta_2$  is the efficiency at  $T_1 = 947^\circ\text{C}$  and  $T_2 = 47^\circ\text{C}$  The ratio  $\frac{\eta_1}{\eta_2}$  will be:

[25-Jul-2022-Shift-2]

Options:

A. 0.41

B. 0.56

C. 0.73

D. 0.70

**Answer: B**

**Solution:**

$$\eta_1 = 1 - \frac{420}{720} = \frac{300}{720}$$

$$\text{And } \eta_2 = 1 - \frac{320}{1220} = \frac{900}{1220}$$

$$\Rightarrow \frac{\eta_1}{\eta_2} = \frac{300}{720} \times \frac{1220}{900}$$

$$\approx 0.56$$

---

## Question141

7 mol of a certain monoatomic ideal gas undergoes a temperature increase of 40K at constant pressure. The increase in the internal energy of the gas in this process is :

(Given  $R = 8.3\text{JK}^{-1}\text{mol}^{-1}$  )

[26-Jul-2022-Shift-1]

Options:

A. 5810J

B. 3486J

C. 11620J

D. 6972J

**Answer: B**

**Solution:**

$$\begin{aligned}\Delta U &= nC_v \Delta T \\ &= 7 \times \frac{3R}{2} \times 40 \\ &= 3486J\end{aligned}$$

---

## Question142

**A monoatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be :  
[26-Jul-2022-Shift-1]**

**Options:**

A. P

B. 8P

C. 32P

D. 64P

**Answer: C**

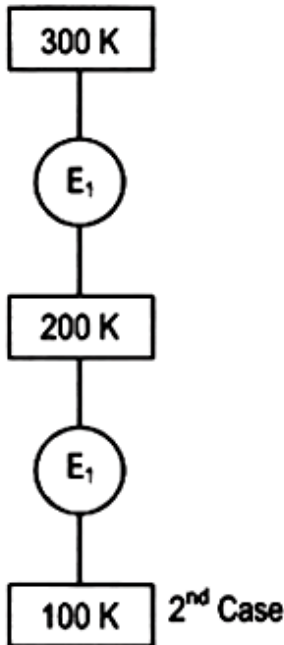
**Solution:**

$$\begin{aligned}PV^\gamma &= \text{constant} \\ \Rightarrow PV^\gamma &= (P') \left( \frac{V}{8} \right)^\gamma \text{ where } \gamma = 5/3 \\ \Rightarrow P' &= 32P\end{aligned}$$

---

## Question143

In 1<sup>st</sup> case, Carnot engine operates between temperatures 300K and 100K. In 2<sup>nd</sup> case, as shown in the figure, a combination of two engines is used. The efficiency of this combination (in 2<sup>nd</sup> case) will be :



[27-Jul-2022-Shift-2]

Options:

- A. same as the 1<sup>st</sup> case.
- B. always greater than the 1<sup>st</sup> case.
- C. always less than the 1<sup>st</sup> case.
- D. may increase or decrease with respect to the 1<sup>st</sup> case.

**Answer: A**

**Solution:**

$$\text{First case : } \eta = 1 - \frac{100}{300} = \frac{2}{3}$$

$$\text{Second case : } \eta_{\text{net}} = \eta_1 + \eta_2 - \eta_1\eta_2$$

$$\eta_1 = 1 - \frac{200}{300} = \frac{1}{3}$$

$$\eta_2 = 1 - \frac{100}{200} = \frac{1}{2}$$

$$\eta_{\text{net}} = \frac{1}{3} + \frac{1}{2} - \frac{1}{6} = \frac{2}{3}$$

$\eta(\text{ first case }) = \eta(\text{ second case })$

---

## Question144

**A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by 400°C, its efficiency increase by 30%. The temperature of the source will be:**

**[28-Jul-2022-Shift-1]**

**Options:**

A. 166.7K

B. 255.1K

C. 266.7K

D. 367.7K

**Answer: C**

**Solution:**

$$\eta = 1 - \frac{T_L}{T_H}$$

$$\frac{1}{2} = 1 - \frac{T_L}{T_H}$$

$$\frac{1}{2}(1 \cdot 3) = 1 - \left( \frac{T_L - 40}{T_H} \right)$$

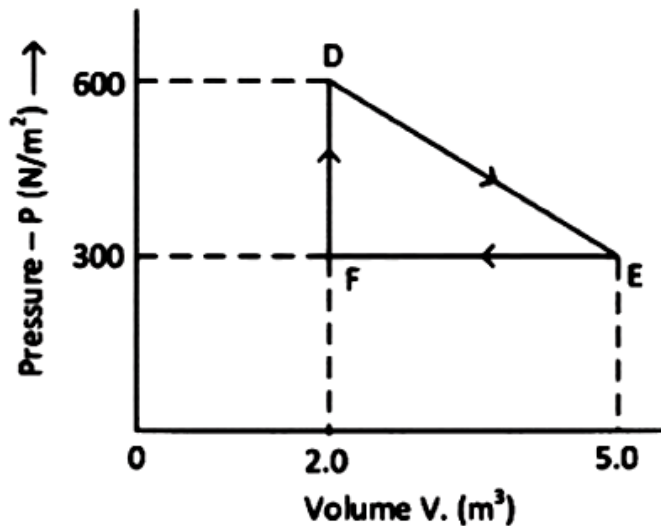
$$\frac{1}{2}(1 \cdot 3) = \frac{1}{2} + \frac{40}{T_H} \quad T_H = 266.7\text{K}$$


---

## Question145

**A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in the figure. Its volume is then reduced to the original volume from E to F by an isobaric process. The total work done by the gas from D to E to F will**

be



[29-Jul-2022-Shift-2]

Options:

- A.  $-450\text{J}$
- B.  $450\text{J}$
- C.  $900\text{J}$
- D.  $1350\text{J}$

**Answer: B**

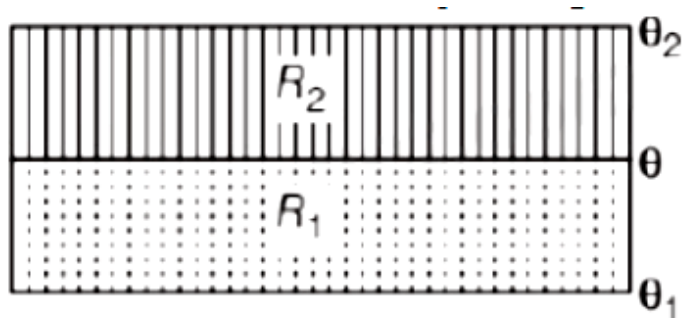
**Solution:**

$$\begin{aligned}W_{DE} &= \frac{1}{2}(600 + 300)3\text{J} \\&= 1350\text{J} \\W_{EF} &= -300 \times 3 = -900\text{J} \\W_{DEF} &= 450\text{J}\end{aligned}$$

---

## Question146

The temperature  $\theta$  at the junction of two insulating sheets, having thermal resistances  $R_1$  and  $R_2$  as well as top and bottom temperatures  $\theta_1$  and  $\theta_2$  (as shown in figure) is given by



[26 Feb 2021 Shift 1]

**Options:**

A.  $\frac{\theta_2 R_2 - \theta_1 R_1}{R_2 - R_1}$

B.  $\frac{\theta_1 R_2 - \theta_2 R_1}{R_2 - R_1}$

C.  $\frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$

D.  $\frac{\theta_1 R_1 + \theta_2 R_2}{R_1 + R_2}$

**Answer: C**

**Solution:**

Let,  $Q$  = heat current,

$k$  = thermal conductivity,

$A$  = area,

$l$  = length of capacitor

and  $\Delta\theta$  = change in temperature .

$$\therefore Q = \frac{kA \Delta\theta}{l} = \frac{\Delta\theta}{R} \quad \left( \because R = \frac{l}{kA} \right)$$

$$\Rightarrow \frac{\theta_2 - \theta}{R_2} = \frac{\theta - \theta_1}{R_1}$$

$$\Rightarrow R_1 \theta_2 - R_1 \theta = R_2 \theta - R_2 \theta_1$$

$$\Rightarrow \theta = \frac{R_1 \theta_2 + R_2 \theta_1}{R_1 + R_2}$$

## Question147

**A reversible heat engine converts one-fourth of the heat input into work. When the temperature of the sink is reduced by 52K , its**

efficiency is doubled. The temperature in kelvin of the source will be .....

[25 Feb 2021 Shift 2]

**Answer: 104**

**Solution:**

Given, initial efficiency ( $\eta_h$ ) =  $1/4$

$$\Rightarrow \eta_h = 1 - \frac{T_2}{T_1} = \frac{1}{4} \Rightarrow \frac{T_2}{T_1} = 1 - \frac{1}{4} = \frac{3}{4}$$

When temperature of sink is reduced by 50K ,

$$\eta_2 = 2\eta_1 = 1 - \frac{T_2 - 52}{T_1}$$

$$\Rightarrow 2 \times \frac{1}{4} = 1 - \frac{T_2}{T_1} + \frac{52}{T_1}$$

$$\Rightarrow \frac{1}{2} = 1 - \frac{3}{4} + \frac{52}{T_1} \Rightarrow \frac{1}{2} - \frac{1}{4} = \frac{52}{T_1}$$

$$\Rightarrow \frac{2}{4} = \frac{52}{T_1}$$

$$\Rightarrow T_1 = 104\text{K}$$

---

## Question148

1 mole of rigid diatomic gas performs a work of  $Q/5$  when heat  $Q$  is supplied to it. The molar heat capacity of the gas during this transformation is  $\frac{xR}{8}$ , The value of  $x$  is .....

[  $R$  = universal gas constant ]

[26 Feb 2021 Shift 2]

**Answer: 25**

**Solution:**



Given, number of mole,  $n = 1$

Heat supplied =  $Q$

Work done =  $Q/5$

By using first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

$$Q = \Delta U + Q/5$$

$$\Delta U = 4Q/5 = nC_v \Delta T \dots (i)$$

where,  $C_v$  is heat coefficient at constant volume and

$\Delta T$  is change in temperature.

Degree of freedom of diatomic gas,  $f = 5$

$$\therefore Q = \frac{5}{4} \frac{f}{2} R \Delta T \text{ [from Eq. (i)]} \Rightarrow \frac{Q}{\Delta T} = C = \frac{5}{4} \times \frac{5}{2} R = \frac{25}{8} R$$

Hence,  $x = 25$

## Question149

**Match List-I with List-II:**

| List-I         | List-II                       |
|----------------|-------------------------------|
| (A) Isothermal | (i) Pressure constant         |
| (B) Isochoric  | (ii) Temperature constant     |
| (C) Adiabatic  | (iii) Volume constant         |
| (D) Isobaric   | (iv) Heat content is constant |

**Choose the correct answer from the options given below:**  
**[24feb2021shift1]**

**Options:**

- A. (A)  $\rightarrow$  (i), (B)  $\rightarrow$  (iii), (C)  $\rightarrow$  (ii), (D)  $\rightarrow$  (iv)
- B. (A)  $\rightarrow$  (ii), (B)  $\rightarrow$  (iii), (C)  $\rightarrow$  (iv), (D)  $\rightarrow$  (i)
- C. (A)  $\rightarrow$  (ii), (B)  $\rightarrow$  (iv), (C)  $\rightarrow$  (iii), (D)  $\rightarrow$  (i)
- D. (A)  $\rightarrow$  (iii), (B)  $\rightarrow$  (ii), (D)  $\rightarrow$  (i), (D)  $\rightarrow$  (iv)

**Answer: B**

**Solution:**

- (A) Isothermal ( $\Delta T = 0$ ) (A)  $\rightarrow$  (ii)  
(B) Isochoric ( $\Delta V = 0$ ) (B)  $\rightarrow$  (iii)  
(C) Adiabatic ( $\Delta Q = 0$ ) (C)  $\rightarrow$  (iv)  
(D) Isobaric ( $\Delta P = 0$ ) (D)  $\rightarrow$  (i)
- 

## Question150

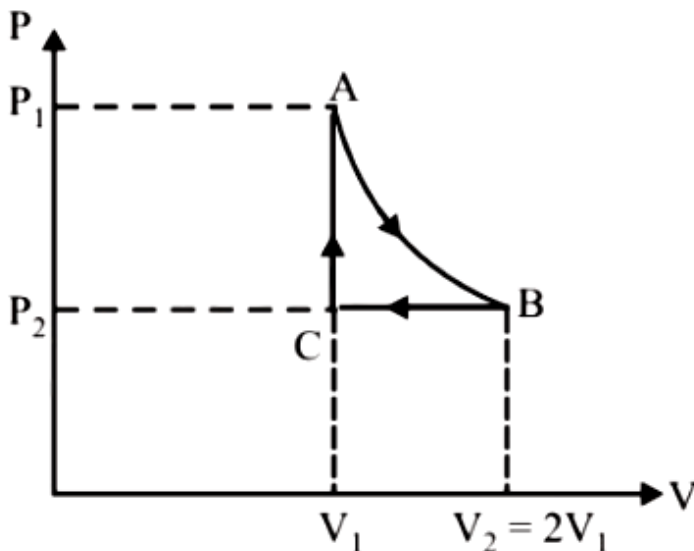
n mole a perfect gas undergoes a cyclic process ABCA (see (figure) consisting of the following processes.

**A  $\rightarrow$  B :** Isothermal expansion at temperature T so that the volume is doubled from  $V_1$  to  $V_2 = 2V_1$  and pressure changes from  $P_1$  to  $P_2$ .

**B  $\rightarrow$  C :** Isobaric compression at pressure  $P_2$  to initial volume  $V_1$ .

**C  $\rightarrow$  A :** Isochoric change leading to change of pressure from  $P_2$  to  $P_1$ .

Total work done in the complete cycle ABCA is:



[24feb2021shift1]

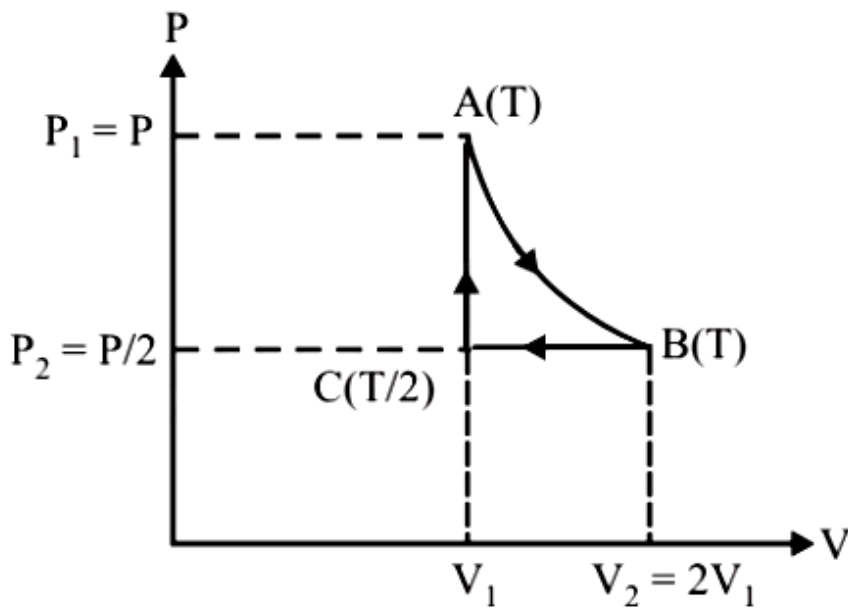
Options:

- A. 0
- B.  $nRT \left( \ln 2 + \frac{1}{2} \right)$
- C.  $nRT \ln 2$
- D.  $nRT \left( \ln 2 - \frac{1}{2} \right)$

**Answer: D**

## Solution:

Work done during isothermal process,



$$W_{AB} = nRT \ln \left( \frac{V_2}{V_1} \right)$$

$$W_{AB} = nRT \ln \left( \frac{2V}{V} \right) = nRT \ln 2$$

$$W_{BC} = nR(T_C - T_B) = nR \left( \frac{T}{2} - T \right) = -nR \frac{T}{2}$$

$$W_{CA} = P \Delta V = 0 \quad (\because \Delta V = 0)$$

$$\Rightarrow W_{\text{net}} = W_{AB} + W_{BC} + W_{CA}$$

$$\Rightarrow W_{\text{net}} = nRT \ln 2 - \frac{nRT}{2}$$

$$\Rightarrow W_{\text{net}} = nRT \left( \ln 2 - \frac{1}{2} \right)$$

---

## Question151

The volume  $V$  of a given mass of monoatomic gas changes with temperature  $T$  according to the relation  $V = kT^{2/3}$ . The work done when temperature changes by  $90\text{K}$  will be  $xR$ . The value of  $x$  is.....

[  $R$  = universal gas constant ]

[26 Feb 2021 Shift 2]

**Answer: 60**

## Solution:

Given,  $V = kT^{2/3}$  ... (i)

where,  $T$  is temperature.

Change in temperature,  $\Delta T = 90\text{K}$

Let  $p$  be the pressure,  $dV$  be the change in volume and work done be  $W$ .

As we know that,

$$W = \int p dV \quad \dots \text{ (ii)}$$

As,  $pV = nRT$

$$\therefore p = \frac{nRT}{V}$$

Substituting this value in Eq. (ii), we get

$$W = \int nRT \frac{dV}{V}$$

$$W = \int nRT \frac{dV}{kT^{2/3}} \quad [\text{using Eq. (i)}] \dots \text{ (iii)}$$

On differentiating Eq. (i) w.r.t. temperature on both sides, we get

$$\begin{aligned} \frac{dV}{dT} &= k \cdot \frac{2}{3} T^{-1/3} \\ &= \frac{2}{3} k T^{-1/3} = \frac{2}{3} \frac{k}{T^{1/3}} \end{aligned}$$

$$\therefore dV = \frac{2}{3} k T^{-1/3} dT$$

Substituting this in Eq. (iii), we get

$$W = \int nRT \frac{\frac{2}{3} k T^{-1/3} dT}{k T^{2/3}}$$

$$\begin{aligned} \Rightarrow W &= \frac{2}{3} nR \int T dT = \frac{2}{3} nR [T]_{T_1}^{T_2} \\ &= \frac{2}{3} nR [T_2 - T_1] = \frac{2}{3} nR \Delta T \\ &= \frac{2}{3} nR 90 = 60nR \end{aligned}$$

Here,  $n = 1$

So, work done will be  $60R$ .

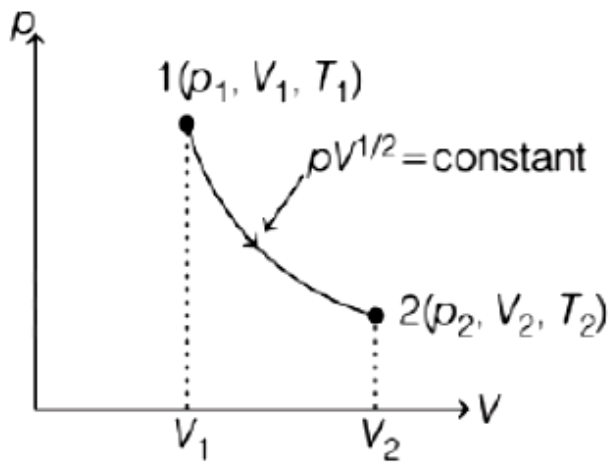
Hence,  $x = 60$ .

---

## Question152

**Thermodynamic process is shown below on a p-V diagram for one mole of an ideal gas.**

**If  $V_2 = 2V_1$ , then the ratio of temperature  $\frac{T_2}{T_1}$  is**



[25 Feb 2021 Shift 2]

Options:

A.  $\frac{1}{\sqrt{2}}$

B.  $\sqrt{2}$

C.  $\frac{1}{2}$

D. 2

**Answer: B**

**Solution:**

Here,  $p_1$  and  $p_2$ ,  $T_1$  and  $T_2$ ,  $V_1$  and  $V_2$  are initial and final pressures, temperatures and volumes, respectively.

Given,  $V_2 = 2V_1$

$pV^{1/2} = \text{constant}$

From graph,  $\gamma = \text{adiabatic constant} = 1/2$

$$\Rightarrow p_1 V_1^{1/2} = p_2 V_2^{1/2} \Rightarrow \frac{p_1}{p_2} = \left( \frac{2V_1}{V_1} \right)^{1/2} = 2^{1/2}$$

Also,  $p^{1-\gamma} T^\gamma = \text{constant}$  (for adiabatic process)

$$\Rightarrow p_1^{1-\gamma} T_1^\gamma = p_2^{1-\gamma} T_2^\gamma$$

$$\Rightarrow \left( \frac{p_1}{p_2} \right)^{1-\gamma} = \left( \frac{T_2}{T_1} \right)^\gamma \Rightarrow \left( \frac{p_1}{p_2} \right)^{\frac{1-\gamma}{\gamma}} = \frac{T_2}{T_1}$$

$$\therefore T_2/T_1 = (2^{1/2})^{\frac{1-0.5}{0.5}} = \sqrt{2}$$

**Question 153**

**In a certain thermodynamical process, the pressure of a gas depends on its volume as  $kV^3$ . The work done when the temperature changes from  $100^\circ\text{C}$  to  $300^\circ\text{C}$  will be .....  $nR$ , where  $n$  denotes number of moles of a gas.**

**[25 Feb 2021 Shift 1]**

**Answer: 50**

**Solution:**

Given, pressure  $(p) \propto kV^3$

$T_1 = 100^\circ\text{C}$ ,  $T_2 = 300^\circ\text{C}$

$\Delta T = T_2 - T_1 = 300 - 100 = 200^\circ\text{C}$

$\Rightarrow$  By using ideal gas equation,

$pV = nRT$

$kV^3 \cdot V = nRT \Rightarrow kV^4 = nRT$

On differentiating both sides w.r.t temperature, we get

$$4kV^3 \frac{dV}{dT} = nR$$

$$\Rightarrow 4kV^3 dV = nR dT \Rightarrow kV^3 dV = nR dT / 4$$

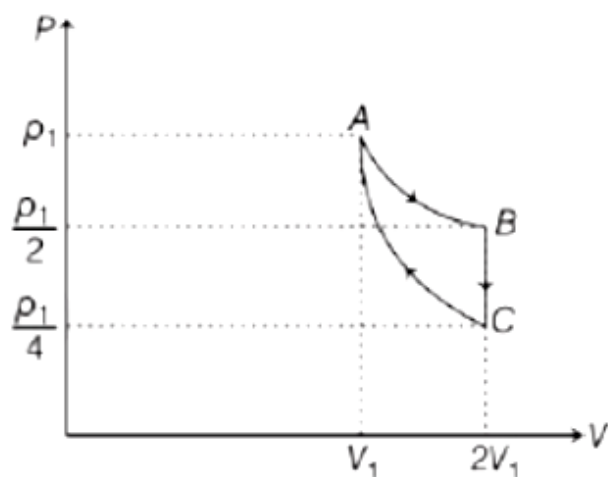
$$\Rightarrow p dV = nR dT / 4$$

As, work done  $(W) = p dV = nR dT / 4$

$$= \frac{nR}{4} \Delta T = \frac{nR}{4} \times 200 = 50nR$$

## Question154

**If one mole of an ideal gas at  $(p_1, V_1)$  is allowed to expand reversibly and isothermally ( A to B ), its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value ( B  $\rightarrow$  C ). Then, it is restored to its initial state by a reversible adiabatic compression ( C to A ). The net work done by the gas is equal to**



**[24 Feb 2021 Shift 2]**

**Options:**

A.  $RT \left( \ln 2 - \frac{1}{2(\gamma-1)} \right)$

B.  $-\frac{RT}{2(\gamma-1)}$

C. 0

D.  $RT \ln 2$

**Answer: A**

**Solution:**

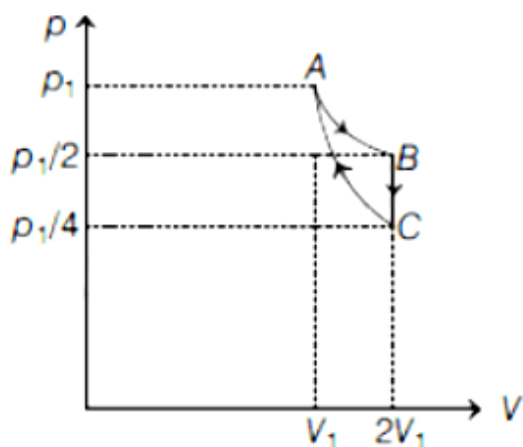
Let  $p_i$ ,  $p_f$ ,  $V_i$  and  $V_f$  be the initial and final pressure and volume.

Given, AB is isothermal ( $\Delta T = 0$ ),

BC is isochoric ( $\Delta V = 0$ )

and CA is adiabatic ( $\Delta Q = 0$ )

Since, isothermal work ( $W_{AB}$ ) =  $p_1 V_1 \ln \frac{V_f}{V_i}$



where,  $V_i$  and  $V_f$  are volume at A and B, respectively.

$$\therefore W_{AB} = p_1 V_1 \ln \frac{2V_1}{V_1} = p_1 V_1 \ln 2$$

Since, at constant volume, work done is zero.

$$\therefore W_{BC} = 0$$

Since,  $W_{CA}$  is an adiabatic work done,

$$\text{i.e. } W_{CA} = \frac{1}{1-\gamma} (p_f V_f - p_i V_i)$$

$$\Rightarrow W_{CA} = \frac{1}{1-\gamma} \left( p_1 V_1 - \frac{p_1}{4} \times 2V_1 \right)$$

$$= \frac{1}{1-\gamma} (p_1 V_1 - p_1 V_1 / 2) = \frac{1}{1-\gamma} \frac{p_1 V_1}{2}$$

$$\therefore \text{Net work done, } W_{\text{net}} = W_{AB} + W_{BC} + W_{CA}$$

$$= p_1 V_1 \ln 2 + 0 + \frac{1}{1-\gamma} \frac{p_1 V_1}{2}$$

$$= p_1 V_1 [\ln 2 + 1/2(1-\gamma)]$$

From ideal gas law,  $pV = nRT$

$$\therefore W_{\text{net}} = RT [\ln 2 + 1/2(\gamma - 1)]$$

$$(\because n = 1)$$

---

## Question 155

**In thermodynamics, heat and work are  
[16 Mar 2021 Shift 1]**

**Options:**

- A. path functions
- B. intensive thermodynamic state variables
- C. extensive thermodynamic state variables
- D. point functions

**Answer: A**

**Solution:**

According to first law of thermodynamics

$$dQ = dU + dW \dots (i)$$

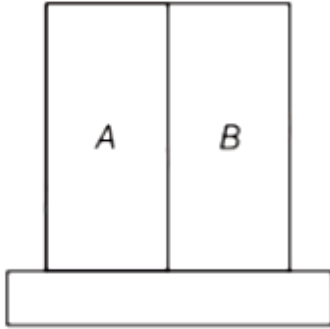
As we know that, work done by the gas depends on the type of process, i.e. path and  $dU$  depends on the initial and final states. So, considering Eq. (i),  $dQ$  will also be dependent on path. It means that, in thermodynamics, heat and work are path functions.

---



## Question156

A bimetallic strip consists of metals A and B. It is mounted rigidly as shown. The metal A has higher coefficient of expansion compared to that of metal B. When the bimetallic strip is placed in a cold bath, it will



[16 Mar 2021 Shift 2]

Options:

- A. bend towards the right
- B. not bend but shrink
- C. Neither bend nor shrink
- D. bend towards the left

**Answer: D**

**Solution:**

As per the question, coefficient of expansion of metal A is higher than that of B.

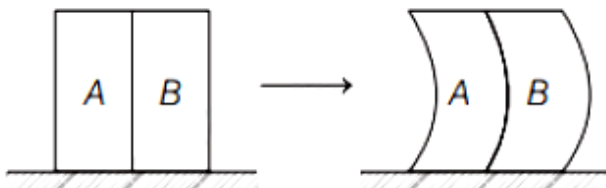
It means,  $\alpha_A > \alpha_B$

If the bimetallic strip is placed in a cold bath, then the length of both strips will decrease.

i.e.,  $L_A > L_B$

$\therefore$  The bimetallic strip will bend towards left.

It can be shown as below.



---

## Question157

**Two identical metal wires of thermal conductivities  $K_1$  and  $K_2$  respectively are connected in series. The effective thermal conductivity of the combination is**  
**[17 Mar 2021 Shift 1]**

**Options:**

A.  $\frac{2K_1K_2}{K_1 + K_2}$

B.  $\frac{K_1 + K_2}{2K_1K_2}$

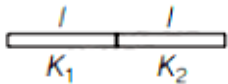
C.  $\frac{K_1 + K_2}{K_1K_2}$

D.  $\frac{K_1K_2}{K_1 + K_2}$

**Answer: A**

**Solution:**

Two identical metal wires of thermal conductivities  $K_1$  and  $K_2$  respectively connected in series are represented as follows



The above figure can also be represented as



Therefore, the effective thermal conductivity of the combination will be given by

$$R_{\text{eff}} = \frac{l}{K_1 A} + \frac{l}{K_2 A} = \frac{2l}{K_{\text{eq}} A}$$

$$\Rightarrow \frac{2l}{K_{\text{eq}} A} = \frac{l}{A} \left( \frac{1}{K_1} + \frac{1}{K_2} \right) \Rightarrow \frac{2l}{K_{\text{eq}} A} = \frac{l}{A} \left( \frac{K_1 + K_2}{K_1 K_2} \right)$$

$$\Rightarrow \frac{2}{K_{\text{eq}}} = \frac{K_1 + K_2}{K_1 K_2} \Rightarrow K_{\text{eq}} = \frac{2K_1 K_2}{K_1 + K_2}$$

## Question158

**An ideal gas in a cylinder is separated by a piston in such a way that the entropy of one part is  $S_1$  and that of the other part is  $S_2$ . Given**

that  $S_1 > S_2$ . If the piston is removed, then the total entropy of the system will be  
[18 Mar 2021 Shift 2]

Options:

A.  $S_1 \times S_2$

B.  $S_1 - S_2$

C.  $\frac{S_1}{S_2}$

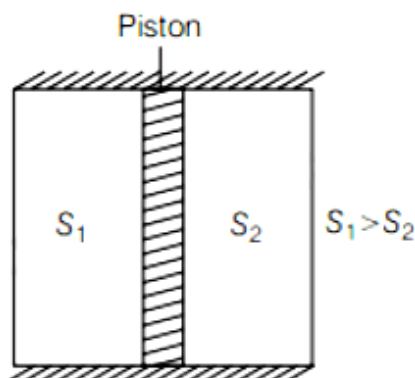
D.  $S_1 + S_2$

Answer: D

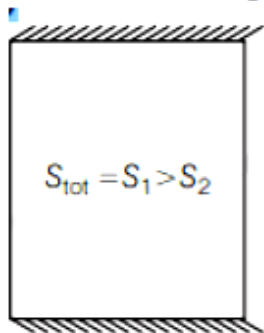
Solution:

Entropy present in one part of the system is  $S_1$ .

Entropy present in other part of the system is  $S_2$ .



After removing piston,



After removing piston,

We know that,

$$\Delta S_{\text{unvi}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

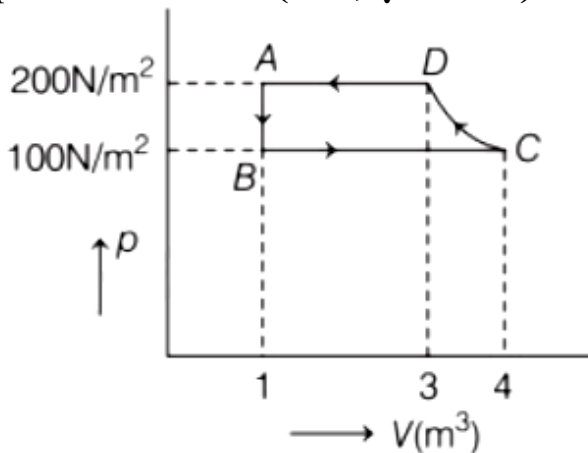
Thus, after removing the piston, the total entropy present in system,

$$S_{\text{tot}} = S_1 + S_2$$

---

## Question159

The  $p - V$  diagram of a diatomic ideal gas system going under cyclic process as shown in figure. The work done during an adiabatic process CD is (use,  $\gamma = 1.4$ )



[18 Mar 2021 Shift 1]

Options:

- A.  $-500\text{J}$
- B.  $-400\text{J}$
- C.  $400\text{J}$
- D.  $200\text{J}$

**Answer: A**

**Solution:**

Given

The final pressure of the adiabatic process,  $p_f = 200\text{N} / \text{m}^2$

The initial pressure of the adiabatic process,  $p_i = 100\text{N} / \text{m}^2$

The initial volume of the adiabatic process,  $V_i = 4\text{m}^3$

The final volume of the adiabatic process,  $V_f = 3\text{m}^3$

We know that, for an adiabatic process

$$W = \frac{p_f V_f - p_i V_i}{1 - \gamma}$$
$$\Rightarrow W = \frac{200(3) - (100)(4)}{1 - 1.4}$$

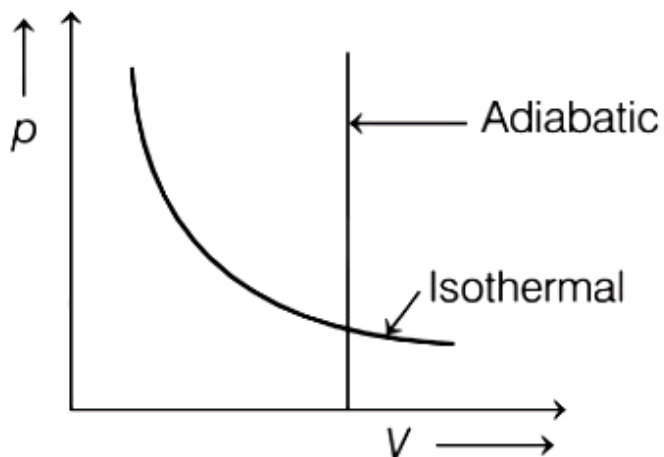
$$\Rightarrow W = -500\text{J}$$

So, the work done by an adiabatic process is  $500\text{J}$  and negative sign represents the work done on the system.

---

## Question160

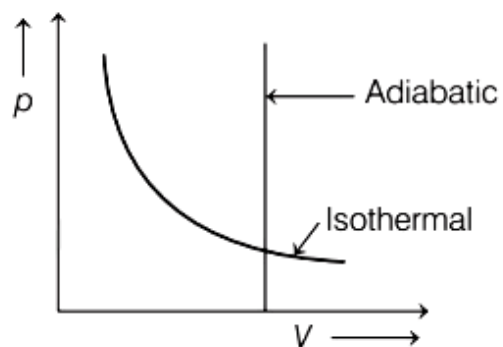
Which one is the correct option for the two different thermodynamic processes?



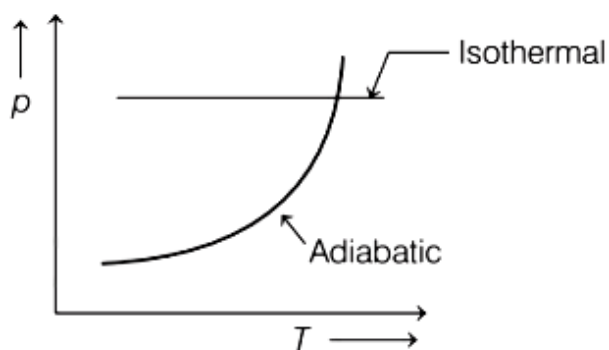
[17 Mar 2021 Shift 2]

Options:

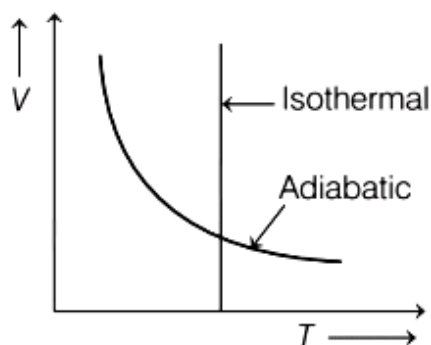
A.



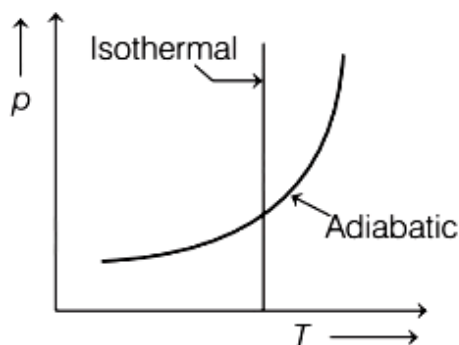
B.



C.



D.



**Answer: B**

**Solution:**

In graph (A) vertical line is isochoric process, so the graph (A) is incorrect.

In graph (B) horizontal line is isobaric process, so the graph (B) is incorrect.

In graph (C) and (D) vertical line is isothermal process and the curve line is adiabatic process.

So, the correct representation of thermodynamic process graph is (C) and (D).

So, the correct option is (b)

## Question161

**A Carnot's engine working between 400K and 800K has a work output of 1200J per cycle. The amount of heat energy supplied to the engine from the source in each cycle is**  
**[17 Mar 2021 Shift 1]**

**Options:**

A. 3200J

B. 1800J

C. 1600J

D. 2400J

**Answer: D**

**Solution:**

Efficiency of Carnot heat engine is given by

$$\eta = 1 - \frac{T_2}{T_1}$$
$$= 1 - \frac{Q_2}{Q_1} = \frac{W}{Q_1} \dots (i)$$

where.  $W$  = net work done by the gas

$Q_1$  = heat absorbed by the gas,

$Q_2$  = heat released by the gas,

$T_1$  = temperature of hot reservoir,

and  $T_2$  = temperature of cold reservoir.

Using Eq. (i), we can write

$$1 - \frac{T_2}{T_1} = \frac{W}{Q_1} \Rightarrow \frac{T_2}{T_1} = 1 - \frac{W}{Q_1}$$
$$\Rightarrow \frac{400}{800} = 1 - \frac{W}{Q_1} \Rightarrow \frac{W}{Q_1} = 1 - \frac{400}{800}$$
$$\Rightarrow \frac{W}{Q_1} = 1 - \frac{1}{2} = \frac{1}{2} \Rightarrow \frac{W}{Q_1} = \frac{1}{2}$$
$$\Rightarrow Q_1 = 2 \times W = 2 \times 1200 [\because W_1 = 1200J]$$
$$= 2400J$$

---

## Question162

**For an ideal heat engine, the temperature of the source is  $127^\circ\text{C}$ . In order to have 60% efficiency the temperature of the sink should be .....  $^\circ\text{C}$ .**

**(Round off to the nearest integer)**

**[16 Mar 2021 Shift 2]**

**Answer: 113**

**Solution:**

Given, temperature of source,  $T_H = 127^\circ\text{C}$

$$= 273 + 127 = 400\text{K}$$

Efficiency,  $\eta = 60\% = 0.6$

The efficiency of Carnot (ideal) heat engine is given by

$$\eta = \left( 1 - \frac{T_L}{T_H} \right)$$

where,  $T_L$  = temperature of sink,

$$0.6 = \left( 1 - \frac{T_L}{T_H} \right) \Rightarrow \frac{T_L}{T_H} = 1 - 0.6$$

$$\frac{T_L}{T_H} = 0.4 \Rightarrow T_L = 0.4 \times T_H$$

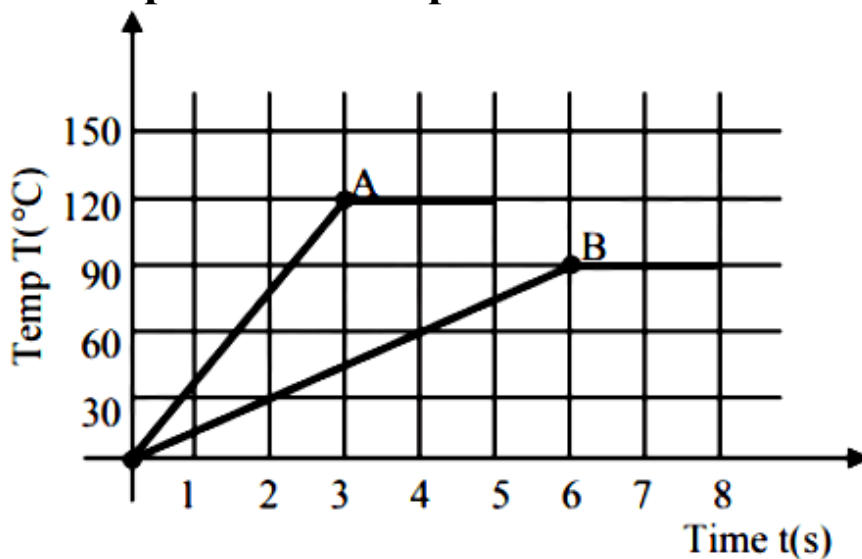
$$= 0.4 \times 400 = 160\text{K}$$

$$= 160 - 273 = -113^\circ\text{C}$$

---

## Question163

Two different metal bodies A and B of equal mass are heated at a uniform rate under similar conditions. The variation of temperature of the bodies is graphically represented as shown in the figure. The ratio of specific heat capacities is :



[25 Jul 2021 Shift 1]

Options:

A.  $\frac{8}{3}$

B.  $\frac{3}{8}$

C.  $\frac{3}{4}$



D.  $\frac{4}{3}$

**Answer: B**

**Solution:**

$$\left(\frac{\Delta Q}{\Delta t}\right)_A = \left(\frac{\Delta Q}{\Delta t}\right)_B$$

$$mS_A \left(\frac{\Delta T}{\Delta t}\right)_A = mS_B \left(\frac{\Delta T}{\Delta t}\right)_B$$

$$\frac{S_A}{S_B} = \frac{\left(\frac{\Delta T}{\Delta t}\right)_A}{\left(\frac{\Delta T}{\Delta t}\right)_B} = \frac{90/6}{120/3} = \frac{15}{40} = \frac{3}{8}$$

---

## Question164

**A body takes 4 min. to cool from 61° C to 59°C. If the temperature of the surroundings is 30°C, the time taken by the body to cool from 51°C to 49° C is :**  
**[27 Jul 2021 Shift 1]**

**Options:**

A. 4 min.

B. 3 min.

C. 8 min.

D. 6 min.

**Answer: D**

**Solution:**

$$\frac{\Delta T}{\Delta t} = K (T_t - T_s)$$

$T_t$  = average temp

$T_s$  = surrounding temp.

$$\frac{61 - 59}{4} = K \left( \frac{61 + 59}{2} - 30 \right) \dots\dots\dots(1)$$

$$\frac{51 - 49}{t} = K \left( \frac{51 + 49}{2} - 30 \right) \dots\dots\dots(2)$$

Divide (1) & (2)

$$\frac{t}{4} = \frac{60 - 30}{50 - 30} = \frac{30}{20}$$

so,  $t = 6$  minutes

## Question165

**In 5 minutes, a body cools from 75°C to 65°C at room temperature of 25°C. The temperature of body at the end of next 5 minutes is**

**\_\_\_\_\_°C**

**[22 Jul 2021 Shift 2]**

**Answer: 57**

**Solution:**

By newton's law of cooling (with approximation)

$$\frac{\Delta T}{\Delta t} = -C(T_{\text{arg}} - T_s)$$

$$1^{\text{st}} \frac{-10^\circ\text{C}}{5\text{min}} = -C(70^\circ\text{C} - 25^\circ\text{C})$$

$$\Rightarrow C = \frac{2}{45} \text{min}^{-1}$$

$$2^{\text{nd}} \frac{T - 65}{5\text{min}} = -C \left( \frac{T + 65}{2} - 25 \right) = - \left( \frac{2}{45} \right) \left( \frac{T + 15}{2} \right) \Rightarrow 9(T - 65) = -(T + 15)$$

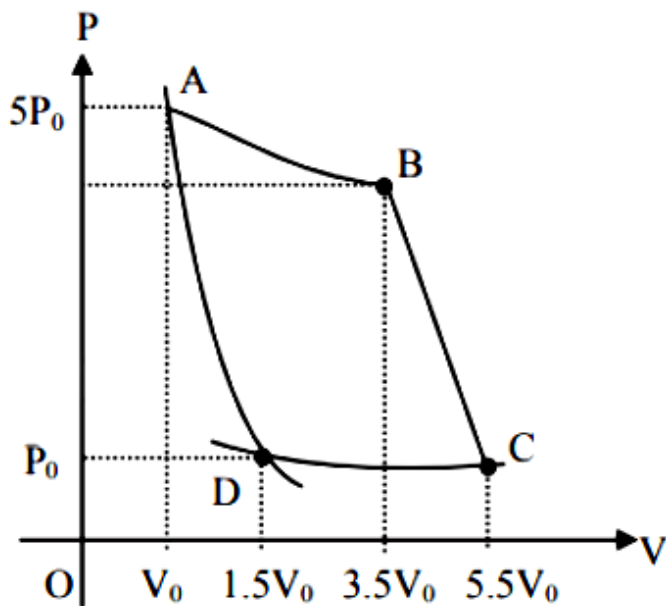
$$\Rightarrow 10T = 570$$

$$\Rightarrow T = 57^\circ\text{C}$$

## Question166

**In the reported figure, there is a cyclic process ABCDA on a sample of 1 mol of a diatomic gas. The temperature of the gas during the**

process  $A \rightarrow B$  and  $C \rightarrow D$  are  $T_1$  and  $T_2$  ( $T_1 > T_2$ ) respectively.



Choose the correct option out of the following for work done if processes BC and DA are adiabatic.

[27 Jul 2021 Shift 1]

Options:

- A.  $W_{AB} = W_{DC}$
- B.  $W_{AD} = W_{BC}$
- C.  $W_{BC} + W_{DA} > 0$
- D.  $W_{AB} < W_{CD}$

**Answer: B**

**Solution:**

$$\text{Work done in adiabatic process} = \frac{-nR}{\gamma-1}(T_f - T_i)$$

$$\therefore W_{AD} = \frac{-nR}{\gamma-1}(T_2 - T_1)$$

$$\text{and } W_{BC} = \frac{-nR}{\gamma-1}(T_2 - T_1)$$

$$\therefore W_{AD} = W_{BC}$$

**Question167**

A monoatomic ideal gas, initially at temperature  $T_1$  is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature  $T_2$  by releasing the piston suddenly. If  $l_1$  and  $l_2$  are the lengths of the gas column, before and after the expansion respectively, then the value of  $\frac{T_1}{T_2}$  will be :

[25 Jul 2021 Shift 1]

Options:

A.  $\left(\frac{l_1}{l_2}\right)^{\frac{2}{3}}$

B.  $\left(\frac{l_2}{l_1}\right)^{\frac{2}{3}}$

C.  $\frac{l_2}{l_1}$

D.  $\frac{l_1}{l_2}$

**Answer: B**

**Solution:**

$$PV^\gamma = \text{const}$$

$$TV^{\gamma-1} = \text{const.}$$

$$T(l)^{\frac{5}{3}-1} = \text{const}$$

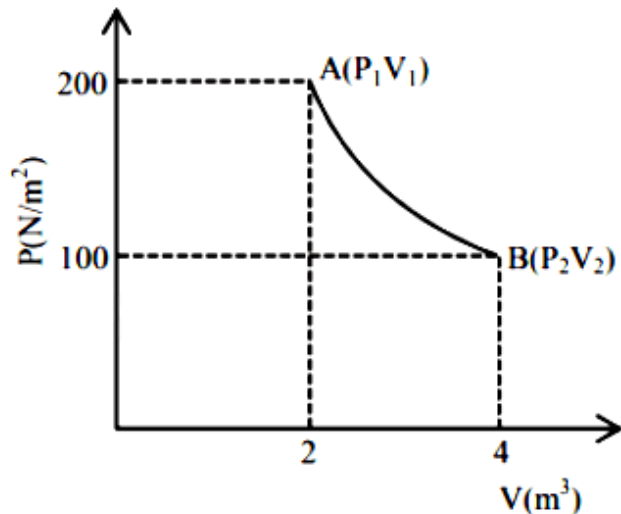
$$\frac{T_1}{T_2} = \left(\frac{l_2}{l_1}\right)^{2/3}$$

## Question168

One mole of an ideal gas at  $27^\circ\text{C}$  is taken from A to B as shown in the given PV indicator diagram.

The work done by the system will be \_\_\_\_\_  $\times 10^{-1} \text{J}$ .  
 [Given:  $R = 8.3 \text{J / mol eK}$ ,  $\ln 2 = 0.6931$ ]

(Round off to the nearest integer)



[20 Jul 2021 Shift 2]

**Answer: 17258**

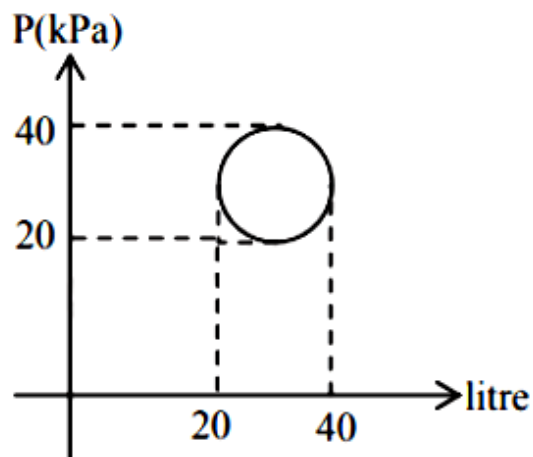
**Solution:**

Process of isothermal

$$\begin{aligned}
 W &= nRT \ln \left( \frac{V_2}{V_1} \right) \\
 &= 1 \times 8.3 \times 300 \times \ln 2 \\
 &= 17258 \times 10^{-1} \text{J}
 \end{aligned}$$

## Question169

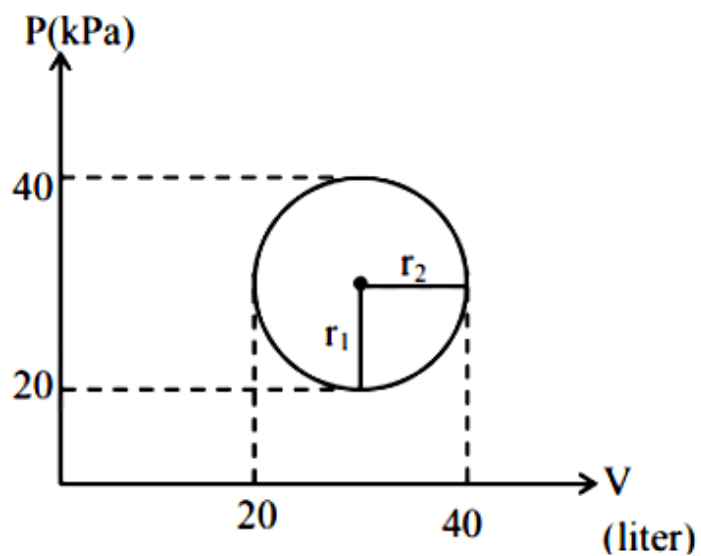
In the reported figure, heat energy absorbed by a system in going through a cyclic process is \_\_\_\_\_  $\pi \text{J}$ .



[20 Jul 2021 Shift 1]

Answer: 100

Solution:



For complete cyclic process

$$\Delta U = 0$$

$$\therefore \text{from } \Delta Q = \Delta U + W$$

$$= 0 + W$$

$$\Delta Q = W$$

$$= \text{Area}$$

$$= \pi r_1 \cdot r_2$$

$$= \pi \times (10 \times 10^3) \times (10 \times 10^{-3})$$

$$\Delta Q = 100\pi$$

Question170

**The amount of heat needed to raise the temperature of 4 moles of a rigid diatomic gas from 0°C to 50°C when no work is done is \_\_\_\_\_. (R is the universal gas constant)**  
**[20 Jul 2021 Shift 1]**

**Options:**

- A. 250 R
- B. 750 R
- C. 175 R
- D. 500 R

**Answer: D**

**Solution:**

$$\Delta Q = \Delta U + \Delta W$$

$$\text{Here } \Delta W = 0$$

$$\Delta Q = \Delta U = nC_V \Delta T$$

$$\Delta Q = 4 \times \frac{5R}{2}(50) = 500R$$

Hence option (4).

-----

## **Question171**

**Two Carnot engines A and B operate in series such that engine A absorbs heat at  $T_1$  and rejects heat to a sink at temperature  $T$ .**

**Engine B absorbs half of the heat rejected by Engine A and rejects heat to the sink at  $T_3$ . When work done in both the cases is equal, to value of  $T$  is :**

**[27 Jul 2021 Shift 2]**

**Options:**

A.  $\frac{2}{3}T_1 + \frac{3}{2}T_3$

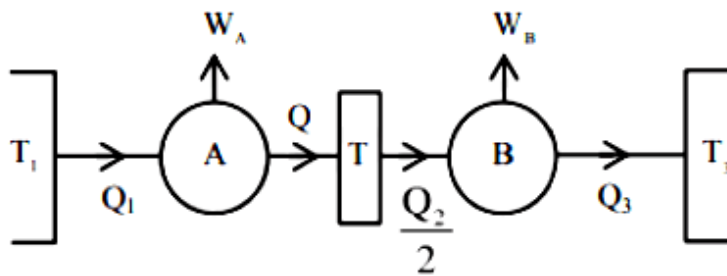
B.  $\frac{1}{3}T_1 + \frac{2}{3}T_3$

C.  $\frac{3}{2}T_1 + \frac{1}{3}T_3$

D.  $\frac{2}{3}T_1 + \frac{1}{3}T_3$

**Answer: D**

**Solution:**



$$W_A = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T}{T_1} \Rightarrow \frac{Q_2}{Q_1} = \frac{T}{T_1}$$

$$W_B = 1 - \frac{Q_3}{(Q_2/2)} = 1 - \frac{T_3}{T} \Rightarrow \frac{2Q_3}{Q_2} = \frac{T_3}{T}$$

$$\text{Now, } W_A = W_B$$

$$Q_1 - Q_2 = \frac{Q_2}{2} - Q_3$$

$$\Rightarrow \frac{2Q_1}{Q_2} + \frac{2Q_3}{Q_2} = 3$$

$$\Rightarrow \frac{2T_1}{T} + \frac{T_3}{T} = 3$$

$$\frac{2T_1}{3} + \frac{T_3}{3} = T$$

## Question172

A heat engine has an efficiency of  $\frac{1}{6}$ . When the temperature of sink is reduced by  $62^\circ\text{C}$ , its efficiency get doubled. The temperature of the source is :

[25 Jul 2021 Shift 2]

**Options:**

A.  $124^\circ\text{C}$

B.  $37^\circ\text{C}$

C.  $62^\circ\text{C}$



D.  $99^{\circ}\text{C}$

**Answer: D**

**Solution:**

$$\eta = 1 - \frac{T_L}{T_H} \dots\dots\dots (i)$$

$$2\eta = 1 - \frac{(T_L - 62)}{T_H} = 1 - \frac{T_L}{T_H} + \frac{62}{T_H}$$

$$\Rightarrow \eta = \frac{62}{T_H} \Rightarrow \frac{1}{6} = \frac{62}{T_H} \Rightarrow T_H = 6 \times 62 = 372\text{K}$$

$$\text{In } ^{\circ}\text{C} \Rightarrow 372 - 273 = 99^{\circ}\text{C}$$

---

## Question173

**The height of victoria falls is 63m. What is the difference in temperature of water at the top and at the bottom of fall ?**

**[Given,  $1 \text{ cal} = 4.2\text{J}$  and specific heat of water  $= 1 \text{ cal g}^{-1} \text{ }^{\circ}\text{C}^{-1}$ ]**

**[27 Aug 2021 Shift 2]**

**Options:**

A.  $0.147^{\circ}\text{C}$

B.  $14.76^{\circ}\text{C}$

C.  $1.476^{\circ}\text{C}$

D.  $0.014^{\circ}\text{C}$

**Answer: A**

**Solution:**

Given, height of fall,  $H = 63\text{m}$

Acceleration due to gravity,  $g = 9.8\text{ms}^{-2}$

Specific heat of water,  $s = 1 \text{ cal g}^{-1} \text{ }^{\circ}\text{C}^{-1}$

$$= 4.2 \times 10^3 \text{J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$$

Since, energy,  $E = mgH = ms \Delta T$

where,  $m = \text{mass}$

and  $\Delta T$  = change in temperature.

$$\therefore \Delta T = \frac{gH}{s} = \frac{9.8 \times 63}{4.2 \times 10^3} = 0.147^\circ\text{C}$$

---

## Question174

The temperature of equal masses of three different liquids x, y and z are  $10^\circ\text{C}$ ,  $20^\circ\text{C}$  and  $30^\circ\text{C}$ , respectively. The temperature of mixture when x is mixed with y is  $16^\circ\text{C}$  and that when y is mixed with z is  $26^\circ\text{C}$ . The temperature of mixture when x and z are mixed will be [26 Aug 2021 Shift 2]

Options:

A.  $28.32^\circ\text{C}$

B.  $25.62^\circ\text{C}$

C.  $23.84^\circ\text{C}$

D.  $20.28^\circ\text{C}$

**Answer: C**

**Solution:**

Given masses of three liquids are equal,

i.e.  $m_x = m_y = m_z = m$

Temperature of mixture of liquids x and y,  $T_1 = 16^\circ\text{C}$

Temperature of mixture of liquids y and z,  $T_2 = 26^\circ\text{C}$

While mixing the liquids x and y, the heat energy lost by liquid y will be equal to the heat energy absorbed by liquid x.

Consider the specific heat capacity of liquid x is  $s_x$ , of liquid y is  $s_y$  and of liquid z is  $s_z$ .

Now, for liquid x and y, using principle of calorimetry,

Heat gained by liquid x = Heat lost by liquid y

$$ms_x(T_1 - 10) = ms_y(20 - T_1)$$

$$\Rightarrow s_x(16 - 10) = s_y(20 - 16)$$

$$\Rightarrow \frac{s_x}{s_y} = \frac{2}{3} \dots (i)$$

Similarly, for liquid y and z, using principle of calorimetry,

Heat gained by liquid y = Heat lost by liquid z

$$ms_y(T_2 - 20) = ms_z(30 - T_2)$$

$$\Rightarrow s_y(26 - 20) = s_z(30 - 26) \dots (ii)$$

$$\Rightarrow \frac{s_y}{s_z} = \frac{2}{3} \dots (i)$$

Multiply Eq. (i) by Eq. (ii), we get

$$\frac{s_x}{s_z} = \frac{4}{9}$$

Consider the mixing of liquids x and z, let the temperature of mixture be  $T_3$ .

Using principle of calorimetry for liquids x and z.

Heat gained by liquid x = Heat lost by liquid z

$$ms_x(T_3 - 10) = ms_z(30 - T_3)$$

$$\Rightarrow s_x(T_3 - 10) = s_z(30 - T_3)$$

$$\Rightarrow \frac{s_x}{s_z} = \frac{30 - T_3}{T_3 - 10}$$

$$\Rightarrow \frac{4}{9} = \frac{30 - T_3}{T_3 - 10}$$

$$\Rightarrow 4T_3 - 40 = 270 - 9T_3$$

$$\Rightarrow T_3 = 23.84^\circ\text{C}$$

## Question 175

Two thin metallic spherical shells of radii  $r_1$  and  $r_2$  ( $r_1 < r_2$ ) are placed with their centres coinciding. A material of thermal conductivity  $K$  is filled in the space between the shells. The inner shell is maintained at temperature  $\theta_1$  and the outer shell at temperature  $\theta_2$  ( $\theta_1 < \theta_2$ ). The rate at which heat flows radially through the material is  
[31 Aug 2021 Shift 2]

Options:

A.  $\frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$

B.  $\frac{\pi r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$

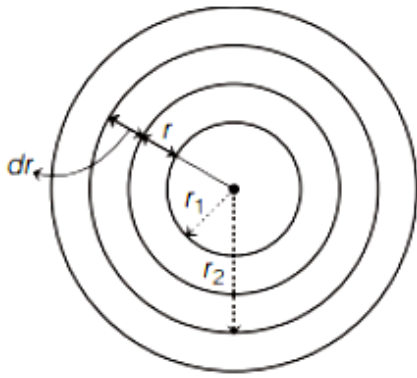
C.  $\frac{K(\theta_2 - \theta_1)}{r_2 - r_1}$

D.  $\frac{K(\theta_2 - \theta_1)(r_2 - r_1)}{4\pi r_1 r_2}$

**Answer: A**

**Solution:**

Given that,  $r_1$  and  $r_2$  be the radii of inner and outer shells.



A material of thermal conductivity ( $K$ ) is filled between region.  
Consider an elementary sphere of radius  $r$  and thickness  $dr$ .

Now, thermal resistance of elementary sphere  $dR = \frac{dr}{K4\pi r^2}$  [ $\because$  inner surface area,  $A = 4\pi r^2$ ]

So, total thermal resistance,

$$R = \int dR = \int_{r_1}^{r_2} \frac{dr}{4\pi K r^2} = -\frac{1}{4\pi K} \left[ \frac{1}{r} \right]_{r_1}^{r_2}$$

$$R = \frac{r_2 - r_1}{4\pi K r_1 r_2}$$

Hence, heat current (or the rate of flow of heat)

$$\frac{dQ}{dt} = \frac{\Delta\theta}{R} = \frac{(\theta_2 - \theta_1)}{\frac{(r_2 - r_1)}{4\pi K r_1 r_2}} = \frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$$

## Question176

**An electric appliance supplies 6000J/min heat to the system. If the system delivers a power of 90W. How long it would take to increase the internal energy by  $2.5 \times 10^3$  J ?**  
[26 Aug 2021 Shift 1]

**Options:**

A.  $2.5 \times 10^2$  s

B.  $4.1 \times 10^1$  s

C.  $24 \times 10^3$  s

D.  $2.5 \times 10^1$  s

**Answer: A**

**Solution:**

Given, heat supplied to the system,

$$\frac{\Delta Q}{\Delta t} = 6000 \text{ J/min} = \frac{6000}{60} \text{ J/s} = 100 \text{ J/s}$$

$$\text{Power delivered, } P = \frac{\Delta W}{t} = 90 \text{ W}$$

Increase in internal energy,  $\Delta U = 2.5 \times 10^3 \text{ J}$

From first law of thermodynamics, we have

$$\Delta Q = \Delta U + \Delta W$$

$$\text{or } \frac{\Delta Q}{\Delta t} = \frac{\Delta U}{\Delta t} + \frac{\Delta W}{\Delta t} \dots (i)$$

Substituting the given values in Eq. (i), we get

$$100 = \frac{2.5 \times 10^3}{\Delta t} + 90$$

$$\Rightarrow 10 = \frac{2.5 \times 10^3}{\Delta t}$$

$$\Rightarrow \Delta t = \frac{2.5 \times 10^3}{10}$$

$$\Rightarrow \Delta t = 2.5 \times 10^2 \text{ s}$$

---

## Question 177

**A sample of gas with  $\gamma = 1.5$  is taken through an adiabatic process in which the volume is compressed from  $1200 \text{ cm}^3$  to  $300 \text{ cm}^3$ . If the initial pressure is  $200 \text{ kPa}$ . The absolute value of the work done by the gas in the process is ..... J.**

**[31 Aug 2021 Shift 2]**

**Answer: 480**

**Solution:**

Given, sample of an ideal gas with ( $\gamma = 1.5$ ) is taken through adiabatic process, where

$$\text{Initial volume, } V_1 = 1200 \text{ cm}^3 = 12 \times 10^{-4} \text{ m}^3$$

$$\text{Final volume, } V_2 = 300 \text{ cm}^3 = 3 \times 10^{-4} \text{ m}^3$$

$$\text{Initial pressure, } p_1 = 200 \text{ kPa} = 2 \times 10^5 \text{ Pa}$$

Let final pressure =  $p_2$

Using equation of adiabatic process,

$$p_1 V_1^\gamma = p_2 V_2^\gamma$$

$$p_2 = p_1 \left( \frac{V_1}{V_2} \right)^\gamma = 2 \times 10^5 \left( \frac{12 \times 10^{-4}}{3 \times 10^{-4}} \right)^{1.5}$$

$$= 16 \times 10^5 \text{ Pa}$$

Work done in adiabatic process,  $W = \frac{p_1 V_1 - p_2 V_2}{\gamma - 1}$

Substituting the values, we get

$$W = \frac{2 \times 10^5 \times 12 \times 10^{-4} - 16 \times 10^5 \times 3 \times 10^{-4}}{1.5 - 1} = -480 \text{ J}$$

Hence, work done in adiabatic compression is 480 J.

---

## Question 178

A reversible engine has an efficiency of  $\frac{1}{4}$ . If the temperature of the sink is reduced by  $58^\circ\text{C}$ , its efficiency becomes double. Calculate the temperature of the sink.

[31 Aug 2021 Shift 1]

**Options:**

A.  $174^\circ\text{C}$

B.  $280^\circ\text{C}$

C.  $180.4^\circ\text{C}$

D.  $382^\circ\text{C}$

**Answer: A**

**Solution:**

Given, initial efficiency of engine,  $\eta_1 = \frac{1}{4}$

Let, initial temperature of sink =  $T_2$

Final temperature of sink =  $T_2 - 58^\circ\text{C}$

Final efficiency of engine,  $\eta_2 = 2\eta_1 = \frac{2}{4} = \frac{1}{2}$

Temperature of source =  $T_1$

Since,  $\eta = 1 - \frac{T_2}{T_1}$

$$\therefore \eta_1 = \frac{1}{4} = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow \frac{T_2}{T_1} = 1 - \frac{1}{4}$$

$$\Rightarrow \frac{T_2}{T_1} = \frac{3}{4}$$

$$\Rightarrow T_1 = \frac{4}{3}T_2 \dots (i)$$

$$\text{and } \eta_2 = \frac{1}{2} = 1 - \frac{T_2 - 58}{T_1}$$

$$\Rightarrow \frac{T_2 - 58}{T_1} = 1 - \frac{1}{2}$$

$$\Rightarrow \frac{T_2 - 58}{\frac{4}{3}T_2} = \frac{1}{2} [\text{From Eq. (i)}]$$

$$\Rightarrow 2T_2 - 116 = \frac{4}{3}T_2$$

$$\Rightarrow \frac{2T_2}{3} = 116$$

$$\Rightarrow T_2 = \frac{116 \times 3}{2} = 58 \times 3 = 174^\circ\text{C}$$

## Question 179

A heat engine operates between a cold reservoir at temperature  $T_2 = 400\text{K}$  and a hot reservoir at temperature  $T_1$ . It takes 300 J of heat from the hot reservoir and delivers 240 J of heat to the cold reservoir in a cycle. The minimum temperature of the hot reservoir has to be ..... K.

[27 Aug 2021 Shift 2]

**Answer: 500**

**Solution:**

Given, temperature of hot reservoir =  $T_1$  Temperature of cold reservoir,  $T_2 = 400\text{K}$

Heat of hot reservoir,  $Q_1 = 300\text{J}$

Heat of cold reservoir,  $Q_2 = 240\text{J}$

Since, efficiency ( $\eta$ ) =  $1 - \frac{T_2}{T_1} = \frac{Q_1 - Q_2}{Q_1}$

$$1 - \frac{T_2}{T_1} = 1 - \frac{Q_2}{Q_1}$$

$$\Rightarrow 1 - \frac{400}{T_1} = 1 - \frac{240}{300}$$

$$\Rightarrow \frac{400}{T_1} = \frac{4}{5} \Rightarrow T_1 = 500\text{K}$$


---

## Question180

**A refrigerator consumes an average 35W power to operate between temperature  $-10^\circ\text{C}$  to  $25^\circ\text{C}$ . If there is no loss of energy, then how much average heat per second does it transfer?**

**[26 Aug 2021 Shift 2]**

**Options:**

A. 263J/s

B. 298J/s

C. 350J/s

D. 35J/s

**Answer: A**

**Solution:**

Given, power consumed by refrigerator

$$P = 35\text{W} = 35\text{Js}^{-1}$$

$\therefore$  Energy consumed by refrigerator per second,  $W = 35\text{J}$

Lower temperature limit,  $T_L = -10^\circ\text{C} = 263\text{K}$

Higher temperature limit,  $T_H = 25^\circ\text{C} = 298\text{K}$

The coefficient of performance of refrigerator is given as

$$\text{COP} = \frac{T_L}{T_H - T_L} = \frac{Q}{W}$$

Here,  $Q$  is the heat transferred by the refrigerator.

Substituting the values, we get

$$\frac{263}{298 - 263} = \frac{Q}{35}$$

$$\Rightarrow Q = \frac{263}{35} \times 35 = 263\text{J/s}$$

Thus, the average heat transfer by refrigerator per second is  $263\text{Js}^{-1}$ .

---

## Question181



Due to cold weather a 1m water pipe of cross-sectional area  $1\text{cm}^2$  is filled with ice at  $-10^\circ\text{C}$ . Resistive heating is used to melt the ice. Current of  $0.5\text{A}$  is passed through  $4\text{k}\Omega$  resistance. Assuming that, all the heat produced is used for melting, what is the minimum time required?

[Given, latent heat of fusion for water/ice  $= 3.33 \times 10^5 \text{Jkg}^{-1}$ , specific heat of ice  $= 2 \times 10^3 \text{Jkg}^{-1}$  and density of ice  $= 10^3 \text{kg/m}^3$ ]  
[1 Sep 2021 Shift 2]

Options:

- A.  $0.353\text{s}$
- B.  $35.3\text{s}$
- C.  $3.53\text{s}$
- D.  $70.6\text{s}$

**Answer: B**

**Solution:**

Given, the length of the water pipe,  $L = 1\text{ m}$

The cross-sectional area of the water pipe,  $A = 1\text{cm}^2 = 10^{-4}\text{m}^2$

The temperature of the ice  $= -10^\circ\text{C}$

Current passing in the conductor,  $I = 0.5\text{ A}$

Resistance of the conductor,  $R = 4\text{ k}\Omega$

The latent heat of fusion for ice,  $L_i = 3.33 \times 10^5 \text{J/kg}$

The density of the ice,  $d = 1000 \text{kg/m}^3$

The specific heat of the ice,  $c_{p, \text{ice}} = 2 \times 10^3 \text{J/kg}$

Heat required to melt the ice at  $10^\circ\text{C}$  to  $0^\circ\text{C}$

$$Q = mc_p\Delta T + mL_i$$

$$Q = dVc_p\Delta T + dVL_f$$

$$= 1000 \times 10^{-4} \times 2 \times 10^3 \times (10) + 1000 \times 10^{-4} \times 3.33 \times 10^5$$

$$(\because V = A \times L)$$

$$= 35300 \text{ J}$$

According to the Joule's law of heating,

$$H = I^2Rt$$

$$\Rightarrow 35300 = (0.5)^2(4000)(t)$$

$$t = 35.3 \text{ s}$$

Thus, the minimum time required to melt the ice is  $35.3 \text{ s}$ .

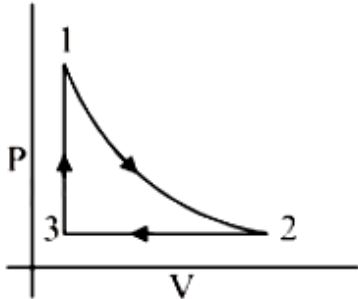
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## Question182

Which of the following is an equivalent cyclic process corresponding to the thermodynamic cyclic given in the figure?

where,  $1 \rightarrow 2$  is adiabatic.

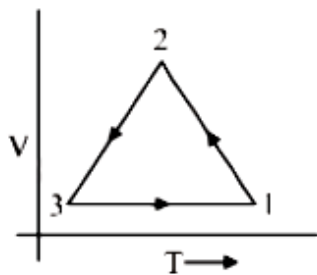
(Graphs are schematic and are not to scale)



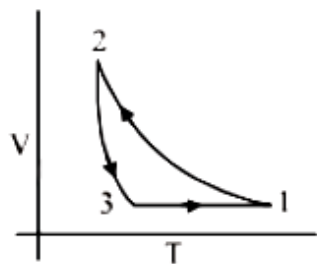
[9 Jan. 2020 I]

Options:

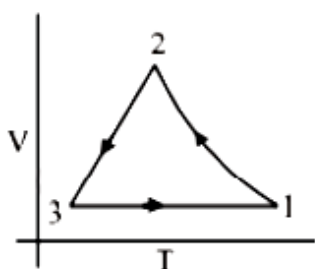
A.



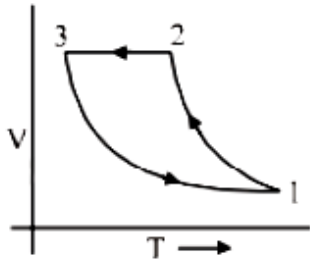
B.



C.



D.



**Answer: C**

**Solution:**

For process  $3 \rightarrow 1$  volume is constant

$\therefore$  Graph given in option (d) is wrong.

And process  $1 \rightarrow 2$  is adiabatic  $\therefore$  graph in option (1) is wrong

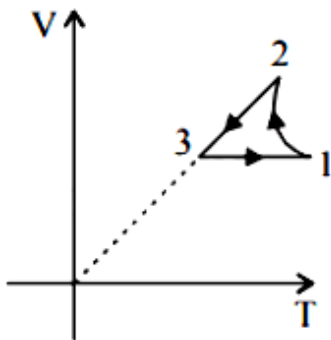
$\therefore v = \text{constant}$

$P \uparrow, T \uparrow$

For Process  $2 \rightarrow 3$  Pressure constant i.e.,  $P = \text{constant}$

$\therefore V \downarrow, T \downarrow$

Hence graph (c) is the correct  $V - T$  graph of given  $P - V$  graph



## Question183

Starting at temperature  $300\text{K}$ , one mole of an ideal diatomic gas

( $\gamma = 1.4$ ) is first compressed adiabatically from volume  $V_1$  to  $V_2 = \frac{V_1}{16}$ .

It is then allowed to expand isobarically to volume  $2V_2$ . If all the processes are the quasi-static then the final temperature of the gas (in  $^\circ\text{K}$ ) is (to the nearest integer) \_\_\_\_\_.

[9 Jan. 2020 II]

**Answer: 1818**

### **Solution:**

For an adiabatic process,

$$TV^{\gamma-1} = \text{constant}$$

$$\therefore T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\Rightarrow T_2 = (300) \times \left( \frac{V_1}{\frac{V_1}{16}} \right)^{1.4-1}$$

$$\Rightarrow T_2 = 300 \times (16)^{0.4}$$

Ideal gas equation,  $PV = nRT$

$$\therefore V = \frac{nRT}{P}$$

$$\Rightarrow V = kT \text{ (since pressure is constant for isobaric process)}$$

So, during isobaric process

$$V_2 = kT_2 \dots (i)$$

$$2V_2 = kT_f \dots (ii)$$

Dividing (i) by (ii)

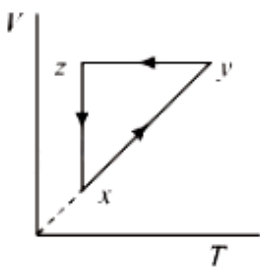
$$\frac{1}{2} = \frac{T_2}{T_f}$$

$$T_f = 2T_2 = 300 \times 2 \times (16)^{0.4} = 1818K$$

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## **Question184**

**A thermodynamic cycle xyzx is shown on a V-T diagram.**

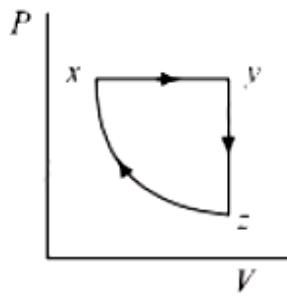


**The P-V diagram that best describes this cycle is: (Diagrams are schematic and not to scale)**

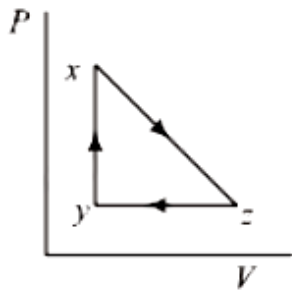
**[8 Jan. 2020 I]**

**Options:**

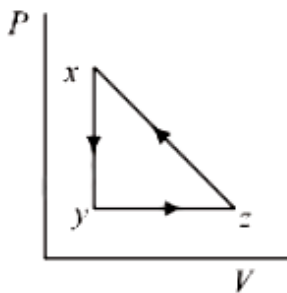
A.



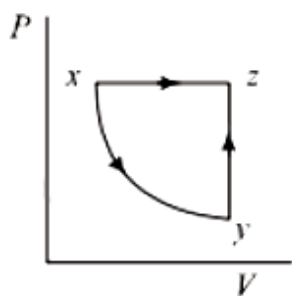
B.



C.



D.

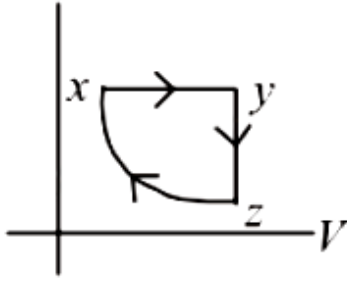


**Answer: A**

**Solution:**

From the corresponding V-T graph given in question,  
 Process  $xy \rightarrow$  Isobaric expansion,  
 Process  $yz \rightarrow$  Isochoric (Pressure decreases)  
 Process  $zx \rightarrow$  Isothermal compression

Therefore, corresponding PV graph is as shown in figure



## Question185

**A litre of dry air at STP expands adiabatically to a volume of 3 litres. If  $\gamma = 1.40$ , the work done by air is: ( $3^{1.4} = 4.6555$ ) [Take air to be an ideal gas]**  
**[7 Jan. 2020 I]**

**Options:**

- A. 60.7 J
- B. 90.5 J
- C. 100.8 J
- D. 48 J

**Answer: B**

**Solution:**

$$V_1 = 1 \Rightarrow \text{initial volume, } V_1 = 1 \times 10^{-3} \text{ m}^3$$

$$V_2 = 3 \Rightarrow \text{final volume, } V_2 = 3 \times 10^{-3} \text{ m}^3$$

At STP:

$$T = 273 \text{ K}$$

$$P_1 = 101.325 \text{ kPa} \Rightarrow \text{initial pressure } \gamma = 1.4$$

$$\text{Work done in adiabatic process, } \Delta W = - \left( \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} \right)$$

$$\text{adiabatic} \Rightarrow PV^\gamma = \text{constant}$$

$$\Rightarrow P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow P_2 = P_1 \left( \frac{V_1}{V_2} \right)^\gamma = 101.325 \left( \frac{1 \times 10^{-3}}{3 \times 10^{-3}} \right)^{1.4}$$

$$\Rightarrow P_2 = 101.325 \times \frac{1}{(3)^{1.4}} = \left( \frac{101.325}{4.6555} \right) \text{ kPa}$$

$$P_2 = 21.7646 \text{ K Pa}$$

$$\Delta W = - \left( \frac{21.7646 \times 10^3 \times 3 \times 10^{-3} - 101.325 \times 10^3 \times 1 \times 10^{-3}}{1.4 - 1} \right)$$

$$\Delta W = - \left( \frac{65.2937 - 101.325}{0.4} \right) = 90.5 \text{ J}$$

Considering only magnitude, the work done by the air will be 90.5J

## Question 186

**Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from  $\tau_1$  to  $\tau_2$ . If  $\frac{C_p}{C_v} = \gamma$  for this gas then a good estimate for  $\frac{\tau_2}{\tau_1}$  is given by:**

**[7 Jan. 2020 I]**

**Options:**

A. 2

B.  $\frac{1}{2}$

C.  $\left(\frac{1}{2}\right)^\gamma$

D.  $\left(\frac{1}{2}\right)^{\frac{\gamma+1}{2}}$

**Answer: D**

**Solution:**

$$\text{Mean free path, } \lambda = \left( \frac{RT}{\sqrt{2} \pi d^2 N_A P} \right)$$

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\therefore \tau = \frac{\lambda}{V_{\text{rms}}} = \frac{RT}{\sqrt{2} \pi d^2 N_A P} \times \sqrt{\frac{M}{3RT}}$$

$$\tau \propto \frac{\sqrt{T}}{P}$$

$$\frac{\tau_1}{\tau_2} = \frac{\sqrt{T_1}}{P_1} \times \frac{P_2}{\sqrt{T_2}} = \left( \frac{P_2}{P_1} \right) \sqrt{\frac{T_1}{T_2}}$$

$$PV = nRT$$

$$\begin{aligned}
\frac{P_1}{P_2} \times \frac{V_1}{V_2} &= \frac{T_1}{T_2} \\
\Rightarrow 2^\gamma \times \frac{V}{2V} &= \frac{T_1}{T_2} \\
\Rightarrow \frac{T_1}{T_2} &= (2^{\gamma-1}) \\
\therefore \frac{\tau_1}{\tau_2} &= \left( \frac{P_2}{P_1} \right) \times \sqrt{\frac{T_1}{T_2}} = \frac{1}{2^\gamma} \times \sqrt{2^{\gamma-1}} = \frac{2^{\frac{\gamma-1}{2}}}{2^\gamma} \\
\frac{\tau_2}{\tau_1} &= \frac{2^\gamma}{2^{\frac{\gamma-1}{2}}} = 2^{\gamma - \left(\frac{\gamma-1}{2}\right)} = 2^{\frac{\gamma+1}{2}} \\
\Rightarrow \frac{\tau_1}{\tau_2} &= \left( \frac{1}{2} \right)^{\frac{\gamma+1}{2}}
\end{aligned}$$


---

## Question 187

**A Carnot engine having an efficiency of  $\frac{1}{10}$  is being used as a refrigerator. If the work done on the refrigerator is 10 J, the amount of heat absorbed from the reservoir at lower temperature is:**  
**[8 Jan. 2020 II]**

**Options:**

- A. 99 J
- B. 100 J
- C. 1 J
- D. 90 J

**Answer: D**

**Solution:**

For Carnot refrigerator

$$\text{Efficiency} = \frac{Q_1 - Q_2}{Q_1}$$

Where,  $Q_1$  = heat lost from surrounding

$Q_2$  = heat absorbed from reservoir at low

$$\text{Also, } \frac{Q_1 - Q_2}{Q_1} = \frac{W}{Q_1}$$



$$\Rightarrow \frac{1}{10} = \frac{w}{Q_1}$$

$$\Rightarrow Q_1 = w \times 10 = 100\text{J}$$

$$\text{So, } Q_1 - Q_2 = w$$

$$\Rightarrow Q_2 = Q_1 - w$$

$$\Rightarrow 100 - 10 = Q_2 = 90\text{J}$$

## Question188

**A Carnot engine operates between two reservoirs of temperatures 900 K and 300 K. The engine performs 1200 J of work per cycle. The heat energy (in J) delivered by the engine to the low temperature reservoir, in a cycle, is \_\_\_\_\_.**

**[NA 7 Jan. 2020 I]**

**Answer: 600**

**Solution:**

Given;  $T_1 = 900\text{K}$ ,  $T_2 = 300\text{K}$ ,  $W = 1200\text{J}$

$$\text{Using, } 1 - \frac{T_2}{T_1} = \frac{W}{Q_1}$$

$$\Rightarrow 1 - \frac{300}{900} = \frac{1200}{Q_1}$$

$$\Rightarrow \frac{2}{3} = \frac{1200}{Q_1} \Rightarrow Q_1 = 1800$$

Therefore heat energy delivered by the engine to the low temperature reservoir,

$$Q_2 = Q_1 - W = 1800 - 1200 = 600.00\text{J}$$

## Question189

**Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperatures,  $T_1$  and  $T_2$ . The temperature of the hot reservoir of the first engine is  $T_1$  and the temperature of the cold reservoir of the**

second engine is  $T_2$ .  $T$  is temperature of the sink of first engine which is also the source for the second engine. How is  $T$  related to  $T_1$  and  $T_2$ , if both the engines perform equal amount of work?

[7 Jan. 2020 II]

Options:

A.  $T = \frac{2T_1T_2}{T_1 + T_2}$

B.  $T = \frac{T_1 + T_2}{2}$

C.  $T = \sqrt{T_1T_2}$

D.  $T = 0$

**Answer: B**

**Solution:**

Let  $Q_H$  = Heat taken by first engine

$Q_L$  = Heat rejected by first engine

$Q_2$  = Heat rejected by second engine

Work done by 1<sup>st</sup> engine = work done by 2<sup>nd</sup> engine

$$W = Q_H - Q_L = Q_L - Q_2 \Rightarrow 2Q_L = Q_H + Q_2$$

$$2 = \frac{Q_H}{Q_L} + \frac{Q_2}{Q_L}$$

Let  $T$  be the temperature of cold reservoir of first engine. Then in carnot engine.

$$\frac{Q_H}{Q_L} = \frac{T_1}{T} \text{ and } \frac{Q_2}{Q_L} = \frac{T}{T_2}$$

$$\Rightarrow 2 = \frac{T_1}{T} + \frac{T}{T_2} \text{ using (i)}$$

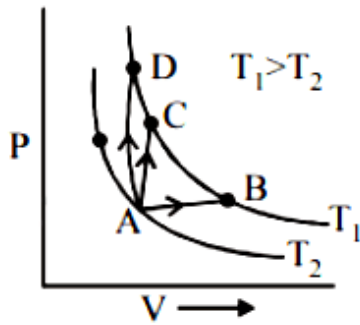
$$\Rightarrow 2T = T_1 + T_2 \Rightarrow T = \frac{T_1 + T_2}{2}$$

## Question190

Three different processes that can occur in an ideal monoatomic gas are shown in the  $P$  vs  $V$  diagram. The paths are labelled as

$A \rightarrow B$ ,  $A \rightarrow C$  and  $A \rightarrow D$ . The change in internal energies during these process are taken as  $E_{AB}$ ,  $E_{AC}$  and  $E_{AD}$  and the work done as

$W_{AB}$ ,  $W_{AC}$  and  $W_{AD}$  The correct relation between these parameters are:



[5 Sep. 2020 (I)]

Options:

- A.  $E_{AB} = E_{AC} < E_{AD}$ ,  $W_{AB} > 0$ ,  $W_{AC} = 0$ ,  $W_{AD} < 0$
- B.  $E_{AB} = E_{AC} = E_{AD}$ ,  $W_{AB} > 0$ ,  $W_{AC} = 0$ ,  $W_{AD} > 0$
- C.  $E_{AB} < E_{AC} < E_{AD}$ ,  $W_{AB} > 0$ ,  $W_{AC} > W_{AD}$
- D.  $E_{AB} > E_{AC} > E_{AD}$ ,  $W_{AB} < W_{AC} < W_{AD}$

**Answer: B**

**Solution:**

Temperature change  $\Delta T$  is same for all three processes

$A \rightarrow B$ ;  $A \rightarrow C$  and  $A \rightarrow D$

$\Delta U = nC_v\Delta T = \text{same}$

$E_{AB} = E_{AC} = E_{AD}$

Work done,  $W = P \times \Delta V$

$AB \rightarrow$  volume is increasing  $\Rightarrow W_{AB} > 0$

$AD \rightarrow$  volume is decreasing  $\Rightarrow W_{AD} < 0$

$AC \rightarrow$  volume is constant  $\Rightarrow W_{AC} = 0$

## Question191

In an adiabatic process, the density of a diatomic gas becomes 32 times its initial value. The final pressure of the gas is found to be  $n$  times the initial pressure. The value of  $n$  is :

[5 Sep. 2020 (II)]

**Options:**

A. 32

B. 326

C. 128

D.  $\frac{1}{32}$

**Answer: C**

**Solution:**

In adiabatic process

$$PV^\gamma = \text{constant}$$

$$\therefore P \left( \frac{m}{\rho} \right)^\gamma = \text{constant} \left( \because V = \frac{m}{\rho} \right)$$

As mass is constant

$$\therefore P \propto \rho^\gamma$$

If  $P_i$  and  $P_f$  be the initial and final pressure of the gas and  $\rho_i$  and  $\rho_f$  be the initial and final density of the gas. Then

$$\frac{P_f}{P_i} = \left( \frac{\rho_f}{\rho_i} \right)^\gamma = (32)^{7/5}$$

$$\Rightarrow \frac{n P_i}{P_i} = (2^5)^{7/5} = 2^7$$

$$\Rightarrow n = 2^7 = 128$$

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## Question192

**Match the thermodynamic processes taking place in a system with the correct conditions. In the table :  $\Delta Q$  is the heat supplied,  $\Delta W$  is the work done and  $\Delta U$  is change in internal energy of the system.**

| Process         | Condition                                               |
|-----------------|---------------------------------------------------------|
| (I) Adiabatic   | (A) $\Delta W = 0$                                      |
| (II) Isothermal | (B) $\Delta Q = 0$                                      |
| (III) Isochoric | (C) $\Delta U \neq 0, \Delta W \neq 0, \Delta Q \neq 0$ |
| IV) Isobaric    | (D) $\Delta U = 0$                                      |

[4 Sep. 2020 (II)]

Options:

- A. (I)-(A), (II)-(B), (III)-(D), (IV)-(D)
- B. (I)-(B), (II)-(A), (III)-(D), (IV)-(C)
- C. (I)-(A), (II)-(A), (III)-(B), (IV)-(C)
- D. (I)-(B), (II)-(D), (III)-(A), (IV)-(C)

Answer: D

Solution:

(I) Adiabatic process : No exchange of heat takes place with surroundings.

$$\Rightarrow \Delta Q = 0$$

(II) Isothermal process : Temperature remains constant

$$\therefore \Delta T = 0 \Rightarrow \Delta U = \frac{f}{2} n R \Delta T \Rightarrow \Delta U = 0$$

No change in internal energy [ $\Delta U = 0$ ]

(III) Isochoric process volume remains constant

$$\Delta V = 0 \Rightarrow W = \int P \cdot dV = 0$$

Hence work done is zero.

(IV) In isobaric process pressure remains constant.

$$W = P \cdot \Delta V \neq 0$$

$$\Delta U = \frac{f}{2} n R \Delta T = \frac{f}{2} [P \Delta V] \neq 0$$

$$\therefore \Delta Q = n C_p \Delta T \neq 0$$

Question193

**A balloon filled with helium (32°C and 1.7 atm.) bursts. Immediately afterwards the expansion of helium can be considered as :**  
**[3 Sep. 2020 (I)]**

**Options:**

- A. irreversible isothermal
- B. irreversible adiabatic
- C. reversible adiabatic
- D. reversible isothermal

**Answer: B**

**Solution:**

Bursting of helium balloon is irreversible and in this process  $\Delta Q = 0$ , so adiabatic.

-----

## **Question 194**

**An engine takes in 5 mole of air at 20°C and 1 atm, and compresses it adiabatically to  $1/10^{\text{th}}$  of the original volume. Assuming air to be a diatomic ideal gas made up of rigid molecules, the change in its internal energy during this process comes out to be X kJ. The value of X to the nearest integer is \_\_\_\_\_.**  
**[NA 2 Sep. 2020 (I)]**

**Answer: 46**

**Solution:**

For adiabatic process,  $T V^{\gamma-1} = \text{constant}$

$$\text{or, } T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_1 = 20^\circ\text{C} \text{ and } \gamma = \frac{7}{5}$$

$$T_1(V_1)^{\gamma-1} = T_2\left(\frac{V_1}{10}\right)^{\gamma-1}$$

$$\Rightarrow 293 = T_2\left(\frac{1}{10}\right)^{2/5} \Rightarrow T_2 = 293(10)^{2/5} \simeq 736\text{K}$$

$$\Delta T = 736 - 293 = 443\text{K}$$

During the process, change in internal energy

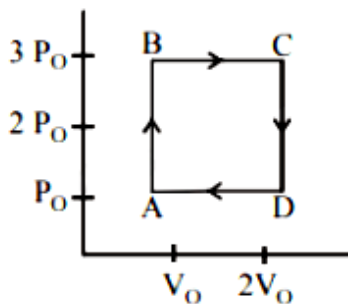
$$\Delta U = N C_V \Delta T = 5 \times \frac{5}{2} \times 8.3 \times 443 \simeq 46 \times 10^3 \text{J} = X \text{kJ}$$

$$\therefore X = 46$$

## Question195

An engine operates by taking a monatomic ideal gas through the cycle shown in the figure. The percentage efficiency of the engine is close is

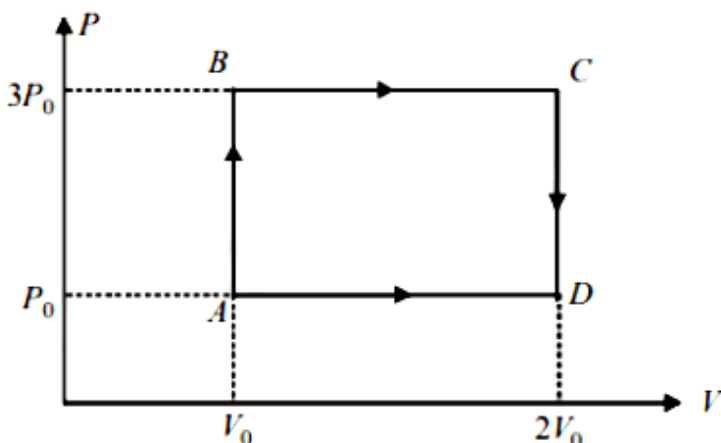
\_\_\_\_\_.



[NA 6 Sep. 2020 (II)]

**Answer: 19**

**Solution:**



From the figure,

$$\text{Work, } W = 2P_0 V_0$$

$$\text{Heat given, } Q_{\text{in}} = W_{AB} + W_{BC} = n \cdot C_V \Delta T_{AB} + n C_P \Delta T_{BC}$$

$$= n \frac{3R}{2} (T_B - T_A) + \frac{n5R}{2} (T_C - T_B) \left( \because C_V = \frac{3R}{2} \text{ and } C_P = \frac{5R}{2} \right)$$

$$= \frac{3}{2} (P_B V_B - P_A V_A) + \frac{5}{2} (P_C V_C - P_B V_B)$$

$$= \frac{3}{2} \times [3P_0 V_0 - P_0 V_0] + \frac{5}{2} [6P_0 V_0 - 3P_0 V_0]$$

$$= 3P_0 V_0 + \frac{15}{2} P_0 V_0 = \frac{21}{2} P_0 V_0$$

$$\text{Efficiency, } \eta = \frac{W}{Q_{\text{in}}} = \frac{2P_0 V_0}{\frac{21}{2} P_0 V_0} = \frac{4}{21}$$

$$\eta\% = \frac{400}{21} \approx 19$$

---

## Question196

**If minimum possible work is done by a refrigerator in converting 100 grams of water at 0°C to ice, how much heat (in calories) is released to the surroundings at temperature 27°C (Latent heat of ice = 80 Cal/gram) to the nearest integer?**  
[NA 3 Sep. 2020 (II)]

**Answer: 8791**

### Solution:

Given,

$$\text{Heat absorbed, } Q_2 = mL = 80 \times 100 = 8000 \text{ Cal}$$

$$\text{Temperature of ice, } T_2 = 273\text{K}$$

Temperature of surrounding,

$$T_1 = 273 + 27 = 300\text{K}$$

$$\text{Efficiency} = \frac{W}{Q_2} = \frac{Q_1 - Q_2}{Q_2} = \frac{T_1 - T_2}{T_2} = \frac{300 - 273}{273}$$

$$\Rightarrow \frac{Q_1 - 8000}{8000} = \frac{27}{273} \Rightarrow Q_1 = 8791 \text{ Cal}$$

---

## Question197



A heat engine is involved with exchange of heat of 1915 J, – 40 J, +125 J and – Q J, during one cycle achieving an efficiency of 50.0%. The value of Q is :  
[2 Sep. 2020 (II)]

Options:

- A. 640 J
- B. 40 J
- C. 980 J
- D. 400 J

Answer: C

Solution:

$$\text{Efficiency, } \eta = \frac{\text{Work done}}{\text{Heat absorbed}} = \frac{W}{\sum Q}$$

$$= \frac{Q_1 + Q_2 + Q_3 + Q_4}{Q_1 + Q_3} = 0.5$$

Here,  $Q_1 = 1915\text{J}$ ,  $Q_2 = -40\text{J}$  and  $Q_3 = 125\text{J}$

$$\therefore \frac{1915 - 40 + 125 + Q_4}{1915 + 125} = 0.5$$

$$\Rightarrow 1915 - 40 + 125 + Q_4 = 1020$$

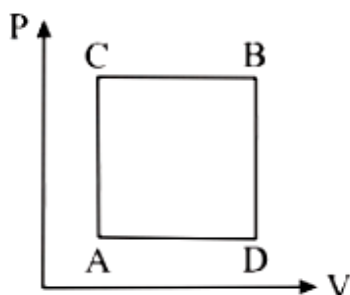
$$\Rightarrow Q_4 = 1020 - 2000$$

$$\Rightarrow Q_4 = -Q = -980\text{J}$$

$$\Rightarrow Q = 980\text{J}$$

## Question198

A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used 60 J of heat flows into the system and 30J of

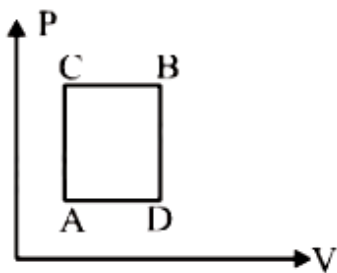
work is done by the system. If path ADB is used work done by the system is 10 J. The heat Flow into the system in path ADB is :  
[9 Jan. 2019 I]

Options:

- A. 40 J
- B. 80 J
- C. 100 J
- D. 20 J

Answer: A

Solution:



$\Delta U$  remains same for both paths ACB and ADB

$$\Delta Q_{ACB} = \Delta W_{ACB} + \Delta U_{ACB}$$

$$\Rightarrow 60\text{J} = 30\text{J} + \Delta U_{ACB}$$

$$\Rightarrow \Delta U_{ACB} = 30\text{J}$$

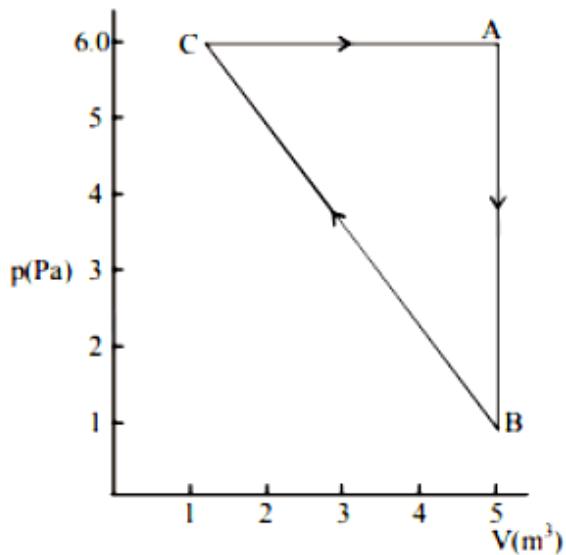
$$\therefore \Delta U_{ADB} = \Delta U_{ACB} = 30\text{J}$$

$$\begin{aligned}\Delta Q_{ADB} &= \Delta U_{ADB} + \Delta W_{ADB} \\ &= 10\text{J} + 30\text{J} = 40\text{J}\end{aligned}$$

---

## Question199

For the given cyclic process CAB as shown for gas, the work done is:



[12 Jan. 2019 I]

Options:

- A. 30 J
- B. 10 J
- C. 1 J
- D. 5 J

**Answer: B**

**Solution:**

Total work done by the gas during the cycle is equal to area of triangle ABC.

$$\therefore \Delta W = \frac{1}{2} \times 4 \times 5 = 10\text{J}$$


---

## Question200

**A rigid diatomic ideal gas undergoes an adiabatic process at room temperature. The relation between temperature and volume for this process is  $T V^x = \text{constant}$ , then x is:**

**[11 Jan. 2019 I]**

Options:

A.  $\frac{3}{5}$

B.  $\frac{2}{5}$

C.  $\frac{2}{3}$

D.  $\frac{5}{3}$

**Answer: B**

**Solution:**

Equation of adiabatic change is

$$T V^{\gamma-1} = \text{constant}$$

Put  $\gamma = \frac{7}{5}$ , we get:  $\gamma - 1 = \frac{7}{5} - 1$

$$\therefore x = \frac{2}{5}$$

---

## Question201

**Half mole of an ideal monoatomic gas is heated at constant pressure of 1 atm from 20°C to 90°C. Work done by gas is close to: (Gas constant R = 8.31 J/mol-K)  
[10 Jan. 2019 II]**

**Options:**

A. 581 J

B. 291 J

C. 146 J

D. 73 J

**Answer: B**

**Solution:**

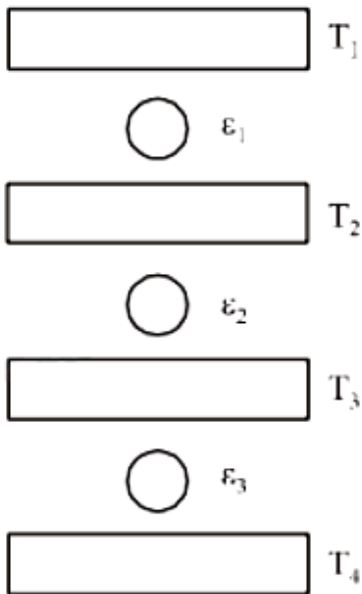
Work done,

$$W = P\Delta V = nR\Delta T = \frac{1}{2} \times 8.31 \times 70 \simeq 291\text{J}$$

---

## Question 202

Three Carnot engines operate in series between a heat source at a temperature  $T_1$  and a heat sink at temperature  $T_4$  (see figure). There are two other reservoirs at temperature  $T_2$  and  $T_3$ , as shown, with  $T_1 > T_2 > T_3 > T_4$ . The three engines are equally efficient if:



[10 Jan. 2019 I]

Options:

- A.  $T_2 = (T_1 T_4)^{1/2}$ ;  $T_3 = (T_1^2 T_4)^{1/3}$
- B.  $T_2 = (T_1^2 T_4)^{1/3}$ ;  $T_3 = (T_1 T_4^2)^{1/3}$
- C.  $T_2 = (T_1 T_4^2)^{1/3}$ ;  $T_3 = (T_1^2 T_4)^{1/3}$
- D.  $T_2 = (T_1^3 T_4)^{1/4}$ ;  $T_3 = (T_1 T_4^3)^{1/4}$

**Answer: B**

**Solution:**

According to question,  $\eta_1 = \eta_2 = \eta_3$

$$\therefore 1 - \frac{T_2}{T_1} = 1 - \frac{T_3}{T_2} = 1 - \frac{T_4}{T_3}$$

[  $\because$  Three engines are equally efficient ]

$$\Rightarrow \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3}$$

$$\Rightarrow T_2 = \sqrt{T_1 T_3}$$

$$T_3 = \sqrt{T_2 T_4}$$

From (i) and (ii)

$$T_2 = (T_1^2 T_4)^{1/3}$$

$$T_3 = (T_1 T_4^2)^{1/3}$$

---

## Question203

**Two Carnot engines A and B are operated in series. The first one, A receives heat at  $T_1$  ( = 600K ) and rejects to a reservoir at temperature  $T_2$ . The second engine B receives heat rejected by the first engine and in turn, rejects to a heat reservoir at  $T_3$  ( = 400K ). Calculate the temperature  $T_2$  if the work outputs of the two engines are equal: [9 Jan. 2019 II]**

**Options:**

A. 600 K

B. 400 K

C. 300 K

D. 500 K

**Answer: D**

**Solution:**

$$\eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1}$$

$$\text{and, } \eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$$

According to question,

$$W_A = W_B$$

$$\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2}$$

$$\therefore T_2 = \frac{T_1 + T_3}{2}$$

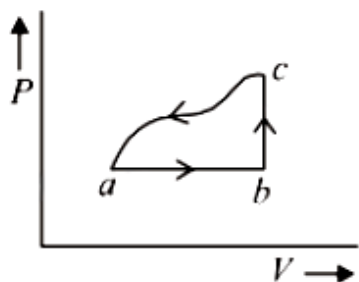
$$= \frac{600 + 400}{2}$$

$$= 500\text{K}$$


---

## Question204

A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is – 180 J, The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is:



[12 Apr. 2019 I]

Options:

- A. 120 J
- B. 130 J
- C. 100 J
- D. 140 J

**Answer: B**

**Solution:**

$$\Delta U_{ac} = -(\Delta U_{ca}) = -(-180) = 180\text{J}$$

$$Q = 250 + 60 = 310\text{J}$$

$$\text{Now } Q = \Delta U + W$$

$$\text{or } 310 = 180 + W$$

$$\text{or } W = 130\text{J}$$


---

## Question205

A cylinder with fixed capacity of 67.2 lit contains helium gas at STP. The amount of heat needed to raise the temperature of the gas by 20oC is : [Given that  $R = 8.31 \text{ Jmol}^{-1} \text{ K}^{-1}$ ]  
[10 Apr. 2019 I]

Options:

- A. 350 J
- B. 374 J
- C. 748 J
- D. 700 J

Answer: C

Solution:

As the process is isochoric so,

$$Q = nc_v \Delta T = \frac{67.2}{22.4} \times \frac{3R}{2} \times 20 = 90R = 90 \times 8.31 \simeq 748\text{J}$$

-----

## Question206

n moles of an ideal gas with constant volume heat capacity  $C_v$  undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is:  
[10 Apr. 2019 I]

Options:

- A.  $\frac{nR}{C_v + nR}$
- B.  $\frac{nR}{C_v - nR}$
- C.  $\frac{4nR}{C_v - nR}$



D.  $\frac{4nR}{C_V + nR}$

**Answer: A**

**Solution:**

At constant volume

Work done

$$(W) = nR\Delta T$$

$$\text{Heat given } Q = C_V\Delta T + nR\Delta T$$

$$\text{So, } \therefore \frac{W}{Q} = \frac{nR\Delta T}{C_V\Delta T + nR\Delta T} = \frac{nR}{C_V + nR}$$

## Question207

One mole of an ideal gas passes through a process where pressure and volume obey the relation  $P = P_0 \left[ 1 - \frac{1}{2} \left( \frac{V_0}{V} \right)^2 \right]$ .

Here  $P$  and  $V_0$  are constants. Calculate the change in the temperature of the gas if its volume changes from  $V_0$  to  $2V_0$ .

[10 Apr. 2019 II]

**Options:**

A.  $\frac{1}{2} \frac{P_0 V_0}{R}$

B.  $\frac{5}{4} \frac{P_0 V_0}{R}$

C.  $\frac{3}{4} \frac{P_0 V_0}{R}$

D.  $\frac{1}{4} \frac{P_0 V_0}{R}$

**Answer: B**

**Solution:**

We have given,

$$P = P_0 \left[ 1 - \frac{1}{2} \left( \frac{V_0}{V} \right)^2 \right]$$

When  $V_1 = V_0$

$$\Rightarrow P_1 = P_0 \left[ 1 - \frac{1}{2} \right] = \frac{P_0}{2}$$

When  $V_2 = 2V_0$

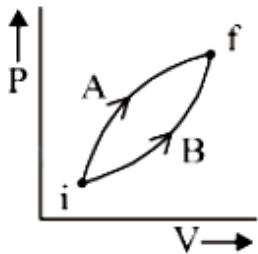
$$\Rightarrow P_2 = P_0 \left[ 1 - \frac{1}{2} \left( \frac{1}{4} \right) \right] = \left( \frac{7P_0}{8} \right)$$

$$\Delta T = T_2 - T_1 = \left| \frac{P_1 V_1}{nR} - \frac{P_2 V_2}{nR} \right| \left[ \because T = \frac{PV}{nR} \right]$$

$$\begin{aligned} \Delta T &= \left| \left( \frac{1}{nR} \right) (P_1 V_1 - P_2 V_2) \right| = \left( \frac{1}{nR} \right) \left| \left( \frac{P_0 V_0}{2} - \frac{7P_0 V_0}{4} \right) \right| \\ &= \frac{5P_0 V_0}{4nR} = \frac{5P_0 V_0}{4R} (\because n = 1) \end{aligned}$$

## Question 208

Following figure shows two processes A and B for a gas. If  $\Delta Q_A$  and  $\Delta Q_B$  are the amount of heat absorbed by the system in two cases, and  $\Delta U_A$  and  $\Delta U_B$  are changes in internal energies, respectively, then:



[9 April 2019 I]

Options:

- A.  $\Delta Q_A < \Delta Q_B$ ,  $\Delta U_A < \Delta U_B$
- B.  $\Delta Q_A > \Delta Q_B$ ,  $\Delta U_A > \Delta U_B$
- C.  $\Delta Q_A > \Delta Q_B$ ,  $\Delta U_A = \Delta U_B$
- D.  $\Delta Q_A = \Delta Q_B$ ;  $\Delta U_A = \Delta U_B$

**Answer: C**

**Solution:**

Internal energy depends only on initial and final state

$$\text{So, } \Delta U_A = \Delta U_B$$

$$\text{Also } \Delta Q = \Delta U + W$$

$$\text{AS } W_A > W_B \Rightarrow \Delta Q_A > \Delta Q_B$$

---

## Question209

**A thermally insulated vessel contains 150g of water at 0°C. Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closed to:(Latent heat of vaporization of water =  $2.10 \times 10^6 \text{ J kg}^{-1}$  and Latent heat of Fusion of water =  $3.36 \times 10^5 \text{ J kg}^{-1}$ )**

**[8 April 2019 I]**

**Options:**

A. 150 g

B. 20 g

C. 130 g

D. 35 g

**Answer: B**

**Solution:**

Suppose amount of water evaporated be M gram.

Then (150 – M) gram water converted into ice.

so, heat consumed in evaporation = Heat released in fusion

$$M \times L_v = (150 - M) \times L$$

$$M \times 2.1 \times 10^6 = (150 - M) \times 3.36 \times 10^5$$

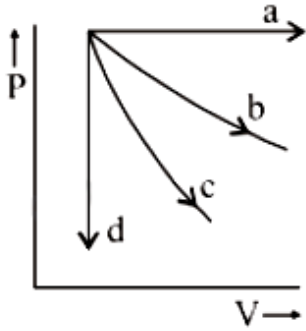
$$\Rightarrow M = 20\text{g}$$

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## Question210

**The given diagram shows four processes i.e., isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in**

the same order is given by :



[8 April 2019 II]

Options:

A. a d b c

B. d a c b

C. a d c b

D. d a b c

**Answer: D**

**Solution:**

a  $\rightarrow$  Isobaric, b  $\rightarrow$  Isothermal, c  $\rightarrow$  Adiabatic, d  $\rightarrow$  Isochoric

-----

## Question211

A Carnot engine has an efficiency of  $1/6$ . When the temperature of the sink is reduced by  $62^\circ\text{C}$ , its efficiency is doubled. The temperatures of the source and the sink are, respectively.

[12 Apr. 2019 II]

Options:

A.  $62^\circ\text{C}$ ,  $124^\circ\text{C}$

B.  $99^\circ\text{C}$ ,  $37^\circ\text{C}$

C.  $124^\circ\text{C}$ ,  $62^\circ\text{C}$

D.  $37^\circ\text{C}$ ,  $99^\circ\text{C}$

**Answer: B**

**Solution:**

$$\text{Using, } n = 1 - \frac{T_2}{T_1}$$

$$n = \frac{1}{6} = 1 - \frac{T_2}{T_1}$$

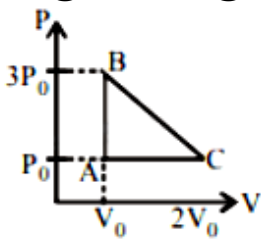
$$\text{and } \frac{T_1}{3} = 1 - \frac{T_2 - 62}{T_1}$$

On solving, we get  $T_1 = 99^\circ\text{C}$  and  $T_2 = 37^\circ\text{C}$

---

## Question212

One mole of an ideal monoatomic gas is taken along the path ABCA as shown in the PV diagram. The maximum temperature attained by the gas along the path BC is given by



[Online April 16, 2018]

**Options:**

A.  $\frac{25P_0V_0}{8R}$

B.  $\frac{25P_0V_0}{4R}$

C.  $\frac{25P_0V_0}{16R}$

D.  $\frac{5P_0V_0}{8R}$

**Answer: A**

**Solution:**

Equation of the BC

$$P = P_0 - \frac{2P_0}{V_0}(V - 2V_0)$$

using  $PV = nRT$

$$\text{Temperature, } T = \frac{P_0 V - \frac{2P_0 V^2}{V_0} + 4P_0 V}{1 \times R}$$

( $\because n = 1$  mole given)

$$T = \frac{P_0}{R} \left[ 5V - \frac{2V^2}{V_0} \right]$$

$$\frac{dT}{dV} = 0 \Rightarrow 5 - \frac{4V}{V_0} = 0 \Rightarrow V = \frac{5}{4}V_0$$

$$T = \frac{P_0}{R} \left[ 5 \times \frac{5V_0}{4} - \frac{2}{V_0} \times \frac{25}{16}V_0^2 \right] = \frac{25P_0 V_0}{8R}$$

---

## Question213

**One mole of an ideal monoatomic gas is compressed isothermally in a rigid vessel to double its pressure at room temperature, 27°C. The work done on the gas will be:**  
**[Online April 15, 2018]**

**Options:**

A.  $300R \ln 6$

B.  $300R$

C.  $300R \ln 7$

D.  $300R \ln 2$

**Answer: D**

**Solution:**

$$\text{Work done on gas} = nRT \ln \left( \frac{P_f}{P_i} \right) = R(300) \ln(2)$$

$$= 300R \ln 2 \left( \because \frac{P_f}{P_i} = 2 \text{ given} \right)$$

---

## Question214

**A Carnot's engine works as a refrigerator between 250 K and 300 K. It receives 500 cal heat from the reservoir at the lower temperature. The amount of work done in each cycle to operate the refrigerator is: [Online April 15, 2018]**

**Options:**

- A. 420 J
- B. 2100 J
- C. 772 J
- D. 2520 J

**Answer: A**

**Solution:**

Given: Temperature of cold body,  $T_2 = 250\text{K}$  temperature of hot body;  $T_1 = 300\text{K}$

Heat received,  $Q_2 = 500$  cal work done,  $W = ?$

$$\text{Efficiency} = 1 - \frac{T_2}{T_1} = \frac{W}{Q_2 + W} \Rightarrow 1 - \frac{250}{300} = \frac{W}{Q_2 + W}$$

$$W = \frac{Q_2}{5} = \frac{500 \times 4.2}{5} \text{J} = 420\text{J}$$

---

## Question215

**Two Carnot engines A and B are operated in series. Engine A receives heat from a reservoir at 600K and rejects heat to a reservoir at temperature T. Engine B receives heat rejected by engine A and in turn rejects it to a reservoir at 100K. If the efficiencies of the two engines A and B are represented by  $\eta_A$  and  $\eta_B$  respectively, then what is the value of  $\frac{\eta_A}{\eta_B}$**

**[Online April 15, 2018]**

**Options:**

- A.  $\frac{12}{7}$

B.  $\frac{12}{5}$

C.  $\frac{5}{12}$

D.  $\frac{7}{12}$

**Answer: D**

**Solution:**

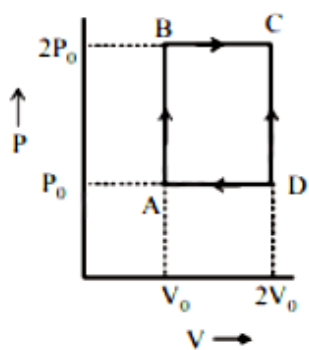
(d) Efficiency of engine A,  $\eta_A = \frac{T_1 - T_2}{T_1}$

and  $\eta_B = \frac{T_2 - T_3}{T_2}$ ;  $T_2 = \frac{T_1 + T_3}{2} = 350\text{K}$

or  $\frac{\eta_A}{\eta_B} = \frac{\frac{600 - 350}{600}}{\frac{350 - 100}{350}} = \frac{7}{12}$

## Question216

**An engine operates by taking n moles of an ideal gas through the cycle ABCDA shown in figure. The thermal efficiency of the engine is :  
(Take  $C_v = 1.5 R$ , where R is gas constant)**



**[Online April 8, 2017]**

**Options:**

A. 0.24

B. 0.15

C. 0.32



D. 0.08

**Answer: B**

**Solution:**

Work-done ( $W$ ) =  $P_0 V_0$

According to principle of calorimetry

Heat given =  $Q_{AB} = Q_{BC}$

=  $nC_v dT_{AB} + nC_p dT_{BC}$

=  $\frac{3}{2}(nRT_B - nRT_A) + \frac{5}{2}(nRT_C - nRT_B)$

=  $\frac{3}{2}(2P_0 V_0 - P_0 V_0) + \frac{5}{2}(4P_0 V_0 - 2P_0 V_0)$

=  $\frac{13}{2}P_0 V_0$

Thermal efficiency of engine ( $\eta$ ) =  $\frac{W}{Q_{\text{given}}} = \frac{2}{13} = 0.15$

---

## Question217

**200g water is heated from 40°C to 60°C. Ignoring the slight expansion of water, the change in its internal energy is close to (Given specific heat of water = 4184 J/kgK):**  
**[Online April 9, 2016]**

**Options:**

A. 167.4 kJ

B. 8.4 kJ

C. 4.2 kJ

D. 16.7 kJ

**Answer: D**

**Solution:**

Volume of water does not change, no work is done on or by the system ( $W = 0$ )

According to first law of thermodynamics

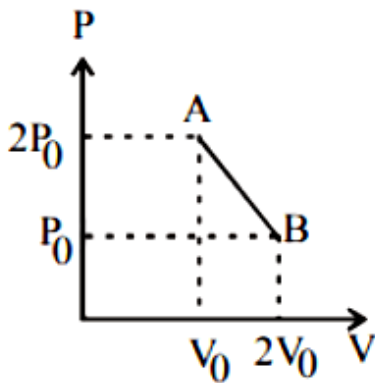
$Q = \Delta U + W$

For Isochoric process  $Q = \Delta U$   
 $\Delta U = \mu c_d T = 2 \times 4184 \times 20 = 16.7 \text{ kJ}$

---

## Question 218

'n' moles of an ideal gas undergoes a process  $A \rightarrow B$  as shown in the figure. The maximum temperature of the gas during the process will be :



[2016]

Options:

A.  $\frac{9P_0 V_0}{2nR}$

B.  $\frac{9P_0 V_0}{nR}$

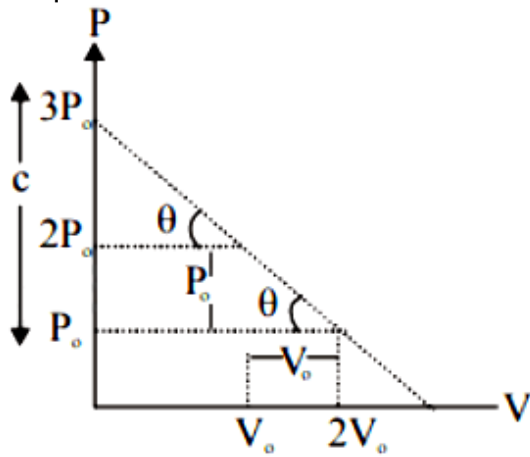
C.  $\frac{9P_0 V_0}{4nR}$

D.  $\frac{3P_0 V_0}{2nR}$

**Answer: C**

**Solution:**

The equation for the line is



$$P = \frac{-P_0}{V_0}V + 3P_0 \left[ \text{slope} = \frac{-P_0}{V_0}, c = 3P_0 \right]$$

$$PV_0 + P_0V = 3P_0V_0 \dots (i)$$

But  $pV = nRT$

$$\therefore P = \frac{nRT}{V} \dots (ii)$$

From (i) & (ii)

$$\frac{nRT}{V}V_0 + P_0V = 3P_0V_0$$

From (i) & (ii)

$$\frac{nRT}{V}V_0 + P_0V = 3P_0V_0$$

$$\therefore nRT V_0 + P_0V^2 = 3P_0V_0V$$

For temperature to be maximum  $\frac{dT}{dV} = 0$

Differentiating e.q. (iii) by 'V' we get

$$nRV_0 \frac{dT}{dV} + P_0(2V) = 3P_0V_0$$

$$\therefore nRV_0 \frac{dT}{dV} = 3P_0V_0 - 2P_0V$$

$$\frac{dT}{dV} = \frac{3P_0V_0 - 2P_0V}{nRV_0} = 0$$

$$V = \frac{3V_0}{2} \quad \therefore P = \frac{3P_0}{2} \left[ \text{From (i)} \right]$$

$$\therefore T_{\max} = \frac{9P_0V_0}{4nR} \left[ \text{From (iii)} \right]$$

## Question 219

The ratio of work done by an ideal monoatomic gas to the heat supplied to it in an isobaric process is :

[Online April 9, 2016]

Options:

A.  $\frac{2}{5}$

B.  $\frac{3}{2}$

C.  $\frac{3}{5}$

D.  $\frac{2}{3}$

**Answer: A**

**Solution:**

Efficiency of heat engine is given by

$$\eta = \frac{W}{Q} = 1 - \frac{C_V}{C_P} = \frac{R}{C_P} = \frac{R}{\frac{5R}{2}} = \frac{2}{5}$$

$$(\because C_P - C_V = R)$$

For monoatomic gas  $C_P = \frac{5}{2}R$ .

---

## Question220

**A Carnot freezer takes heat from water at  $0^\circ\text{C}$  inside it and rejects it to the room at a temperature of  $27^\circ\text{C}$ . The latent heat of ice is  $336 \times 10^3 \text{ J kg}^{-1}$ . If 5kg of water at  $0^\circ\text{C}$  is converted into ice at  $0^\circ\text{C}$  by the freezer, then the energy consumed by the freezer is close to : [Online April 10, 2016]**

**Options:**

A.  $1.51 \times 10^5 \text{ J}$

B.  $1.68 \times 10^6 \text{ J}$

C.  $1.71 \times 10^7 \text{ J}$

D.  $1.67 \times 10^5 \text{ J}$

**Answer: D**

**Solution:**

$$\Delta H = mL = 5 \times 336 \times 10^3 = Q_{\text{sink}}$$

$$\frac{Q_{\text{sink}}}{Q_{\text{source}}} = \frac{T_{\text{sink}}}{T_{\text{source}}}$$

$$\therefore Q_{\text{source}} = \frac{T_{\text{source}}}{T_{\text{sink}}} \times Q_{\text{sink}}$$

Energy consumed by freezer

$$\therefore W_{\text{output}} = Q_{\text{source}} - Q_{\text{sink}} = Q_{\text{sink}} \left( \frac{T_{\text{source}}}{T_{\text{sink}}} - 1 \right)$$

$$\text{Given: } T_{\text{source}} = 27^\circ\text{C} + 273 = 300\text{K}$$

$$T_{\text{sink}} = 0^\circ\text{C} + 273 = 273\text{k}$$

$$W_{\text{output}} = 5 \times 336 \times 10^3 \left( \frac{300}{273} - 1 \right) = 1.67 \times 10^5 \text{J}$$

## Question221

**Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as  $V^q$  where  $V$  is the volume of the gas.**

**The value of  $q$  is:  $\left( \gamma = \frac{C_p}{C_v} \right)$**

**[2015]**

**Options:**

A.  $\frac{\gamma + 1}{2}$

B.  $\frac{\gamma - 1}{2}$

C.  $\frac{3\gamma + 5}{6}$

D.  $\frac{3\gamma - 5}{6}$

**Answer: A**

**Solution:**

$$\tau = \frac{1}{\sqrt{2}\pi d^2 \left(\frac{N}{V}\right) \sqrt{\frac{3RT}{M}}}$$

$$\mu \frac{V}{\sqrt{T}}$$

$$\text{As, } T V^{\gamma-1} = K$$

$$\text{So, } \tau \propto V^{\gamma+1/2}$$

$$\text{Therefore, } q = \frac{\gamma+1}{2}$$

## Question222

Consider a spherical shell of radius  $R$  at temperature  $T$ . The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume  $u = \frac{U}{V} \propto T^4$  and pressure

$p = \frac{1}{3} \left( \frac{U}{V} \right)$ . If the shell now undergoes an adiabatic expansion the relation between  $T$  and  $R$  is:  
[2015]

Options:

A.  $T \propto \frac{1}{R}$

B.  $T \propto \frac{1}{R^3}$

C.  $T \propto e^{-R}$

D.  $T \propto e^{-3R}$

**Answer: A**

**Solution:**

$$\text{As, } P = \frac{1}{3} \left( \frac{U}{V} \right)$$

$$\text{But } \frac{U}{V} = K T^4$$

$$\text{So, } P = \frac{1}{3} K T^4$$

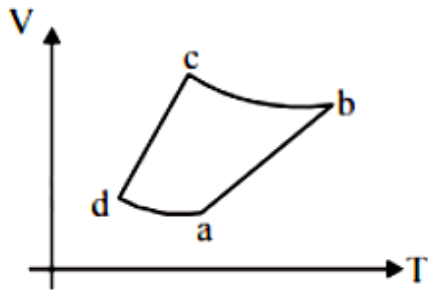
$$\text{or } \frac{uRT}{V} = \frac{1}{3} K T^4 \left[ \text{As } PV = uRT \right]$$

$$\frac{4}{3}pR^3T^3 = \text{constant}$$

$$\text{Therefore, } T \propto \frac{1}{R}$$

## Question223

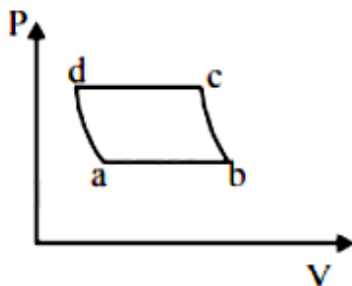
An ideal gas goes through a reversible cycle  $a \rightarrow b \rightarrow c \rightarrow d$  has the  $V - T$  diagram shown below. Process  $d \rightarrow a$  and  $b \rightarrow c$  are adiabatic.



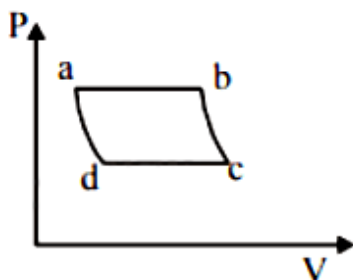
The corresponding  $P - V$  diagram for the process is (all figures are schematic and not drawn to scale) :  
[Online April 10, 2015]

Options:

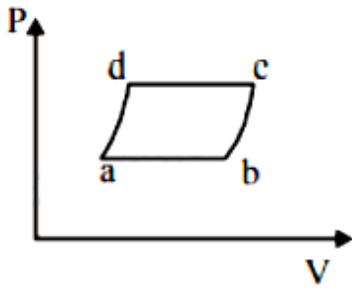
A.



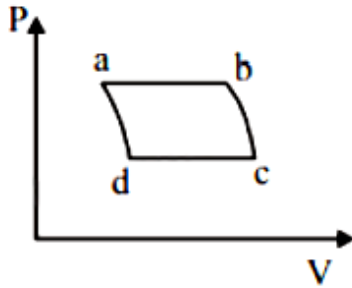
B.



C.



D.



**Answer: B**

**Solution:**

In VT graph

ab-process : Isobaric, temperature increases.

bc process : Adiabatic, pressure decreases.

cd process : Isobaric, volume decreases.

da process : Adiabatic, pressure increases.

The above processes correctly represented in P-V diagram (b).

## Question224

**A solid body of constant heat capacity  $1 \text{ J/}^\circ\text{C}$  is being heated by keeping it in contact with reservoirs in two ways :**

**(i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplies same amount of heat.**

**(ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat.**

**In both the cases body is brought from initial temperature  $100^\circ\text{C}$  to final temperature  $200^\circ\text{C}$ . Entropy change of the body in the two cases respectively is :**

**[2015]**

**Options:**



- A.  $\ln 2, 2\ln 2$
- B.  $2\ln 2, 8\ln 2$
- C.  $\ln 2, 4\ln 2$
- D.  $\ln 2, \ln 2$

**Answer: D**

### **Solution:**

The entropy change of the body

---

## **Question225**

**A gas is compressed from a volume of  $2\text{m}^3$  to a volume of  $1\text{m}^3$  at a constant pressure of  $100 \text{ N/m}^2$ . Then it is heated at constant volume by supplying  $150 \text{ J}$  of energy. As a result, the internal energy of the gas:**

**[Online April 19, 2014]**

### **Options:**

- A. increases by  $250 \text{ J}$
- B. decreases by  $250 \text{ J}$
- C. increases by  $50 \text{ J}$
- D. decreases by  $50 \text{ J}$

**Answer: A**

### **Solution:**

As we know,

$$\Delta Q = \Delta u + \Delta w \text{ (1st law of thermodynamics)}$$

$$\Rightarrow \Delta Q = \Delta u + P\Delta v$$

$$\text{or } 150 = \Delta u + 100(1 - 2)$$

$$= \Delta u - 100$$

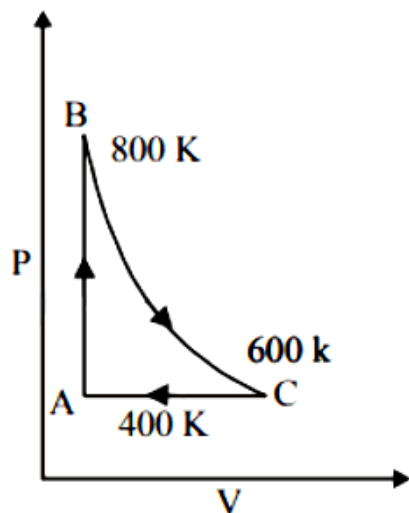
$$\therefore \Delta u = 150 + 100 = 250\text{J}$$

Thus the internal energy of the gas increases by  $250\text{J}$

---

## Question 226

One mole of a diatomic ideal gas undergoes a cyclic process ABC as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are 400 K, 800 K and 600 K respectively. Choose the correct statement:



[2014]

Options:

- A. The change in internal energy in whole cyclic process is 250 R.
- B. The change in internal energy in the process CA is 700 R.
- C. The change in internal energy in the process AB is - 350 R.
- D. The change in internal energy in the process BC is – 500 R.

**Answer: D**

**Solution:**

In cyclic process, change in total internal energy is zero.

$$\Delta U_{\text{cyclic}} = 0$$

$$\Delta U_{BC} = nC_v\Delta T = 1 \times \frac{5R}{2}\Delta T$$

Where,  $C_v$  = molar specific heat at constant volume.

For BC,  $\Delta T = -200\text{K}$

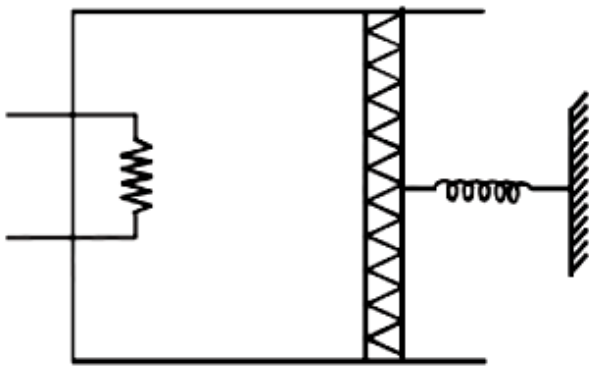
$$\therefore \Delta U_{BC} = -500R$$

---

## Question227

An ideal monoatomic gas is confined in a cylinder by a spring loaded piston of cross section  $8.0 \times 10^{-3} \text{ m}^2$ . Initially the gas is at 300K and occupies a volume of  $2.4 \times 10^{-3} \text{ m}^3$  and the spring is in its relaxed state as shown in figure. The gas is heated by a small heater until the piston moves out slowly by 0.1m. The force constant of the spring is  $8000 \text{ N/m}$  and the atmospheric pressure is  $1.0 \times 10^5 \text{ N/m}^2$ . The cylinder and the piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and the cylinder. The final temperature of the gas will be:

(Neglect the heat loss through the lead wires of the heater. The heat capacity of the heater coil is also negligible).



[Online April 11, 2014]

Options:

- A. 300 K
- B. 800 K
- C. 500 K
- D. 1000 K

**Answer: C**

---

## Question228

During an adiabatic compression, 830J of work is done on 2 moles of a diatomic ideal gas to reduce its volume by 50%. The change in its temperature is nearly:

( $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ )

[Online April 11, 2014]

Options:

A. 40 K

B. 33 K

C. 20 K

D. 14 K

Answer: C

Solution:

Given : work done,  $W = 830 \text{ J}$

No. of moles of gas,  $\mu = 2$

For diatomic gas  $\gamma = 1.4$

Work done during an adiabatic change

$$W = \frac{\mu R (T_1 - T_2)}{\gamma - 1}$$

$$\Rightarrow 830 = \frac{2 \times 8.3 (\Delta T)}{1.4 - 1} = \frac{2 \times 8.3 (\Delta T)}{0.4}$$

$$\Rightarrow \Delta T = \frac{830 \times 0.4}{2 \times 8.3} = 20 \text{ K}$$

---

## Question229

The equation of state for a gas is given by  $PV = nRT + \alpha V$ , where  $n$  is the number of moles and  $\alpha$  is a positive constant. The initial temperature and pressure of one mole of the gas contained in a cylinder are  $T_0$  and  $P_0$  respectively. The work done by the gas when its temperature doubles isobarically will be:

[Online April 9, 2014]

**Options:**

A.  $\frac{P_0 T_0 R}{P_0 - \alpha}$

B.  $\frac{P_0 T_0 R}{P_0 + \alpha}$

C.  $P_0 T_0 R \ln 2$

D.  $P_0 T_0 R$

**Answer: A**

---

## Question230

**A Carnot engine absorbs 1000 J of heat energy from a reservoir at 127°C and rejects 600 J of heat energy during each cycle. The efficiency of engine and temperature of sink will be:  
[Online April 12, 2014]**

**Options:**

A. 20% and – 43°C

B. 40% and – 33°C

C. 50% and – 20°C

D. 70% and – 10°C

**Answer: B**

**Solution:**

Given :

$$Q_1 = 1000\text{J}$$

$$Q_2 = 600\text{J}$$

$$T_1 = 127^\circ\text{C} = 400\text{K}$$

$$T_2 = ?$$

$$\eta = ?$$

Efficiency of carnot engine,

$$\eta = \frac{W}{Q_1} \times 100\%$$

$$\text{or, } \eta = \frac{Q_2 - Q_1}{Q_1} \times 100\%$$

$$\text{or, } \eta = \frac{1000 - 600}{1000} \times 100\%$$

$$\eta = 40\%$$

$$\text{Now, for carnot cycle } \frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

$$\frac{600}{1000} = \frac{T_2}{400}$$

$$T_2 = 600 \times 400 / 1000$$

$$= 240\text{K}$$

$$= 240 - 273$$

$$\therefore T_2 = -33^\circ\text{C}$$

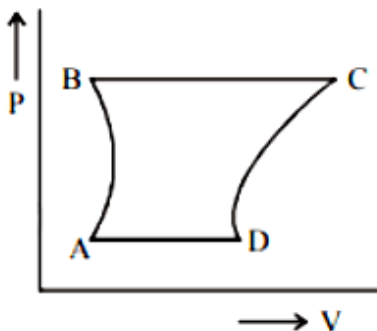
## Question 231

A certain amount of gas is taken through a cyclic process (A B C D A) that has two isobars, one isochore and one isothermal. The cycle can be represented on a P-V indicator diagram as :

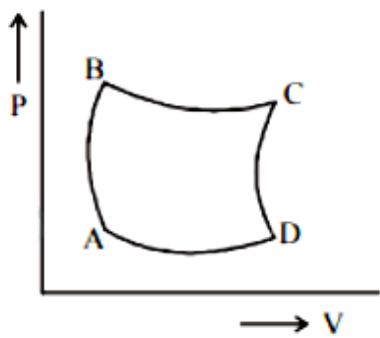
[Online April 22, 2013]

Options:

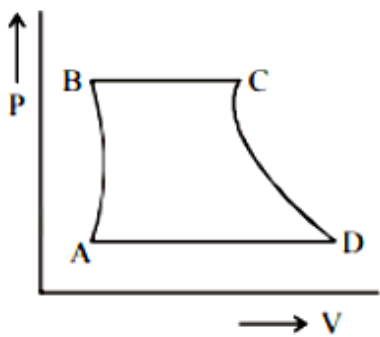
A.



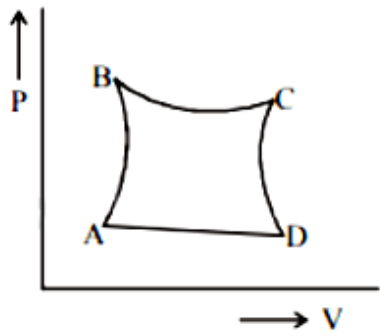
B.



C.



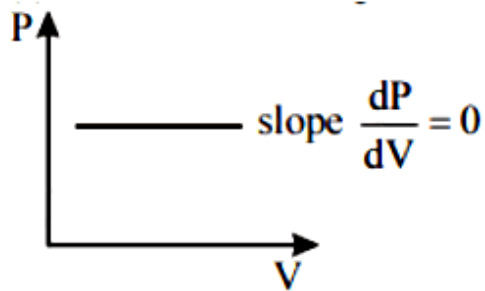
D.



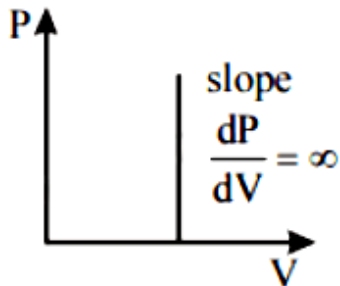
**Answer: C**

**Solution:**

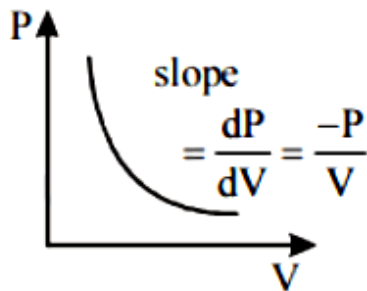
P-V indicator diagram for isobaric



P-V indicator diagram for isochoric process



P-V indicator diagram for isothermal process



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## Question232

An ideal gas at atmospheric pressure is adiabatically compressed so that its density becomes 32 times of its initial value. If the final pressure of gas is 128 atmospheres, the value of ' $\gamma$ ' of the gas is :  
[Online April 22, 2013]

Options:

- A. 1.5
- B. 1.4
- C. 1.3
- D. 1.6

**Answer: B**



## Solution:

Volume of the gas  $v = \frac{m}{d}$  and

Using  $PV^\gamma = \text{constant}$

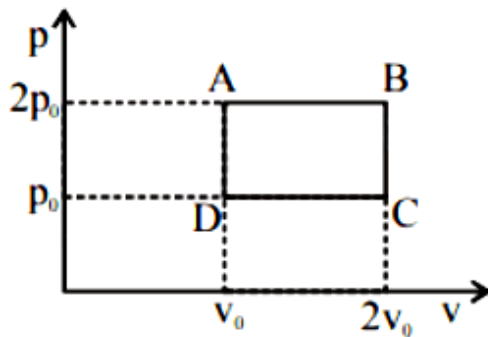
$$\frac{P'}{P} = \frac{V}{V'} = \left(\frac{d'}{d}\right)^\gamma$$

$$\text{or } 128 = (32)^\gamma$$

$$\therefore \gamma = \frac{7}{5} = 1.4$$

---

## Question 233



The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monatomic gas. The amount of heat, extracted from the source in a single cycle is  
[2013]

Options:

A.  $p_0 v_0$

B.  $\left(\frac{13}{2}\right) p_0 v_0$

C.  $\left(\frac{11}{2}\right) p_0 v_0$

D.  $4 p_0 v_0$

**Answer: B**

**Solution:**

Heat is extracted from the source in path DA and AB is

$$\Delta Q = \frac{3}{2}R \left( \frac{P_0 V_0}{R} \right) + \frac{5}{2}R \left( \frac{2P_0 V_0}{R} \right)$$
$$\Rightarrow \frac{3}{2}P_0 V_0 + \frac{5}{2}2P_0 V_0 = \left( \frac{13}{2} \right) P_0 V_0$$

---

## Question 234

**This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.**

**Statement 1: An inventor claims to have constructed an engine that has an efficiency of 30% when operated between the boiling and freezing points of water. This is not possible.**

**Statement 2: The efficiency of a real engine is always less than the efficiency of a Carnot engine operating between the same two temperatures.**

**[Online May 19, 2012]**

**Options:**

- A. Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
- B. Statement 1 is true, Statement 2 is false.
- C. Statement 1 is false, Statement 2 is true.
- D. Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

**Answer: D**

**Solution:**

According to Carnot's theorem - no heat engine working between two given temperatures of source and sink can be more efficient than a perfectly reversible engine i.e. Carnot engine working between the same two temperatures.

Efficiency of Carnot's engine,  $\eta = 1 - \frac{T_2}{T_1}$

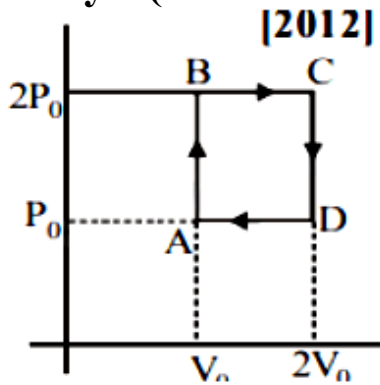
where,  $T_1$  = temperature of source

$T_2$  = temperature of sink

---

## Question 235

Helium gas goes through a cycle ABCDA (consisting of two isochoric and isobaric lines) as shown in figure. The efficiency of this cycle is nearly : (Assume the gas to be close to ideal gas)



**[2012]**

**Options:**

A. 15.4 %

B. 9.1 %

C. 10.5%

D. 12.5 %

**Answer: A**

**Solution:**

The efficiency  $\eta = \frac{\text{output work}}{\text{heat given to the system}}$

$$= n \frac{3}{2} R \Delta T = \frac{3}{2} V_0 \Delta P = \frac{3}{2} P_0 V_0$$

$$W_i = \frac{n}{2} (P_0 V_0) + \frac{n}{2} (2P_0 V_0) + 2P_0 V_0$$

Heat given in going B to C =  $nC_p \Delta T$

$$= n \left( \frac{5}{2} R \right) \Delta T = \frac{5}{2} (2P_0) \Delta V$$

$$= 5P_0 V_0$$

and  $W_0 =$  area under PV diagram  $P_0 V_0$

$$\eta = \frac{W}{Q} = \frac{P_0 V_0}{\frac{13}{2} P_0 V_0} = \frac{2}{13}$$

Efficiency in %

$$\eta = \frac{2}{13} \times 100 = \frac{200}{13} \approx 15.4\%$$

## Question236

An ideal monatomic gas with pressure  $P$ , volume  $V$  and temperature  $T$  is expanded isothermally to a volume  $2V$  and a final pressure  $P_i$ . If the same gas is expanded adiabatically to a volume  $2V$ , the final pressure is  $P_a$ . The ratio  $\frac{P_a}{P_i}$  is

[Online May 26, 2012]

**Options:**

A.  $2^{-1/3}$

B.  $2^{1/3}$

C.  $2^{2/3}$

D.  $2^{-2/3}$

**Answer: D**

**Solution:**

For isothermal process :

$$PV = P_i 2V$$

$$P = 2P_i \dots\dots(i)$$

For adiabatic process

$$PV^\gamma = P_a (2V)^\gamma$$

( $\because$  for monatomic gas  $\gamma = 5/3$ )

$$\text{or, } 2P_i V^{\frac{5}{3}} = P_a (2V)^{\frac{5}{3}} \text{ [From (i)]}$$

$$\Rightarrow \frac{P_a}{P_i} = \frac{2}{2^{\frac{5}{3}}}$$

$$\Rightarrow \frac{P_a}{P_i} = 2^{-\frac{2}{3}}$$

---

## Question237

The pressure of an ideal gas varies with volume as  $P = \alpha V$ , where  $\alpha$  is a constant. One mole of the gas is allowed to undergo expansion such that its volume becomes ' m ' times its initial volume. The work done by the gas in the process is  
[Online May 19, 2012]

Options:

A.  $\frac{\alpha V}{2}(m^2 - 1)$

B.  $\frac{\alpha^2 V^2}{2}(m^2 - 1)$

C.  $\frac{\alpha}{2}(m^2 - 1)$

D.  $\frac{\alpha V^2}{2}(m^2 - 1)$

**Answer: D**

**Solution:**

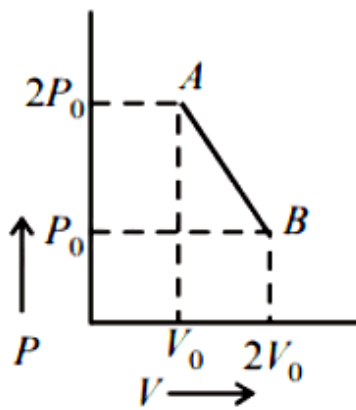
Given  $P = \alpha V$

$$\begin{aligned}\text{Work done, } w &= \int_V^{mV} P dV \\ &= \int_V^{mV} \alpha V dV = \frac{\alpha V^2}{2}(m^2 - 1).\end{aligned}$$

---

## Question238

n moles of an ideal gas undergo a process  $A \rightarrow B$  as shown in the figure. Maximum temperature of the gas during the process is



[Online May 12, 2012]

Options:

A.  $\frac{9P_0V_0}{nR}$

B.  $\frac{3P_0V_0}{2nR}$

C.  $\frac{9P_0V_0}{2nR}$

D.  $\frac{9P_0V_0}{4nR}$

**Answer: B**

**Solution:**

Work done during the process  $A \rightarrow B$

= Area of trapezium (= area bounded by indicator diagram with V-axis)

$$= \frac{1}{2}(2P_0 + P_0)(2V_0 - V_0) = \frac{3}{2}P_0V_0$$

Ideal gas eqn :  $PV = nRT$

$$\Rightarrow T = \frac{PV}{nR} = \frac{3P_0V_0}{2nR}$$

## Question239

**This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.**

**Statement 1: In an adiabatic process, change in internal energy of a**

**gas is equal to work done on/by the gas in the process.**

**Statement 2: The temperature of a gas remains constant in an adiabatic process.**

**[Online May 7, 2012]**

**Options:**

- A. Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.
- B. Statement 1 is true, Statement 2 is false
- C. Statement 1 is false, Statement 2 is true.
- D. Statement 1 is false, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

**Answer: B**

**Solution:**

In an adiabatic process,  $\delta H = 0$   
And according to first law of thermodynamics  
 $\delta H = \delta U + W$   
 $\therefore W = -\delta U$

-----

## **Question240**

**A Carnot engine, whose efficiency is 40%, takes in heat from a source maintained at a temperature of 500K. It is desired to have an engine of efficiency 60%. Then, the intake temperature for the same exhaust (sink) temperature must be :  
[2012]**

**Options:**

- A. efficiency of Carnot engine cannot be made larger than 50%
- B. 1200 K
- C. 750 K
- D. 600 K

**Answer: C**

### **Solution:**

The efficiency of the carnot's heat engine is given as

$$\eta = \left( 1 - \frac{T_2}{T_1} \right) \times 100$$

When efficiency is 40%,

$$T_1 = 500\text{K}; \eta = 40$$

$$40 = \left( 1 - \frac{T_2}{500} \right) \times 100$$

$$\Rightarrow \frac{40}{100} = 1 - \frac{T_2}{500}$$

$$\Rightarrow \frac{T_2}{500} = \frac{60}{100} \Rightarrow T_2 = 300\text{K}$$

When efficiency is 60%, then

$$\frac{60}{100} = \left( 1 - \frac{300}{T_2} \right) \Rightarrow \frac{300}{T_2} = \frac{40}{100}$$

$$\Rightarrow T_2 = \frac{100 \times 300}{40} \Rightarrow T_2 = 750\text{K}$$

---

## **Question241**

**The door of a working refrigerator is left open in a well insulated room. The temperature of air in the room will**  
**[Online May 26, 2012]**

**Options:**

- A. decrease
- B. increase in winters and decrease in summers
- C. remain the same
- D. increase

**Answer: D**

### **Solution:**

In a refrigerator, the heat dissipated in the atmosphere is more than that taken from the cooling chamber, therefore the room is heated. If the door of a refrigerator is kept open.

---



## Question242

A container with insulating walls is divided into equal parts by a partition fitted with a valve. One part is filled with an ideal gas at a pressure  $P$  and temperature  $T$ , whereas the other part is completely evacuated. If the valve is suddenly opened, the pressure and temperature of the gas will be :  
[2011 RS]

Options:

A.  $\frac{P}{2}, \frac{T}{2}$

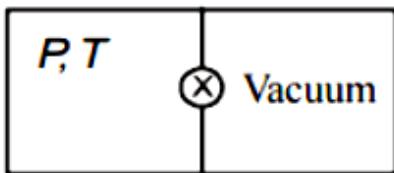
B.  $P, T$

C.  $P, \frac{T}{2}$

D.  $\frac{P}{2}, T$

Answer: D

Solution:



It is the free expansion  
 $\therefore$  So,  $T$  remains constant

$$\Rightarrow P_1 V_1 = P_2 V_2$$

$$\Rightarrow P \frac{V}{2} = P_2 (V)$$

$$P_2 = \left( \frac{P}{2} \right)$$

---

## Question243

A Carnot engine operating between temperatures  $T_1$  and  $T_2$  has efficiency  $\frac{1}{6}$ . When  $T_2$  is lowered by 62K its efficiency increases to  $\frac{1}{3}$ .

**Then  $T_1$  and  $T_2$  are, respectively:**

**[2011]**

**Options:**

A. 372 K and 310 K

B. 330 K and 268 K

C. 310 K and 248 K

D. 372 K and 310 K

**Answer: D**

**Solution:**

Efficiency of engine

$$\eta_1 = 1 - \frac{T_2}{T_1} = \frac{1}{6}$$

$$\Rightarrow \frac{T_2}{T_1} = \frac{5}{6} \dots\dots\dots(i)$$

When  $T_2$  is lowered by 62K, then

$$\text{Again, } \eta_2 = 1 - \frac{T_2 - 62}{T_1}$$

$$= 1 - \frac{T_2}{T_1} + \frac{62}{T_1} = \frac{1}{3} \dots\dots\dots(ii)$$

Solving (i) and (ii), we get,

$$T_1 = 372\text{K} \text{ and } T_2 = \frac{5}{6} \times 372 = 310\text{K}$$

---

## Question244

**A diatomic ideal gas is used in a Carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from  $V$  to  $32V$ , the efficiency of the engine is**

**[2010]**

**Options:**

A. 0.5

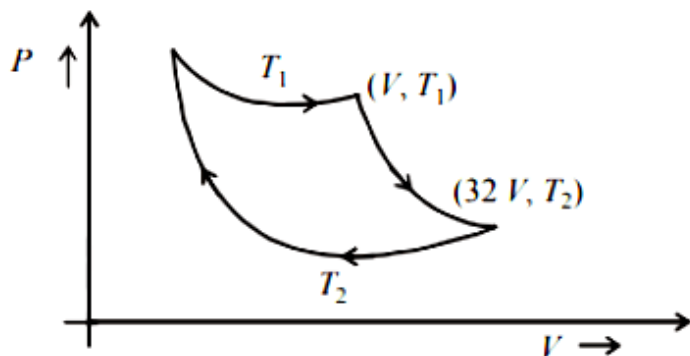
B. 0.75

C. 0.99

D. 0.25

**Answer: B**

**Solution:**



For adiabatic expansion  $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

$$\Rightarrow T_1 V^{\gamma-1} = T_2 (32V)^{\gamma-1}$$

$$\Rightarrow \frac{T_1}{T_2} = (32)^{\gamma-1}$$

For diatomic gas,  $\gamma = \frac{7}{5}$

$$\therefore \gamma - 1 = \frac{2}{5}$$

$$\therefore \frac{T_1}{T_2} = (32)^{\frac{2}{5}} \Rightarrow T_1 = 4T_2$$

Now, efficiency  $= 1 - \frac{T_2}{T_1}$

$$= 1 - \frac{T_2}{4T_2} = 1 - \frac{1}{4} = \frac{3}{4} = 0.75.$$

---

## Question 245

Assuming the gas to be ideal the work done on the gas intaking it from A to B is

[2009]

**Options:**

A. 300 R

B. 400 R

C. 500 R

D. 200 R

**Answer: B**

**Solution:**

The process  $A \rightarrow B$  is isobaric.

$$\begin{aligned}\therefore \text{work done } W_{AB} &= nR(T_2 - T_1) \\ &= 2R(500 - 300) = 400R\end{aligned}$$

---

## Question 246

**The work done on the gas in taking it from D to A is [2009]**

**Options:**

A. + 414 R

B. – 690 R

C. + 690 R

D. – 414 R

**Answer: A**

**Solution:**

The process D to A is isothermal as temperature is constant.

$$\text{Work done, } W_{DA} = 2.303nRT \log_{10} \frac{P_D}{P_A}$$

$$= 2.303 \times 2R \times 300$$

$$\log_{10} \frac{1 \times 10^5}{2 \times 10^5} - 414R$$

Therefore, work done on the gas is +414R.

---

## Question 247

**The net work done on the gas in the cycle ABCDA is  
[2009]**

**Options:**

- A. 279 R
- B. 1076 R
- C. 1904 R
- D. zero

**Answer: A**

**Solution:**

The net work in the cycle ABCDA is

$$\begin{aligned} W &= W_{AB} + W_{BC} + W_{CD} + W_{DA} \\ &= 400R + 2.303nRT \log \frac{P_B}{P_C} + (-400R) - 414R \\ &= 2.303 \times 2R \times 500 \log \frac{2 \times 10^5}{1 \times 10^5} - 414R \\ &= 693.2R - 414R = 279.2R \end{aligned}$$

-----

## Question 248

**An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume  $V_1$  and contains ideal gas at pressure  $P_1$  and temperature  $T_1$ . The other chamber has volume  $V_2$  and contains ideal gas at pressure  $P_2$  and temperature  $T_2$ . If the partition is removed without doing any work on the gas, the final equilibrium temperature of the gas in the container will be  
[2008]**

**Options:**

- A.  $\frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_2 + P_2 V_2 T_1}$
- B.  $\frac{P_1 V_1 T_1 + P_2 V_2 T_2}{P_1 V_1 + P_2 V_2}$

$$C. \frac{P_1 V_1 T_2 + P_2 V_2 T_1}{P_1 V_1 + P_2 V_2}$$

$$D. \frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_1 + P_2 V_2 T_2}$$

**Answer: A**

**Solution:**

Here  $Q = 0$  and  $W = 0$ . Therefore from first law of thermodynamics  $\Delta U = Q + W = 0$

Internal energy of first vessel + Internal energy of second vessel = Internal energy of combined vessel

$$n_1 C_v T_1 + n_2 C_v T_2 = (n_1 + n_2) C_v T$$

$$\therefore T = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$$

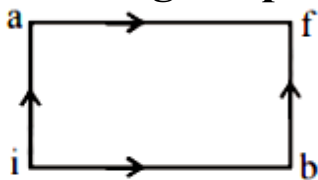
$$\text{For first vessel } n_1 = \frac{P_1 V_1}{RT_1} \text{ and for second vessel } n_2 = \frac{P_2 V_2}{RT_2}$$

$$\therefore T = \frac{\frac{P_1 V_1}{RT_1} \times T_1 + \frac{P_2 V_2}{RT_2} \times T_2}{\frac{P_1 V_1}{RT_1} + \frac{P_2 V_2}{RT_2}}$$

$$= \frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_2 + P_2 V_2 T_1}$$

## Question 249

When a system is taken from state i to state f along the path iaf, it is found that  $Q = 50$  cal and  $W = 20$  cal. Along the path ibf  $Q = 36$  cal.  $W$  along the path ibf is



[2007]

**Options:**

A. 14 cal

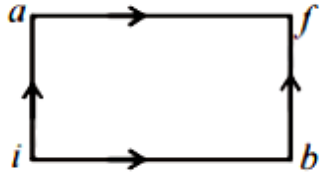
B. 6 cal

C. 16 cal

D. 66 cal

**Answer: B**

**Solution:**



(b) For path iaf,

$$Q_1 = 50\text{cal}, W_1 = 20\text{cal}$$

By first law of thermodynamics,

$$\Delta U = Q_1 - W_1 = 50 - 20 = 30\text{cal}$$

For path ibf

$$Q_2 = 36\text{cal}$$

$$W_2 = ?$$

$$\Delta U_{ibf} = Q_2 - W_2$$

Since, the change in internal energy does not depend on the path, therefore  $\Delta U_{iaf} = \Delta U_{ibf}$

$$\Delta U_{iaf} = \Delta U_{ibf}$$

$$\Rightarrow 30 = Q_2 - W_2$$

$$\Rightarrow W_2 = 36 - 30 = 6 \text{ cal.}$$

---

## Question250

**A Carnot engine, having an efficiency of  $\eta = 1/10$  as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is [2007]**

**Options:**

A. 100 J

B. 99 J

C. 90 J

D. 1 J

**Answer: C**

**Solution:**

The efficiency ( $\eta$ ) of a Carnot engine and the coefficient of performance ( $\beta$ ) of a refrigerator are related as

$$\beta = \frac{1 - \eta}{\eta}$$

$$\text{Also, } \beta = \frac{Q_2}{W}$$

$$\therefore \beta = \frac{1 - \eta}{\eta} = \frac{Q_2}{W}$$

$$\therefore \beta = \frac{1 - \frac{1}{10}}{\left(\frac{1}{10}\right)} = \frac{Q_2}{W}$$

is independent of path taken by the process.

$$\Rightarrow 9 = \frac{Q_2}{10}$$

$$\Rightarrow Q_2 = 90\text{J}.$$

## Question251

**The work of 146 kJ is performed in order to compress one kilo mole of gas adiabatically and in this process the temperature of the gas increases by 7°C. The gas is**

**( $R = 8.3\text{J mol}^{-1}\text{K}^{-1}$ )**

**[2006]**

**Options:**

- A. diatomic
- B. triatomic
- C. a mixture of monoatomic and diatomic
- D. monoatomic

**Answer: A**

**Solution:**

Work done in adiabatic compression is given by

$$W = \frac{nR\Delta T}{1 - \gamma}$$

$$\Rightarrow -146000 = \frac{1000 \times 8.3 \times 7}{1 - \gamma}$$



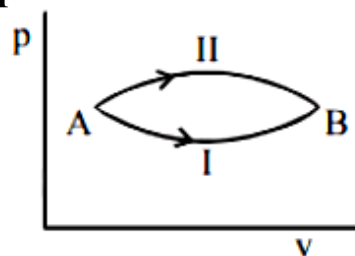
$$\text{or } 1 - \gamma = -\frac{58.1}{146} \Rightarrow \gamma = 1 + \frac{58.1}{146} = 1.4$$

Hence the gas is diatomic.

---

## Question252

A system goes from A to B via two processes I and II as shown in figure. If  $\Delta U_1$  and  $\Delta U_2$  are the changes in internal energies in the processes I and II respectively, then



[2005]

Options:

- A. relation between  $\Delta U_1$  and  $\Delta U_2$  can not be determined
- B.  $\Delta U_1 = \Delta U_2$
- C.  $\Delta U_2 < \Delta U_1$
- D.  $\Delta U_2 > \Delta U_1$

**Answer: B**

**Solution:**

Change in internal energy is independent of path taken by the process. It only depends on initial and final states i.e.,

$$\Delta U_1 = \Delta U_2$$

---

## Question253

Which of the following is incorrect regarding the first law of thermodynamics?

[2005]

**Options:**

- A. It is a restatement of the principle of conservation of energy
- B. It is not applicable to any cyclic process
- C. It does not introduces the concept of the entropy
- D. It introduces the concept of the internal energy

**Answer: 0**

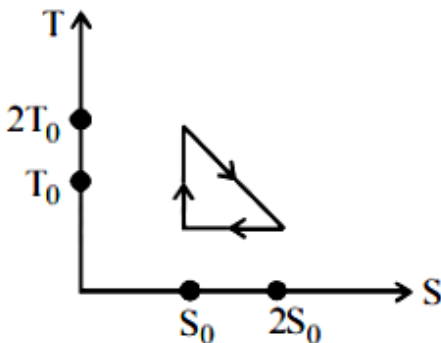
**Solution:**

First law is applicable to a cyclic process. Concept of entropy is introduced by the second law of thermodynamics.

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## Question254

**The temperature-entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is**



**[2005]**

**Options:**

- A.  $\frac{1}{4}$
- B.  $\frac{1}{2}$
- C.  $\frac{2}{3}$
- D.  $\frac{1}{3}$

**Answer: D**

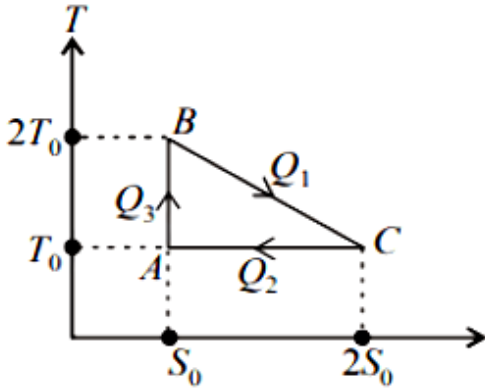
## Solution:

$$Q_1 = \text{area under BC} = T_0 S_0 + \frac{1}{2} T_0 S_0$$

$$Q_2 = \text{area under AC} = T_0 (2S_0 - S_0) = T_0 S_0$$

$$\text{and } Q_3 = 0$$

$$\text{Efficiency, } \eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$



$$= 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_0 S_0}{\frac{3}{2} T_0 S_0} = \frac{1}{3}$$

---

## Question255

Which of the following statements is correct for any thermodynamic system ?

[2004]

Options:

- A. The change in entropy can never be zero
- B. Internal energy and entropy are state functions
- C. The internal energy changes in all processes
- D. The work done in an adiabatic process is always zero.

**Answer: B**

## Solution:

Internal energy and entropy are state function, they are independent of path taken.

---

## Question256

**Which of the following parameters does not characterize the thermodynamic state of matter?**  
**[2003]**

**Options:**

- A. Temperature
- B. Pressure
- C. Work
- D. Volume

**Answer: C**

**Solution:**

Work is not a state function. The remaining three parameters are state function.

-----

## Question257

**“Heat cannot by itself flow from a body at lower temperature to a body at higher temperature” is a statement or consequence of**  
**[2003]**

**Options:**

- A. second law of thermodynamics
- B. conservation of momentum
- C. conservation of mass
- D. first law of thermodynamics

**Answer: A**

**Solution:**

This is a consequence of second law of thermodynamics

---

## Question258

**A Carnot engine takes  $3 \times 10^6$  cal of heat from a reservoir at  $627^\circ\text{C}$ , and gives it to a sink at  $27^\circ\text{C}$ . The work done by the engine is [2003]**

**Options:**

A.  $4.2 \times 10^6 \text{J}$

B.  $8.4 \times 10^6 \text{J}$

C.  $16.8 \times 10^6 \text{J}$

D. zero

**Answer: B**

**Solution:**

Here,  $T_1 = 627 + 273 = 900\text{K}$

$T_2 = 27 + 273 = 300\text{K}$

Efficiency,  $\eta = 1 - \frac{T_2}{T_1}$

$= 1 - \frac{300}{900} = 1 - \frac{1}{3} = \frac{2}{3}$

But  $\eta = \frac{W}{Q}$

$\therefore \frac{W}{Q} = \frac{2}{3} \Rightarrow W = \frac{2}{3} \times Q = \frac{2}{3} \times 3 \times 10^6$

$= 2 \times 10^6 \text{ cal}$

$= 2 \times 10^6 \times 4.2 \text{J} = 8.4 \times 10^6 \text{J}$

---

## Question259

**Which statement is incorrect? [2002]**

**Options:**

A. All reversible cycles have same efficiency

- B. Reversible cycle has more efficiency than an irreversible one
- C. Carnot cycle is a reversible one
- D. Carnot cycle has the maximum efficiency in all cycles

**Answer: A**

### **Solution:**

All reversible engines have same efficiencies if they are working for the same temperature of source and sink.

If the temperatures are different, the efficiency is different.

---

## **Question260**

**Even Carnot engine cannot give 100% efficiency because we cannot [2002]**

### **Options:**

- A. prevent radiation
- B. find ideal sources
- C. reach absolute zero temperature
- D. eliminate friction

**Answer: C**

### **Solution:**

In Carnot's cycle we assume frictionless piston, absolute insulation and ideal source and sink (reservoirs).

The efficiency of carnot's cycle  $\eta = 1 - \frac{T_2}{T_1}$

The efficiency of carnot engine will be 100% when its sink( $T_2$ ) is at 0K .

The temperature of 0K (absolute zero) cannot be realised in practice so, efficiency is never 100%.

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